

Mrabulo Session

Problem statement

Create a secured, database-driven web application that will enable users to interact with a classlist. The application must have three roles, an **HoD**, **lecturer** and **student assistant**. The roles must be capable of doing the following:

HoD

- Remove a student from the list.

Lecturer

- Add a student.
- Find a student.
- Edit a student.
- Display the classlist count.
- Display class composition. This entails:
 - Determining the number of male and female students and displaying them.
 - Determining and displaying the class average age.
- Display class list according to the requested age group/range.

Student assistant

- Add a student to the list (shared functionality with the lecturer).
- Find a student (shared functionality with the lecturer).

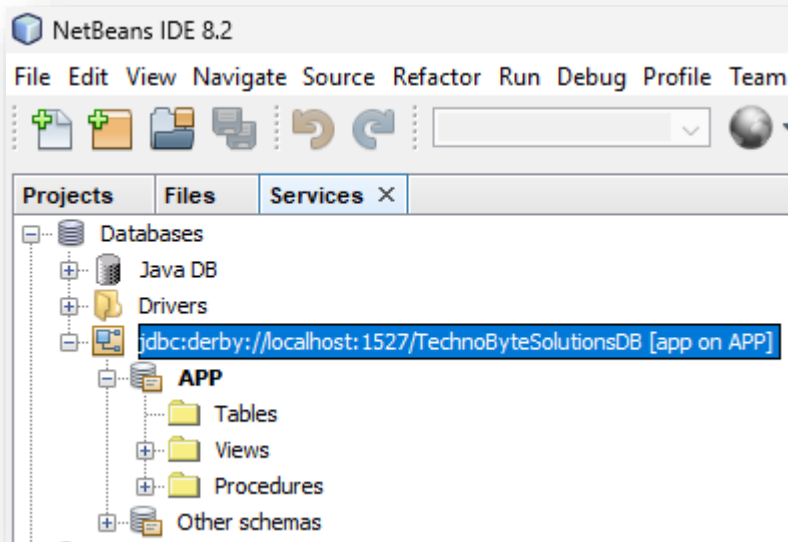
Also note that anyone (authenticated or not authenticated) can view the classlist. Furthermore, create the following specific system users with the assigned roles.

No.	Username	Password	Role(s)
1	Nyiko	123	hod
2	Vuyi	456	lecturer, assistant
3	Thato	6,7,8	assistant

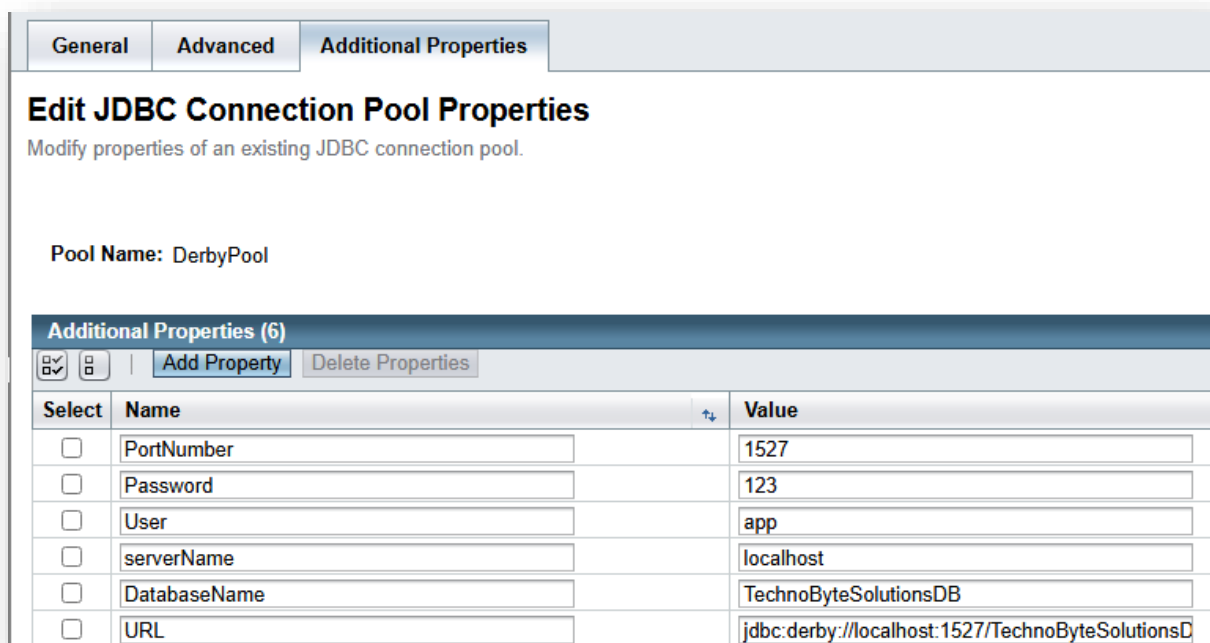
Step 1: Create database, connection pools and data resources.

Create database called **TechnoByteSolutionsDB** with the following credentials:

- **Username:** app
- **Password:** 123



Create the **connection pools**



Ping the **connection pool** to verify that it works

General Advanced Additional Properties

Ping Succeeded

Edit JDBC Connection Pool

Modify an existing JDBC connection pool. A JDBC connection pool is a group of reusable connections for a particular database.

Load Defaults Flush Ping

General Settings

Pool Name: DerbyPool

Resource Type:

Must be specified if the datasource class implements more than 1 of the interface.

Datasource Classname:

Vendor-specific classname that implements the DataSource and/or XADataSource APIs

Driver Classname:

Vendor-specific classname that implements the java.sql.Driver interface.

Ping: ☐ Enabled

When enabled, the pool is pinged during creation or reconfiguration to identify and warn of any errors.

Deployment Order:

Specifies the loading order of the resource at server startup. Lower numbers are loaded first.

Description:

Create/edit the data source, and save.

Edit JDBC Resource

Edit an existing JDBC data source.

Load Defaults

JNDI Name: jdbc/_default

Logical JNDI Name: java:comp/DefaultDataSource

Pool Name:

Use the [JDBC Connection Pools](#) page to create new pools

Deployment Order:

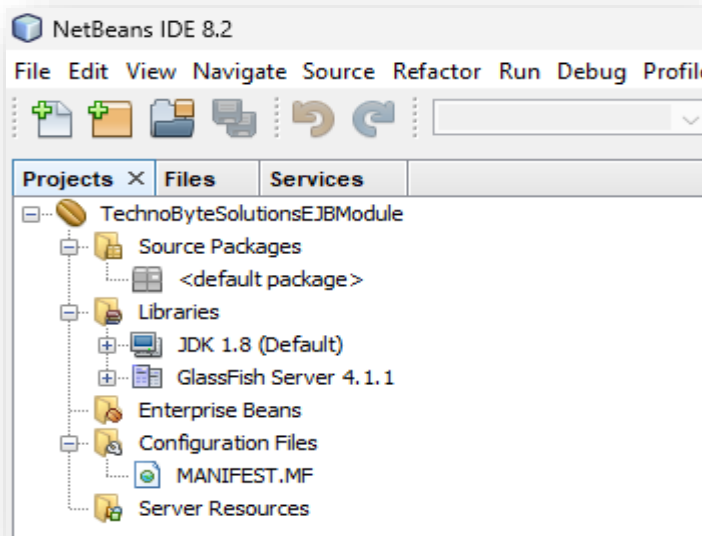
Specifies the loading order of the resource at server startup. Lower numbers are loaded first.

Description:

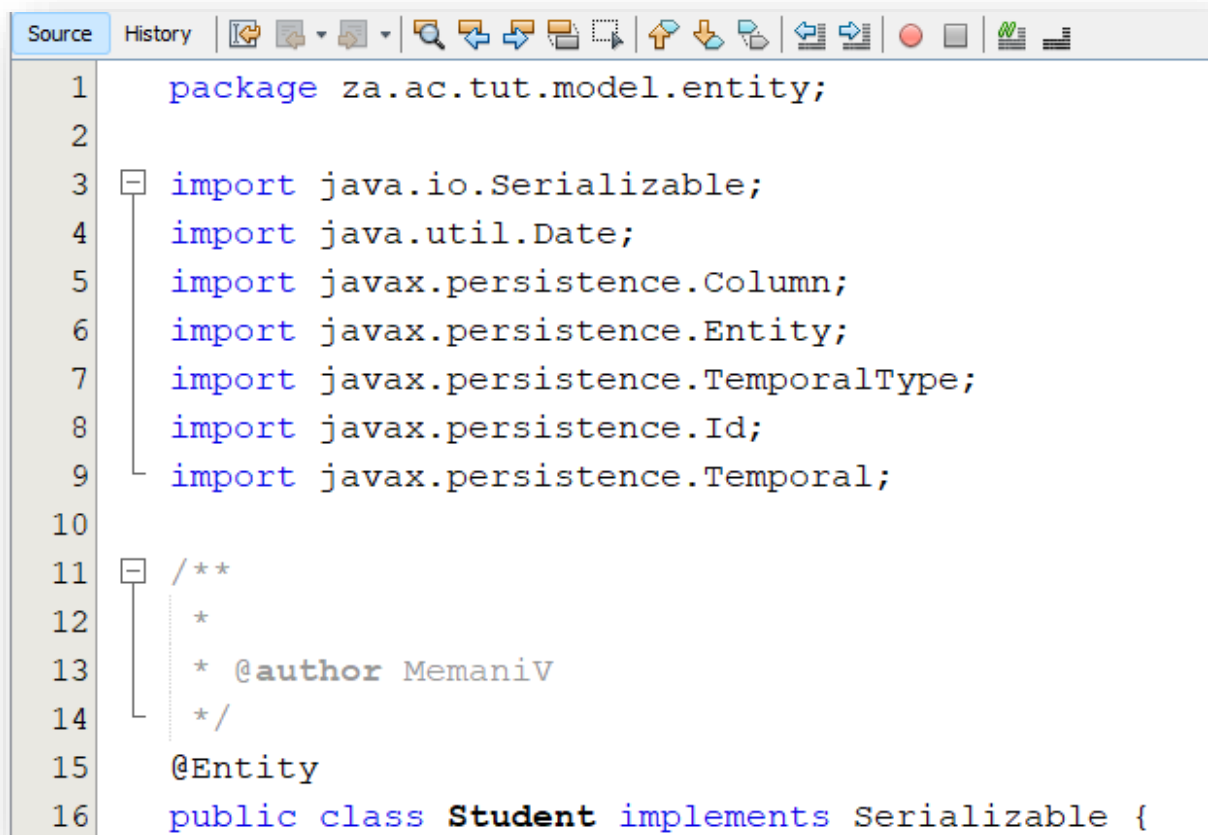
Status: ☒ Enabled

Step 2: Create the EJB module

Create the EJB project called **TechnoByteSolutionsEJBModule**



Create an entity called **Student**



```
18      @Id
19      @Column(name="STUDENT_NUMBER")
20      private Long id;
21      private String name;
22      private String surname;
23      private Integer age;
24      private Character gender;
25      @Temporal(TemporalType.TIMESTAMP)
26      private Date creationDate;
27
28      public Student() {
29      }
30
31      public String getName() {
32          return name;
33      }
34
```

```
35  [ ] public void setName(String name) {  
36      |     this.name = name;  
37      | }  
38  
39  [ ] public String getSurname() {  
40      |     return surname;  
41      | }  
42  
43  [ ] public void setSurname(String surname) {  
44      |     this.surname = surname;  
45      | }  
46  
47  [ ] public Integer getAge() {  
48      |     return age;  
49      | }  
50
```

```
51  [ ] public void setAge(Integer age) {  
52      |     this.age = age;  
53      | }  
54  
55  [ ] public Character getGender() {  
56      |     return gender;  
57      | }  
58  
59  [ ] public void setGender(Character gender) {  
60      |     this.gender = gender;  
61      | }  
62  
63  [ ] public Date getCreationDate() {  
64      |     return creationDate;  
65      | }  
66
```

```

67  public void setCreationDate(Date creationDate) {
68      this.creationDate = creationDate;
69  }
70
71  public Long getId() {
72      return id;
73  }
74
75  public void setId(Long id) {
76      this.id = id;
77  }
78

```

```

79  @Override
80  public int hashCode() {
81      int hash = 0;
82      hash += (id != null ? id.hashCode() : 0);
83      return hash;
84  }
85
86  @Override
87  public boolean equals(Object object) {
88      if (!(object instanceof Student)) {
89          return false;
90      }
91      Student other = (Student) object;
92      if ((this.id == null && other.id != null) ||
93          (this.id != null && !this.id.equals(other.id))) {
94          return false;
95      }
96      return true;
97  }

```

```

99  @Override
100  public String toString() {
101      return "za.ac.tut.web.Student[ id=" + id + " ]";
102  }
103
104  }
105

```

Create session beans for the entity.

- **StudentFacadeLocal**

```
1  package za.ac.tut.model.bl;
2
3  import java.util.List;
4  import javax.ejb.Local;
5  import za.ac.tut.web.Student;
6  /**
7   *
8   * @author MemaniV
9   */
10 @Local
11 public interface StudentFacadeLocal {
12     void create(Student student);
13     void edit(Student student);
14     void remove(Student student);
15     Student find(Object id);
16     List<Student> findAll();
17     List<Student> findRange(int[] range);
18     int count();
19 }
```


- StudentFacade

```

1  package za.ac.tut.model.bl;
2
3  import javax.ejb.Stateless;
4  import javax.persistence.EntityManager;
5  import javax.persistence.PersistenceContext;
6  import za.ac.tut.web.Student;
7
8  /**
9   *
10   * @author MemaniV
11   */
12  @Stateless
13  public class StudentFacade extends AbstractFacade<Student> implements StudentFacadeLocal {
14

```

```

15      @PersistenceContext(unitName = "TechnoByteSolutionsEJBModulePU")
16      private EntityManager em;
17
18      @Override
19      protected EntityManager getEntityManager() {
20          return em;
21      }
22
23      public StudentFacade() {
24          super(Student.class);
25      }
26
27  }

```

Modify the **interface** and session bean to include **other** functionalities

- StudentFacadeLocal

```
10  @Local
11  public interface StudentFacadeLocal {
12      List<Student> findRange(int[] range);
13      List<Student> findAll();
14
15      void remove(Student student);
16
17      void edit(Student student);
18      Long getNumberOfStudentsPerGender(Character gender);
19      List<Student> getStudentsPerGender(Character gender);
20      Long getNumberOfStudentsPerAgeRange(Integer minAge, Integer maxAge);
21      List<Student> getStudentsPerAgeRange(Integer minAge, Integer maxAge);
22      Double getAverageAge();
23
24      void create(Student student);
25      Student find(Object id);
26      int count();
27  }
28
```

- StudentFacade

```
14  @Stateless
15  public class StudentFacade extends AbstractFacade<Student> implements StudentFacadeLocal {
16      @PersistenceContext(unitName = "TechnoByteSolutionsEJBModulePU")
17      private EntityManager em;
18
19      @Override
20      protected EntityManager getEntityManager() {
21          return em;
22      }
23
24      public StudentFacade() {
25          super(Student.class);
26      }
27
```

```

28      @Override
29      public Long getNumberOfStudentsPerGender(Character gender) {
30          String queryStr = "SELECT count(s) FROM Student s WHERE s.gender = ?1";
31          Query query = em.createQuery(queryStr);
32          query.setParameter(1, gender);
33          Long cnt = (Long)query.getSingleResult();
34          return cnt;
35      }
36
37      @Override
38      public List<Student> getStudentsPerGender(Character gender) {
39          String queryStr = "SELECT s FROM Student s WHERE s.gender = ?1";
40          Query query = em.createQuery(queryStr);
41          query.setParameter(1, gender);
42          List<Student> students = (List<Student>)query.getResultList();
43          return students;
44      }
45

```

```

46      @Override
47      public Double getAverageAge() {
48          String queryStr = "SELECT avg(s.age) FROM Student s";
49          Query query = em.createQuery(queryStr);
50          Double cnt = (Double)query.getSingleResult();
51          return cnt;
52      }
53
54      @Override
55      public Long getNumberOfStudentsPerAgeRange(Integer minAge, Integer maxAge) {
56          String queryStr = "SELECT count(s) FROM Student s WHERE s.age >= ?1 AND s.age <= ?2";
57          Query query = em.createQuery(queryStr);
58          query.setParameter(1, minAge);
59          query.setParameter(2, maxAge);
60          Long cnt = (Long)query.getSingleResult();
61          return cnt;
62      }
63

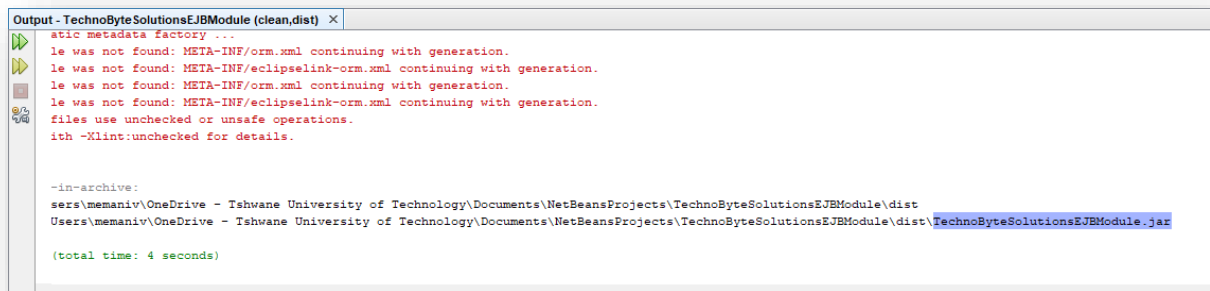
```

```

64      @Override
65      public List<Student> getStudentsPerAgeRange(Integer minAge, Integer maxAge) {
66          String queryStr = "SELECT s FROM Student s WHERE s.age >= ?1 AND s.age <= ?2";
67          Query query = em.createQuery(queryStr);
68          query.setParameter(1, minAge);
69          query.setParameter(2, maxAge);
70          List<Student> students = (List<Student>)query.getResultList();
71          return students;
72      }
73
74  }
75

```

Clean and build the EJB module



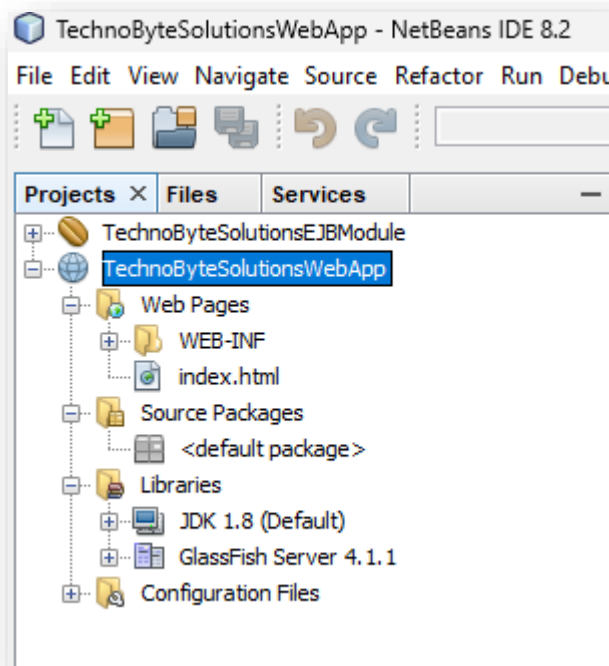
```
Output - TechnoByteSolutionsEJBModule (clean,dist) x
atic metadata factory ...
le was not found: META-INF/orm.xml continuing with generation.
le was not found: META-INF/eclipselink-orm.xml continuing with generation.
le was not found: META-INF/orm.xml continuing with generation.
le was not found: META-INF/eclipselink-orm.xml continuing with generation.
files use unchecked or unsafe operations.
ith -Xlint:unchecked for details.

-in-archive:
sers\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsEJBModule\dist
Users\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsEJBModule\dist\TechnoByteSolutionsEJBModule.jar

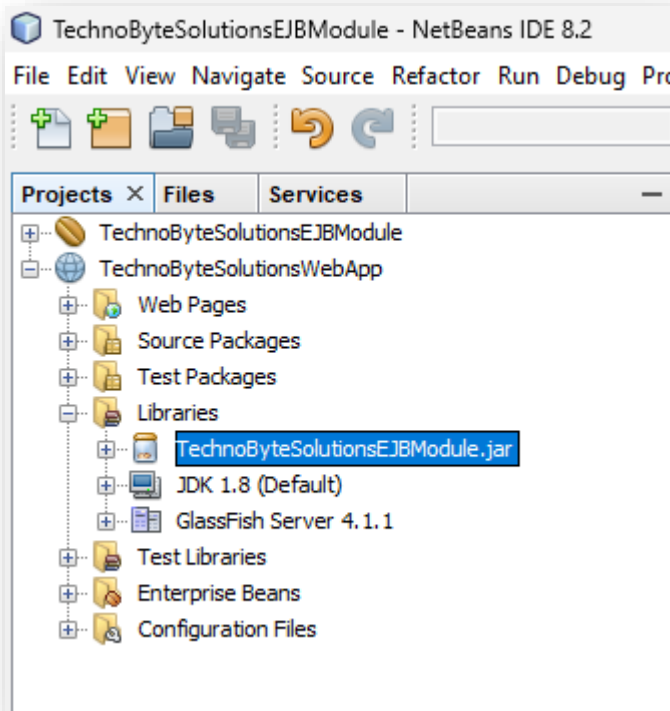
(total time: 4 seconds)
```

Step 3: Create a WEB module

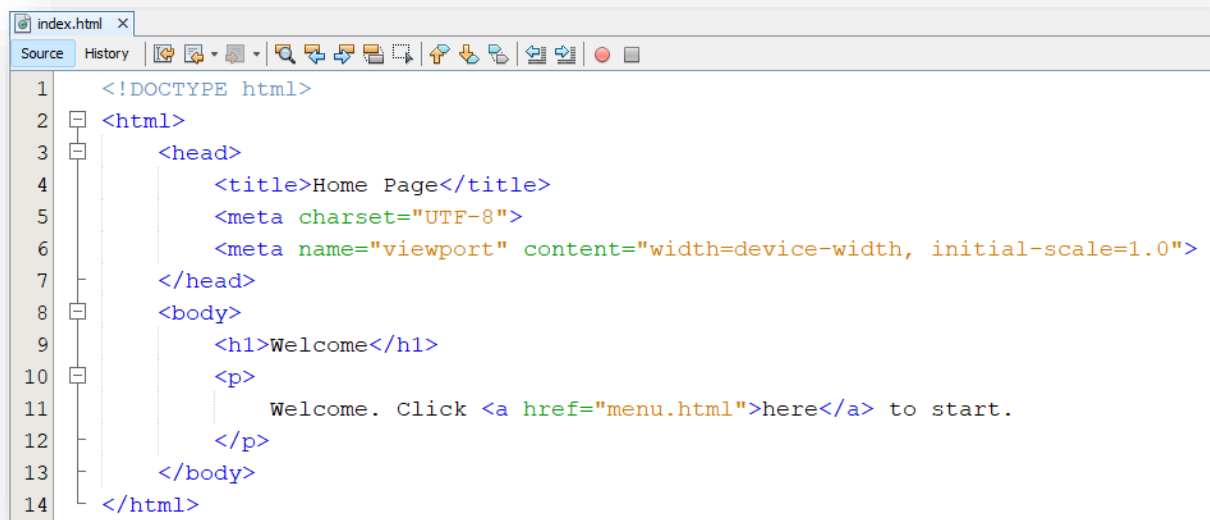
Create a Web project called **TechnoByteSolutionsWebApp**



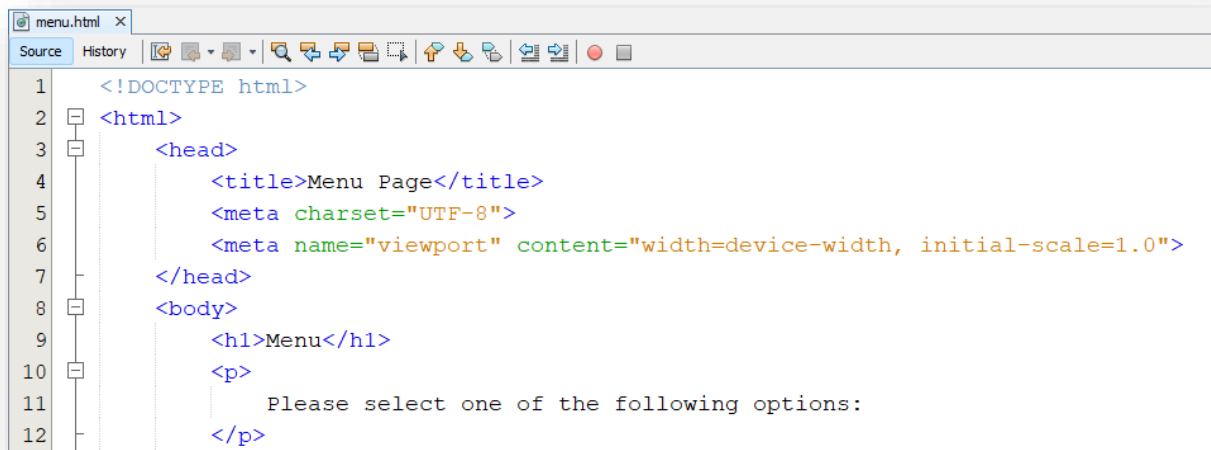
Add the **jar** file of the **EJB module** to this web module.



Modify the **index** page



Create the **menu** page.

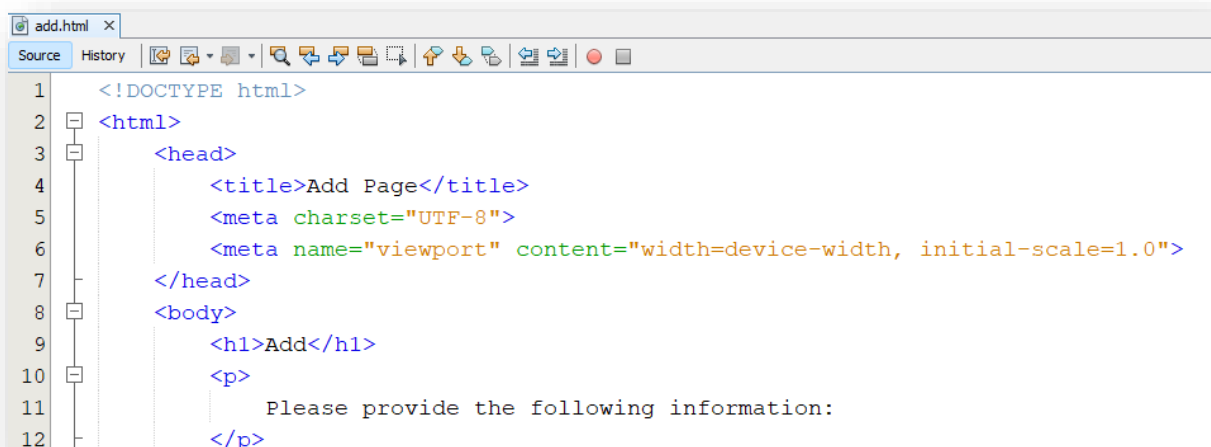


```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Menu Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Menu</h1>
10    <p>
11      Please select one of the following options:
12    </p>
```



```
13    <ol>
14      <li><a href="add.html">Add a student</a></li>
15      <li><a href="find.html">Find a student</a></li>
16      <li><a href="edit.html">Edit a student</a></li>
17      <li><a href="remove.html">Remove a student</a></li>
18      <li><a href="count.html">Display class count</a></li>
19      <li><a href="get_classlist.html">Display class list</a></li>
20      <li><a href="get_class_composition.html">Display class composition information</a></li>
21      <li><a href="get_class_per_age_group.html">Display class within the given age group</a></li>
22    </ol>
23  </body>
24 </html>
```

Create the **add** page



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Add Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Add</h1>
10    <p>
11      Please provide the following information:
12    </p>
```

```

13 <form action="AddStudentServlet.do" method="POST">
14   <table>
15     <tr>
16       <td>Name:</td>
17       <td><input type="text" name="name" required=""/></td>
18     </tr>
19     <tr>
20       <td>Surname:</td>
21       <td><input type="text" name="surname" required=""/></td>
22     </tr>
23     <tr>
24       <td>Student number:</td>
25       <td><input type="text" name="studno" required=""/></td>
26     </tr>
27     <tr>
28       <td>Age:</td>
29       <td><input type="text" name="age" required=""/></td>
30     </tr>

```

```

31     <tr>
32       <td>Gender:</td>
33       <td>
34         <select name="gender">
35           <option value="F">Female</option>
36           <option value="M">Male</option>
37         </select>
38       </td>
39     </tr>
40     <tr>
41       <td></td>
42       <td><input type="submit" value="ADD STUDENT" /></td>
43     </tr>
44   </table>
45 </form>
46 </body>
47 </html>
48

```

Create the **AddStudentServlet** class

```
AddStudentServlet.java x
Source History
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import java.util.Date;
5 import javax.ejb.EJB;
6 import javax.servlet.RequestDispatcher;
7 import javax.servlet.ServletException;
8 import javax.servlet.http.HttpServlet;
9 import javax.servlet.http.HttpServletRequest;
10 import javax.servlet.http.HttpServletResponse;
11 import za.ac.tut.model.bl.StudentFacadeLocal;
12
13 /**
14  *
15  * @author MemaniV
16  */
17 public class AddStudentServlet extends HttpServlet {
18     @EJB StudentFacadeLocal sfl;
```

```
20 @Override
21 protected void doPost(HttpServletRequest request, HttpServletResponse response)
22     throws ServletException, IOException {
23     String name = request.getParameter("name");
24     String surname = request.getParameter("surname");
25     Long studno = Long.parseLong(request.getParameter("studno"));
26     Integer age = Integer.parseInt(request.getParameter("age"));
27     Character gender = request.getParameter("gender").charAt(0);
28
29     Student student = createStudent(name, surname, studno, age, gender);
30     sfl.create(student);
31
32     RequestDispatcher disp = request.getRequestDispatcher("add_outcome.jsp");
33     disp.forward(request, response);
34 }
35
```



```

36     private Student createStudent(String name, String surname, Long studno,
37         Integer age, Character gender) {
38         Student student = new Student();
39         student.setName(name);
40         student.setSurname(surname);
41         student.setGender(gender);
42         student.setAge(age);
43         student.setId(studno);
44         student.setCreationDate(new Date());
45         return student;
46     }
47
48 }
49

```

Create the **add_outcome** page

```

add_outcome.jsp
Source History
2      Document    : add_outcome
3      Created on  : 23 May 2025, 7:21:51 PM
4      Author     : MemaniV
5      --%>
6
7      <%@page contentType="text/html" pageEncoding="UTF-8"%>
8      <!DOCTYPE html>
9      <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Add Outcome Page</title>
13     </head>
14     <body>
15         <h1>Add outcome</h1>
16         <p>
17             The student has been successfully added.
18         </p>

```

```

19         <ul>
20             <li><a href="menu.html">Menu Page</a></li>
21             <li><a href="index.html">Home Page</a></li>
22         </ul>
23     </body>
24 </html>
25

```

Create the **find** page

```
find.html x
Source History
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Find Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Find</h1>
10    <p>
11      Please provide student number below:
12    </p>
```

```
13    <form action="FindStudentServlet.do" method="POST">
14      <table>
15        <tr>
16          <td>Student number:</td>
17          <td><input type="text" name="studno" required=""/></td>
18        </tr>
19        <tr>
20          <td></td>
21          <td><input type="submit" value="FIND STUDENT" /></td>
22        </tr>
23      </table>
24    </form>
25  </body>
26</html>
```

Create the **FindStudentServlet** class



```
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import javax.ejb.EJB;
5 import javax.servlet.RequestDispatcher;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import za.ac.tut.model.bl.StudentFacadeLocal;
11
12 /**
13  *
14  * @author MemaniV
15  */
16 public class FindStudentServlet extends HttpServlet {
17     @EJB StudentFacadeLocal sfl;
18 }
```



```
19 @Override
20 protected void doPost(HttpServletRequest request, HttpServletResponse response)
21     throws ServletException, IOException {
22     Long studno = Long.parseLong(request.getParameter("studno"));
23
24     Student student = sfl.find(studno);
25     request.setAttribute("student", student);
26
27     RequestDispatcher disp = request.getRequestDispatcher("find_outcome.jsp");
28     disp.forward(request, response);
29 }
30
31 }
32 }
```

Create the **find_outcome** page

```
find_outcome.jsp
Source History
1 <%--
2     Document    : find_outcome
3     Created on  : 23 May 2025, 7:54:25 PM
4     Author     : MemaniV
5 --%>
6
7 <%@page import="za.ac.tut.model.entity.Student"%>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13     <title>Find Outcome Page</title>
14 </head>
15 <body>
16     <h1>Find outcome</h1>
```

```
17 <%
18     Student student = (Student)request.getAttribute("student");
19     String name = student.getName();
20     String surname = student.getSurname();
21     Long studno = student.getId();
22     Integer age = student.getAge();
23     Character gender = student.getGender();
24 %>
25 <p>
26     Below are the details of the student.
27 </p>
28 <table>
29 <tr>
30 <td>Name:</td>
31 <td><%=name%></td>
32 </tr>
```

```

33 | <tr>
34 |     <td>Surname:</td>
35 |     <td><%=surname%></td>
36 | </tr>
37 | <tr>
38 |     <td>Student number:</td>
39 |     <td><%=studno%></td>
40 | </tr>
41 | <tr>
42 |     <td>Age:</td>
43 |     <td><%=age%></td>
44 | </tr>
45 | <tr>
46 |     <td>Gender:</td>
47 |     <td><%=gender%></td>
48 | </tr>
49 | </table>

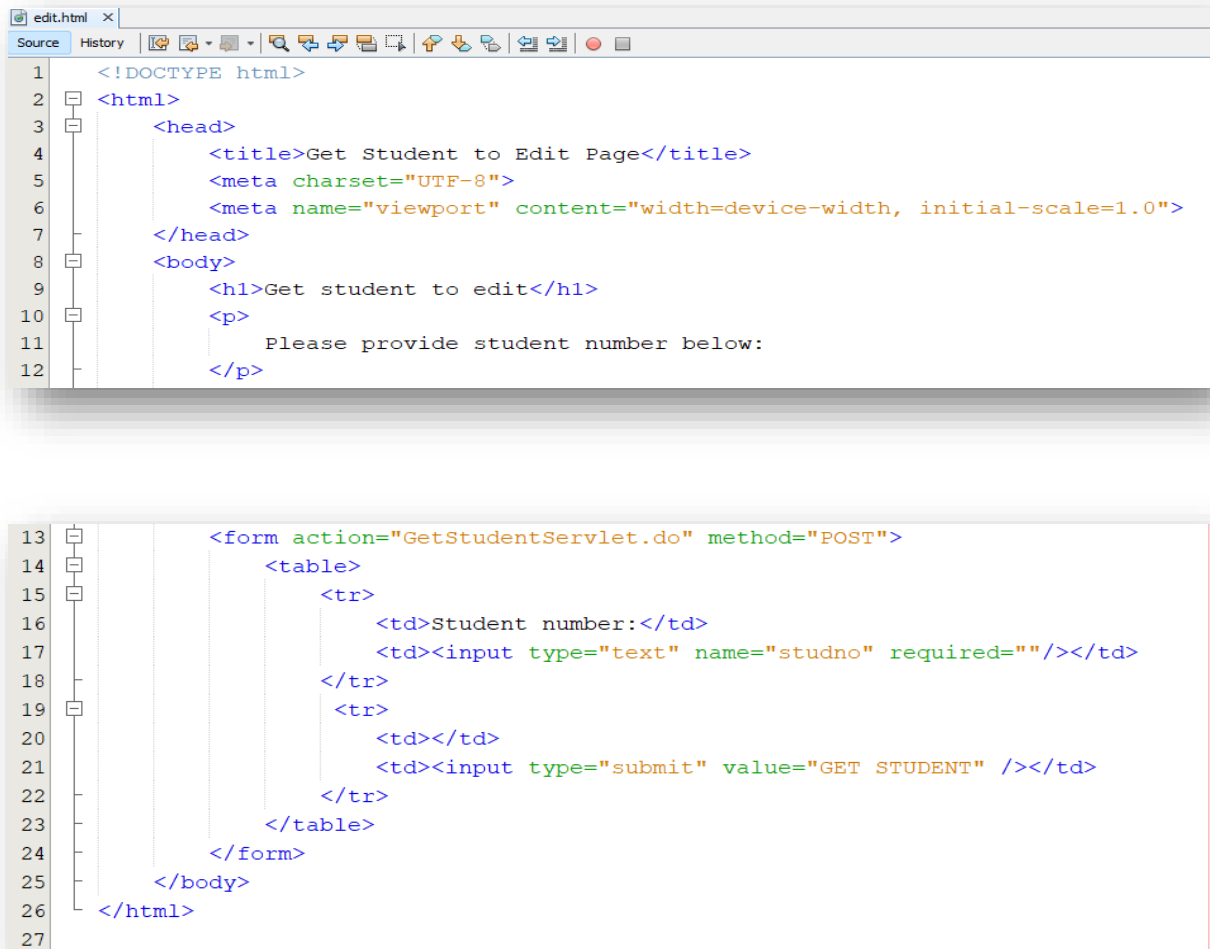
```

```

50 | <ul>
51 |     <li><a href="menu.html">Menu Page</a></li>
52 |     <li><a href="index.html">Home Page</a></li>
53 | </ul>
54 | </body>
55 | </html>
56 |

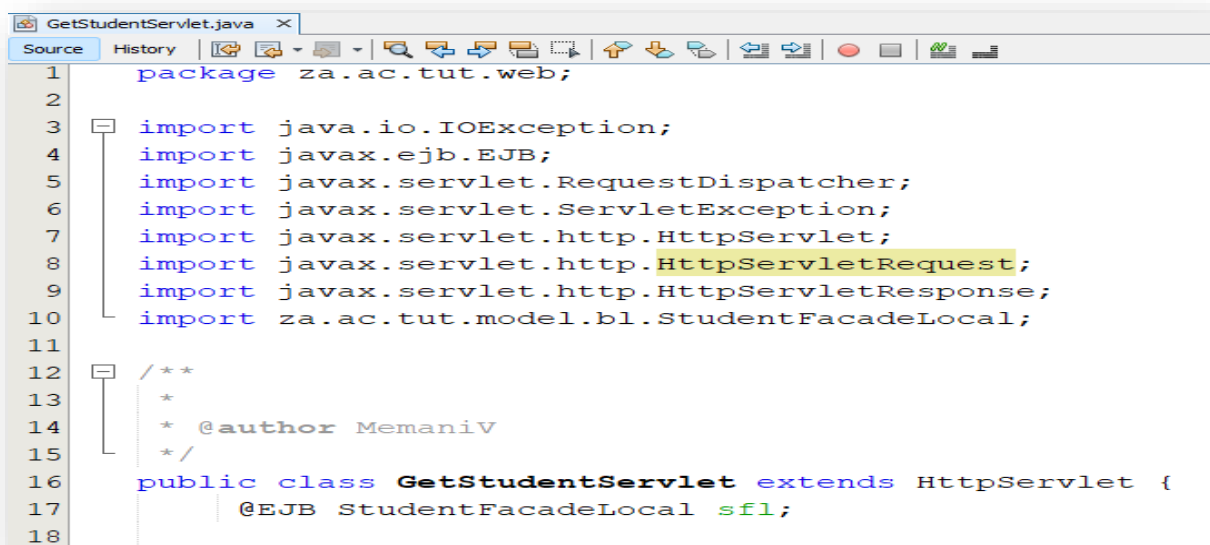
```

Create the edit page



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Get Student to Edit Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 </head>
8 <body>
9     <h1>Get student to edit</h1>
10    <p>
11        Please provide student number below:
12    </p>
13
14    <form action="GetStudentServlet.do" method="POST">
15        <table>
16            <tr>
17                <td>Student number:</td>
18                <td><input type="text" name="studno" required="" /></td>
19            </tr>
20            <tr>
21                <td></td>
22                <td><input type="submit" value="GET STUDENT" /></td>
23            </tr>
24        </table>
25    </form>
26 </body>
27 </html>
```

Create the **GetStudentServlet** class



```
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import javax.ejb.EJB;
5 import javax.servlet.RequestDispatcher;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import za.ac.tut.model.bl.StudentFacadeLocal;
11
12 /**
13  *
14  * @author MemaniV
15  */
16 public class GetStudentServlet extends HttpServlet {
17     @EJB StudentFacadeLocal sfl;
18 }
```

```

19     @Override
20     protected void doPost(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         Long studno = Long.parseLong(request.getParameter("studno"));
23
24         Student student = sfl.find(studno);
25         request.setAttribute("student", student);
26
27         RequestDispatcher disp = request.getRequestDispatcher("edit_student.jsp");
28         disp.forward(request, response);
29     }
30
31 }
32

```

Create the **edit_student** page

```

edit_student.jsp
Source History
1 <!--
2     Document    : edit_student
3     Created on  : 23 May 2025, 8:15:17 PM
4     Author     : MemaniV
5 -->
6
7 <%@page import="za.ac.tut.model.entity.Student"%>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13     <title>Edit Page</title>
14 </head>
15 <body>
16     <h1>Edit</h1>

```

```

17 <%
18     Student student = (Student)request.getAttribute("student");
19     String name = student.getName();
20     String surname = student.getSurname();
21     Long studno = student.getId();
22     Integer age = student.getAge();
23     Character gender = student.getGender();
24 %>
25 <p>
26     Below are the details of the student.
27 </p>

```

```

28 <form action="EditStudentServlet.do" method="POST">
29   <table>
30     <tr>
31       <td>Name:</td>
32       <td><input type="text" value="<%=name%>" disabled=""/></td>
33       <td><input type="text" name="name"/></td>
34     </tr>
35     <tr>
36       <td>Surname:</td>
37       <td><input type="text" value="<%=surname%>" disabled=""/></td>
38       <td><input type="text" name="surname"/></td>
39     </tr>
40     <tr>
41       <td>Student number:</td>
42       <td><input type="text" value="<%=studno%>" disabled=""/></td>
43       <td><input type="text" value="<%=studno%>" name="studno" readonly=""/></td>
44     </tr>

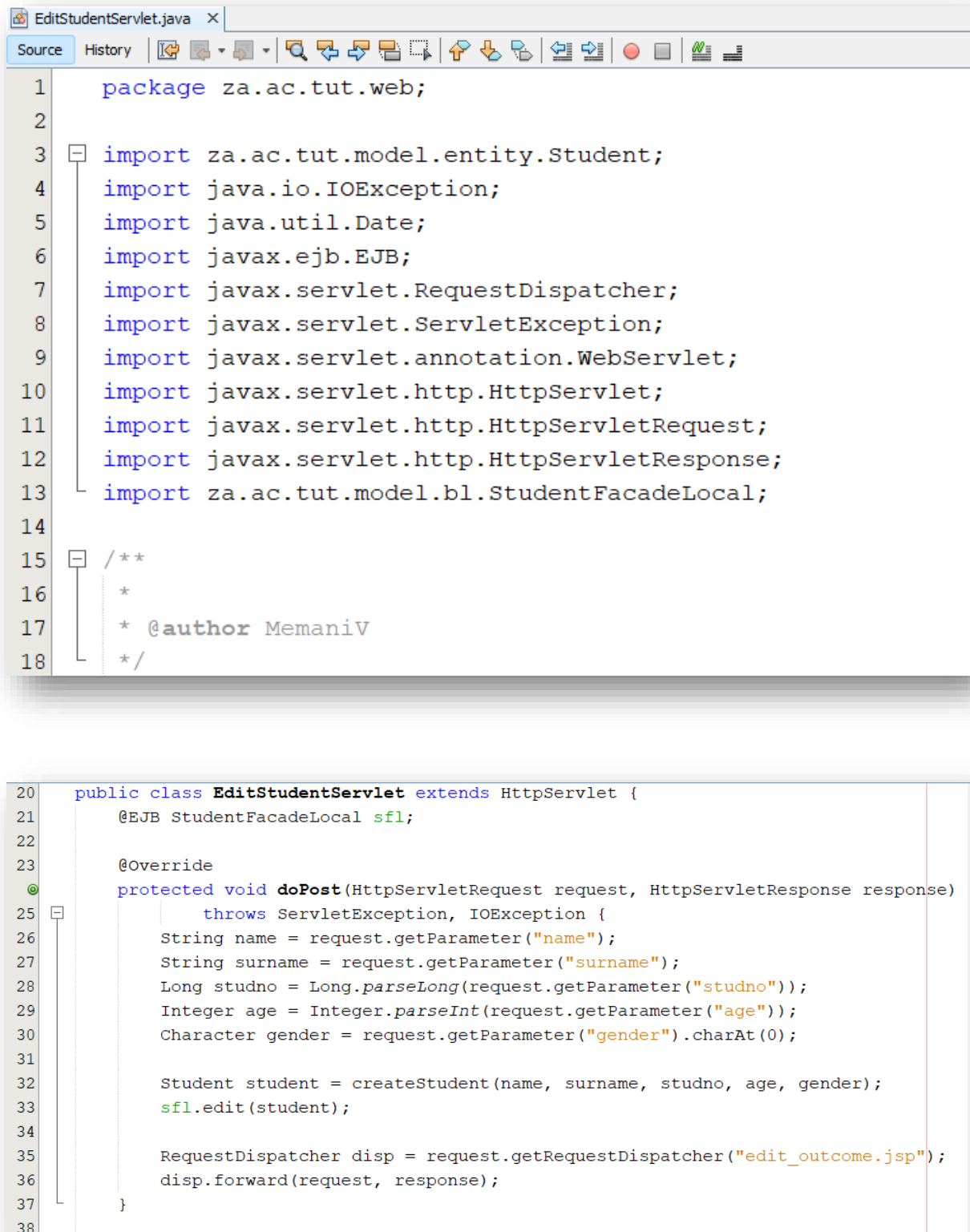
```

```

45   <tr>
46     <td>Age:</td>
47     <td><input type="text" value="<%=age%>" disabled=""/></td>
48     <td><input type="text" name="age"/></td>
49   </tr>
50   <tr>
51     <td>Gender:</td>
52     <td><input type="text" value="<%=gender%>" disabled=""/></td>
53     <td><input type="text" name="gender"/></td>
54   </tr>
55   <tr>
56     <td></td>
57     <td><input type="submit" value="EDIT"/></td>
58   </tr>
59 </table>
60 </form>
61 </body>
62 </html>
63

```


Create the **EditStudentServlet** class



```
1 package za.ac.tut.web;
2
3 import za.ac.tut.model.entity.Student;
4 import java.io.IOException;
5 import java.util.Date;
6 import javax.ejb.EJB;
7 import javax.servlet.RequestDispatcher;
8 import javax.servlet.ServletException;
9 import javax.servlet.annotation.WebServlet;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13 import za.ac.tut.model.bl.StudentFacadeLocal;
14
15 /**
16  *
17  * @author MemaniV
18  */
19
20 public class EditStudentServlet extends HttpServlet {
21     @EJB StudentFacadeLocal sfl;
22
23     @Override
24     protected void doPost(HttpServletRequest request, HttpServletResponse response)
25         throws ServletException, IOException {
26         String name = request.getParameter("name");
27         String surname = request.getParameter("surname");
28         Long studno = Long.parseLong(request.getParameter("studno"));
29         Integer age = Integer.parseInt(request.getParameter("age"));
30         Character gender = request.getParameter("gender").charAt(0);
31
32         Student student = createStudent(name, surname, studno, age, gender);
33         sfl.edit(student);
34
35         RequestDispatcher disp = request.getRequestDispatcher("edit_outcome.jsp");
36         disp.forward(request, response);
37     }
38 }
```

```

39     private Student createStudent(String name, String surname, Long studno,
40                                   Integer age, Character gender) {
41         Student student = new Student();
42         student.setName(name);
43         student.setSurname(surname);
44         student.setGender(gender);
45         student.setAge(age);
46         student.setId(studno);
47         student.setCreationDate(new Date());
48         return student;
49     }
50
51 }
52

```

Create the **edit_outcome** page

```

edit_outcome.jsp
Source History
1  <!--
2      Document      : edit_outcome
3      Created on   : 23 May 2025, 8:20:58 PM
4      Author      : MemaniV
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Edit Outcome Page</title>
13     </head>
14     <body>
15         <h1>Edit outcome</h1>

```

```

16     <p>
17         The student has been successfully edited.
18     </p>
19     <ul>
20         <li><a href="menu.html">Menu Page</a></li>
21         <li><a href="index.html">Home Page</a></li>
22     </ul>
23 </body>
24 </html>
25

```

Create the **remove** page

```
remove.html X
Source History
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Remove Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Remove</h1>
10    <p>
11      Please provide student number below:
12    </p>
```

```
13    <form action="RemoveStudentServlet.do" method="POST">
14      <table>
15        <tr>
16          <td>Student number:</td>
17          <td><input type="text" name="studno" required="" /></td>
18        </tr>
19        <tr>
20          <td></td>
21          <td><input type="submit" value="REMOVE STUDENT" /></td>
22        </tr>
23      </table>
24    </form>
25  </body>
26 </html>
27
```

Create the **RemoveStudentServlet**

```
RemoveStudentServlet.java x
Source History
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import javax.ejb.EJB;
5 import javax.servlet.RequestDispatcher;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import za.ac.tut.model.bl.StudentFacadeLocal;
11 import za.ac.tut.model.entity.Student;
12
13 /**
14  *
15  * @author MemaniV
16  */
```

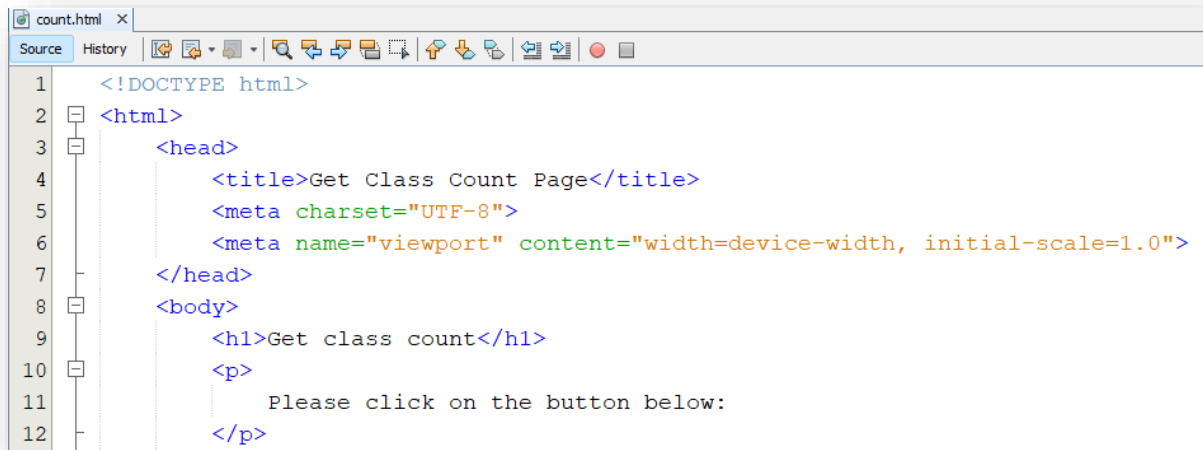
```
17 public class RemoveStudentServlet extends HttpServlet {
18     @EJB StudentFacadeLocal sfl;
19     @Override
20     protected void doPost(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         Long studno = Long.parseLong(request.getParameter("studno"));
23         Student student = createStudent(studno);
24         sfl.remove(student);
25
26         RequestDispatcher disp = request.getRequestDispatcher("remove_outcome.jsp");
27         disp.forward(request, response);
28     }
29
30     private Student createStudent(Long studno) {
31         Student student = new Student();
32         student.setId(studno);
33         return student;
34     }
35 }
```

Create the **remove_outcome** page

```
remove_outcome.jsp
Source History
2      Document    : remove_outcome
3      Created on  : 24 May 2025, 12:06:51 PM
4      Author     : MemaniV
5      --%>
6
7      <%@page contentType="text/html" pageEncoding="UTF-8"%>
8      <!DOCTYPE html>
9      <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Remove Outcome Page</title>
13     </head>
14     <body>
15         <h1>Remove outcome</h1>
```

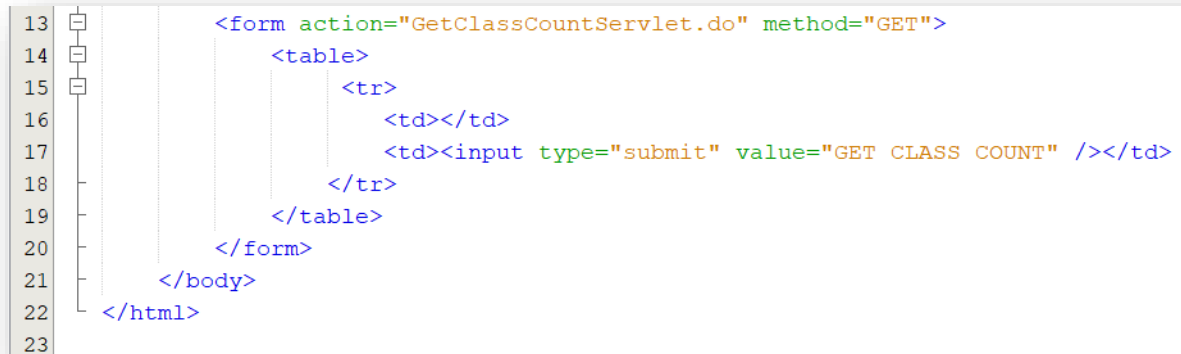
```
16         <p>
17             The student has been successfully removed.
18         </p>
19         <ul>
20             <li><a href="menu.html">Menu Page</a></li>
21             <li><a href="index.html">Home Page</a></li>
22         </ul>
23     </body>
24 </html>
25
```

Create the **count** page



A screenshot of a code editor window titled 'count.html'. The editor shows the first 12 lines of an HTML document. The code includes a DOCTYPE declaration, an html tag, a head section with a title 'Get Class Count Page', a meta charset of 'UTF-8', and a viewport meta tag. The body section starts with an h1 tag 'Get class count' and a paragraph 'Please click on the button below:'. The editor has a 'Source' tab selected and a toolbar with various icons.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Get Class Count Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Get class count</h1>
10    <p>
11      Please click on the button below:
12    </p>
```



A screenshot of a code editor window showing the continuation of the HTML document from line 13 to 23. The code defines a form with an action of 'GetClassCountServlet.do' and a method of 'GET'. Inside the form is a table with two columns: an empty column and a column containing a submit button with the value 'GET CLASS COUNT'. The editor has a 'Source' tab selected and a toolbar with various icons.

```
13   <form action="GetClassCountServlet.do" method="GET">
14     <table>
15       <tr>
16         <td></td>
17         <td><input type="submit" value="GET CLASS COUNT" /></td>
18       </tr>
19     </table>
20   </form>
21 </body>
22 </html>
23
```

Create the **GetClassCountServlet** class

```
GetClassCountServlet.java x
Source History
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import javax.ejb.EJB;
5 import javax.servlet.RequestDispatcher;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import za.ac.tut.model.bl.StudentFacadeLocal;
11
12 /**
13  *
14  * @author MemaniV
15  */
16 public class GetClassCountServlet extends HttpServlet {
```

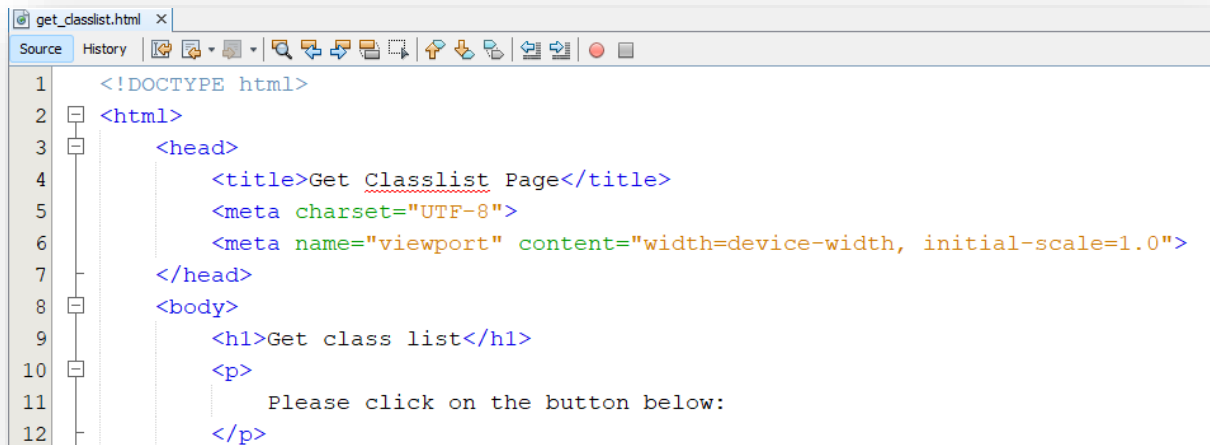
```
17     @EJB StudentFacadeLocal sfl;
18
19     @Override
20     protected void doGet(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         Integer cnt = sfl.count();
23         request.setAttribute("cnt", cnt);
24
25         RequestDispatcher disp = request.getRequestDispatcher("get_class_count_outcome.jsp");
26         disp.forward(request, response);
27     }
28
29 }
30
```

Create the `get_class_count_outcome` page

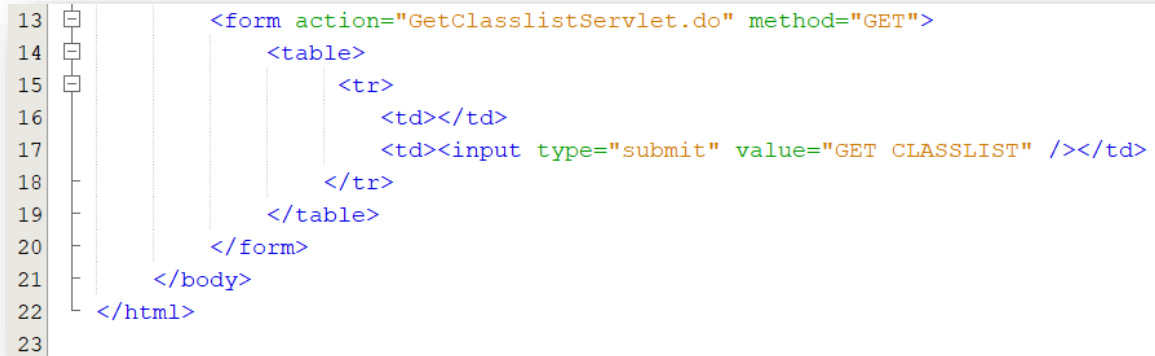
```
get_class_count_outcome.jsp
Source History
1 <%--
2     Document    : get_class_count_outcome
3     Created on  : 24 May 2025, 12:16:30 PM
4     Author     : MemaniV
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12     <title>Class Count Outcome Page</title>
13 </head>
14 <body>
15     <h1>Class Count outcome</h1>
```

```
16 <%
17     Integer cnt = (Integer)request.getAttribute("cnt");
18 %>
19 <p>
20     There are <b><%=cnt%></b> students in the class list.
21 </p>
22 <ul>
23     <li><a href="menu.html">Menu Page</a></li>
24     <li><a href="index.html">Home Page</a></li>
25 </ul>
26 </body>
27 </html>
28
```


Create the **get_classlist** page



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Get Classlist Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Get class list</h1>
10    <p>
11      Please click on the button below:
12    </p>
```



```
13    <form action="GetClasslistServlet.do" method="GET">
14      <table>
15        <tr>
16          <td></td>
17          <td><input type="submit" value="GET CLASSLIST" /></td>
18        </tr>
19      </table>
20    </form>
21  </body>
22 </html>
23
```

Create the **GetClasslistServlet** class

```
GetClasslistServlet.java x
Source History
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import java.util.List;
5 import javax.ejb.EJB;
6 import javax.servlet.RequestDispatcher;
7 import javax.servlet.ServletException;
8 import javax.servlet.http.HttpServlet;
9 import javax.servlet.http.HttpServletRequest;
10 import javax.servlet.http.HttpServletResponse;
11 import za.ac.tut.model.bl.StudentFacadeLocal;
12
13 /**
14  *
15  * @author MemaniV
16  */
17 public class GetClasslistServlet extends HttpServlet {
18     @EJB StudentFacadeLocal sfl;
19 }
```

```
20 @Override
21 protected void doPost(HttpServletRequest request, HttpServletResponse response)
22     throws ServletException, IOException {
23     List<Student> students = sfl.findAll();
24     request.setAttribute("students", students);
25
26     RequestDispatcher disp = request.getRequestDispatcher("get_classlist_outcome.jsp");
27     disp.forward(request, response);
28 }
29
30 }
31
```

Create the **get_classlist_outcome** page

```
get_classlist_outcome.jsp
Source History
1 <%--
2     Document    : get_classlist_outcome
3     Created on  : 23 May 2025, 8:47:50 PM
4     Author     : MemaniV
5 --%>
6
7 <%@page import="java.util.List"%>
8 <%@page import="za.ac.tut.model.entity.Student"%>
9 <%@page contentType="text/html" pageEncoding="UTF-8"%>
10 <!DOCTYPE html>
11 <html>
12 <head>
13     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
14     <title>Get Classlist Outcome Page</title>
15 </head>
16 <body>
17     <h1>Get classlist outcome</h1>
```

```
18 <%
19     List<Student> students = (List<Student>)request.getAttribute("students");
20     String name, surname;
21     Long studno;
22     Integer age;
23     Character gender;
24 %>
25 <p>
26     Below is the classlist.
27 </p>
28 <table border="1">
29     <th>No.</th>
30     <th>Name</th>
31     <th>Surname</th>
32     <th>Student number</th>
33     <th>Age</th>
34     <th>Gender</th>
```

```

35 <%
36     Student student;
37     for(int i = 0; i < students.size();i++){
38         student = students.get(i);
39         name = student.getName();
40         surname = student.getSurname();
41         studno = student.getId();
42         age = student.getAge();
43         gender = student.getGender();
44     }
45     <tr>
46         <td><input type="text" value="<%=i+1%>" readonly="" /></td>
47         <td><input type="text" value="<%=name%>" readonly="" /></td>
48         <td><input type="text" value="<%=surname%>" readonly="" /></td>
49         <td><input type="text" value="<%=studno%>" readonly="" /></td>
50         <td><input type="text" value="<%=age%>" readonly="" /></td>
51         <td><input type="text" value="<%=gender%>" readonly="" /></td>
52     </tr>
53 <%
54     }
55 >
56 </table>

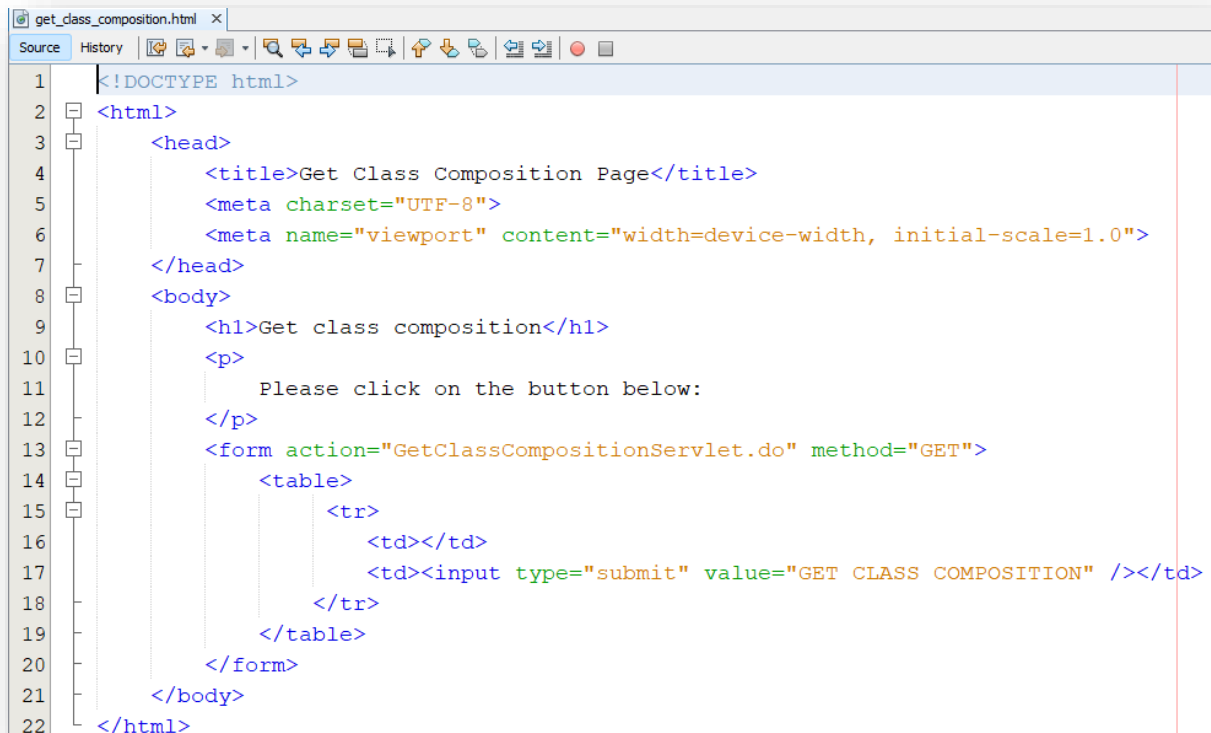
```

```

57 <ul>
58     <li><a href="menu.html">Menu Page</a></li>
59     <li><a href="index.html">Home Page</a></li>
60 </ul>
61 </body>
62 </html>
63

```

Create the **get_class_composition** page

A screenshot of a web browser's source code view for a file named 'get_class_composition.html'. The code is HTML and is displayed with line numbers from 1 to 22 on the left. The HTML structure includes a head section with a title 'Get Class Composition Page', a UTF-8 charset meta tag, and a viewport meta tag. The body section contains an h1 heading 'Get class composition', a paragraph 'Please click on the button below:', and a form. The form has an action of 'GetClassCompositionServlet.do' and a method of 'GET'. Inside the form is a table with one row containing a submit button with the value 'GET CLASS COMPOSITION'.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Get Class Composition Page</title>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   </head>
8   <body>
9     <h1>Get class composition</h1>
10    <p>
11      Please click on the button below:
12    </p>
13    <form action="GetClassCompositionServlet.do" method="GET">
14      <table>
15        <tr>
16          <td></td>
17          <td><input type="submit" value="GET CLASS COMPOSITION" /></td>
18        </tr>
19      </table>
20    </form>
21  </body>
22 </html>
```

Create the **GetClassCompositionServlet**

```
GetClassCompositionServlet.java x
Source History
1 package za.ac.tut.web;
2
3 import za.ac.tut.model.entity.Student;
4 import java.io.IOException;
5 import java.util.List;
6 import javax.ejb.EJB;
7 import javax.servlet.RequestDispatcher;
8 import javax.servlet.ServletException;
9 import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import za.ac.tut.model.bl.StudentFacadeLocal;
13
14 /**
15  *
16  * @author MemaniV
17  */
18 public class GetClassCompositionServlet extends HttpServlet {
```

```
19 @EJB StudentFacadeLocal sfl;
20 @Override
21 protected void doGet(HttpServletRequest request, HttpServletResponse response)
22     throws ServletException, IOException {
23     Double avgAge = sfl.getAverageAge();
24     Long numMales = sfl.getNumberOfStudentsPerGender('M');
25     Long numFemales = sfl.getNumberOfStudentsPerGender('F');
26     List<Student> maleStudents = sfl.getStudentsPerGender('M');
27     List<Student> femaleStudents = sfl.getStudentsPerGender('F');
28
29     request.setAttribute("avgAge", avgAge);
30     request.setAttribute("numMales", numMales);
31     request.setAttribute("numFemales", numFemales);
32     request.setAttribute("maleStudents", maleStudents);
33     request.setAttribute("femaleStudents", femaleStudents);
34
35     RequestDispatcher disp = request.getRequestDispatcher("get_class_composition_outcome.jsp");
36     disp.forward(request, response);
37 }
38 }
```

Create the **get_class_composition_outcome** page

```
get_class_composition_outcome.jsp
Source History
1 <%--
2     Document    : get_class_composition_outcome
3     Created on  : 23 May 2025, 10:17:02 PM
4     Author     : MemaniV
5 --%>
6
7 <%@page import="java.util.List"%>
8 <%@page import="za.ac.tut.model.entity.Student"%>
9 <%@page contentType="text/html" pageEncoding="UTF-8"%>
10 <!DOCTYPE html>
11 <html>
12     <head>
13         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
14         <title>Get Class Composition Outcome Page</title>
15     </head>
16     <body>
17         <h1>Get Class Composition Outcome</h1>
```

```
18 <%
19     List<Student> maleStudents = (List<Student>)request.getAttribute("maleStudents");
20     List<Student> femaleStudents = (List<Student>)request.getAttribute("femaleStudents");
21     Double avgAge = (Double)request.getAttribute("avgAge");
22     Long numMales = (Long)request.getAttribute("numMales");
23     Long numFemales = (Long)request.getAttribute("numFemales");
24     String name, surname;
25     Long studno;
26     Integer age;
27     Character gender;
28 %>
29 <p>
30     Below are the class statistics:
31 </p>
```

```

32 <table>
33 <tr>
34 <td>Average age</td>
35 <td>%=avgAge%</td>
36 </tr>
37 <tr>
38 <td>Number of male students</td>
39 <td>%=numMales%</td>
40 </tr>
41 <tr>
42 <td>Number of female students</td>
43 <td>%=numFemales%</td>
44 </tr>
45 </table>

```

```

47 <%
48     if(numMales > 0){
49         %>
50         <table border="1">
51             <th>No.</th>
52             <th>Name</th>
53             <th>Surname</th>
54             <th>Student number</th>
55             <th>Age</th>
56             <th>Gender</th>
57         <%
58             Student student;
59             for(int i = 0; i < maleStudents.size();i++){
60                 student = maleStudents.get(i);
61                 name = student.getName();
62                 surname = student.getSurname();
63                 studno = student.getId();
64                 age = student.getAge();
65                 gender = student.getGender();
66             %>

```



```

67 <tr>
68 <td><input type="text" value="<%=i+1%>" readonly="" /></td>
69 <td><input type="text" value="<%=name%>" readonly="" /></td>
70 <td><input type="text" value="<%=surname%>" readonly="" /></td>
71 <td><input type="text" value="<%=studno%>" readonly="" /></td>
72 <td><input type="text" value="<%=age%>" readonly="" /></td>
73 <td><input type="text" value="<%=gender%>" readonly="" /></td>
74 </tr>
75 <%
76     }
77 %>
78 </table>
79 <%
80     }
81 %>
82

```

```

83 <%
84     if(numFemales > 0){
85         %>
86         <table border="1">
87             <th>No.</th>
88             <th>Name</th>
89             <th>Surname</th>
90             <th>Student number</th>
91             <th>Age</th>
92             <th>Gender</th>
93         <%
94             Student student;
95             for(int i = 0; i < femaleStudents.size(); i++){
96                 student = femaleStudents.get(i);
97                 name = student.getName();
98                 surname = student.getSurname();
99                 studno = student.getId();
100                 age = student.getAge();
101                 gender = student.getGender();
102             %>

```

```

103 <tr>
104 <td><input type="text" value="<%=i+1%>" readonly=""/></td>
105 <td><input type="text" value="<%=name%>" readonly=""/></td>
106 <td><input type="text" value="<%=surname%>" readonly=""/></td>
107 <td><input type="text" value="<%=studno%>" readonly=""/></td>
108 <td><input type="text" value="<%=age%>" readonly=""/></td>
109 <td><input type="text" value="<%=gender%>" readonly=""/></td>
110 </tr>
111 <%
112     }
113 %>
114 </table>
115 <%
116     }
117 %>
118 <ul>
119 <li><a href="menu.html">Menu Page</a></li>
120 <li><a href="index.html">Home Page</a></li>
121 </ul>
122 </body>
123 </html>

```

Create **get_class_per_age_group** page

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Get Classlist Per Age Group Page</title>
5 <meta charset="UTF-8">
6 <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 </head>
8 <body>
9 <h1>Get Classlist Per Age Group</h1>
10 <p>
11 Please provide the following information:
12 </p>

```

```

13      <form action="GetClasslistPerAgeGroupServlet.do" method="POST">
14          <table>
15              <tr>
16                  <td>Minimum age:</td>
17                  <td><input type="text" name="min_age" required="" /></td>
18              </tr>
19              <tr>
20                  <td>Maximum age:</td>
21                  <td><input type="text" name="max_age" required="" /></td>
22              </tr>
23              <tr>
24                  <td></td>
25                  <td><input type="submit" value="GET CLASSLIST" /></td>
26              </tr>
27          </table>
28      </form>
29  </body>
30 </html>
31

```

Create the **GetClasslistPerAgeGroupServlet** class

```

GetClasslistPerAgeGroupServlet.java x
Source History
1  package za.ac.tut.web;
2
3  import za.ac.tut.model.entity.Student;
4  import java.io.IOException;
5  import java.util.List;
6  import javax.ejb.EJB;
7  import javax.servlet.RequestDispatcher;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import za.ac.tut.model.bl.StudentFacadeLocal;
13
14 /**
15  *
16  * @author MemaniV
17  */
18 public class GetClasslistPerAgeGroupServlet extends HttpServlet {

```

```

19     @EJB StudentFacadeLocal sfl;
20
21     @Override
22     protected void doPost(HttpServletRequest request, HttpServletResponse response)
23         throws ServletException, IOException {
24         Integer minAge = Integer.parseInt(request.getParameter("min_age"));
25         Integer maxAge = Integer.parseInt(request.getParameter("max_age"));
26
27         Long numStudentsPerAgeRange = sfl.getNumberOfStudentsPerAgeRange(minAge, maxAge);
28         List<Student> students = sfl.getStudentsPerAgeRange(minAge, maxAge);
29
30         request.setAttribute("students", students);
31         request.setAttribute("numStudentsPerAgeRange", numStudentsPerAgeRange);
32
33         RequestDispatcher disp = request.getRequestDispatcher("get_classlist_per_age_outcome.jsp");
34         disp.forward(request, response);
35     }
36
37 }
38

```

Create the **get_classlist_per_age_outcome** page

```

get_classlist_per_age_outcome.jsp
Source History
1  <%--
2      Document      : get_classlist_per_age_outcome
3      Created on    : 23 May 2025, 9:38:38 PM
4      Author       : MemaniV
5  --%>
6
7  <%@page import="java.util.List"%>
8  <%@page import="za.ac.tut.model.entity.Student"%>
9  <%@page contentType="text/html" pageEncoding="UTF-8"%>
10 <!DOCTYPE html>
11 <html>
12 <head>
13     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
14     <title>Get Classlist Outcome Page</title>
15 </head>
16 <body>
17     <h1>Get classlist outcome</h1>

```

```

18  <%
19      List<Student> students = (List<Student>)request.getAttribute("students");
20      Long numStudentsPerAgeRange = (Long)request.getAttribute("numStudentsPerAgeRange");
21      String name, surname;
22      Long studno;
23      Integer age;
24      Character gender;
25  %>
26  <p>
27      There are <b><%=numStudentsPerAgeRange%></b>
28      students falling under the specified age group.
29  </p>

```

```

30  <%
31      if(numStudentsPerAgeRange > 0){
32          %>
33          <table border="1">
34              <th>No.</th>
35              <th>Name</th>
36              <th>Surname</th>
37              <th>Student number</th>
38              <th>Age</th>
39              <th>Gender</th>
40          <%
41              Student student;
42              for(int i = 0; i < students.size();i++){
43                  student = students.get(i);
44                  name = student.getName();
45                  surname = student.getSurname();
46                  studno = student.getId();
47                  age = student.getAge();
48                  gender = student.getGender();
49              %>

```

```

50      <tr>
51          <td><input type="text" value="<%=i+1%>" readonly="" /></td>
52          <td><input type="text" value="<%=name%>" readonly="" /></td>
53          <td><input type="text" value="<%=surname%>" readonly="" /></td>
54          <td><input type="text" value="<%=studno%>" readonly="" /></td>
55          <td><input type="text" value="<%=age%>" readonly="" /></td>
56          <td><input type="text" value="<%=gender%>" readonly="" /></td>
57      </tr>
58      <%
59      %>
60      %>
61  </table>
62  <%
63  %>
64  %>
65  <ul>
66      <li><a href="menu.html">Menu Page</a></li>
67      <li><a href="index.html">Home Page</a></li>
68  </ul>
69  </body>
70 </html>

```

Clean and build the web project

```

Output - TechnoByteSolutionsWebApp (clean,dist) x
Compiling 9 source files to C:\Users\memani\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\web\WEB-INF\classes
compile:
compile-jsp:
Created dir: C:\Users\memani\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist
Building jar: C:\Users\memani\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist\TechnoByteSolutionsWebApp.war
do-dist:
dist:
BUILD SUCCESSFUL (total time: 3 seconds)

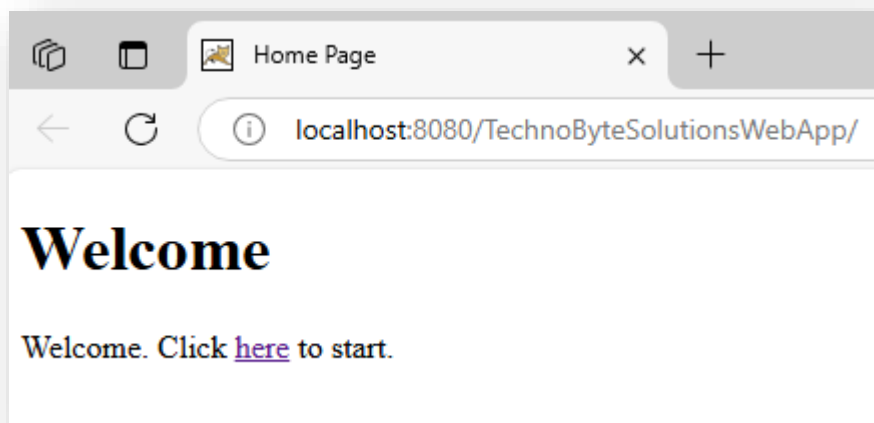
```

Step 4: Test the web application

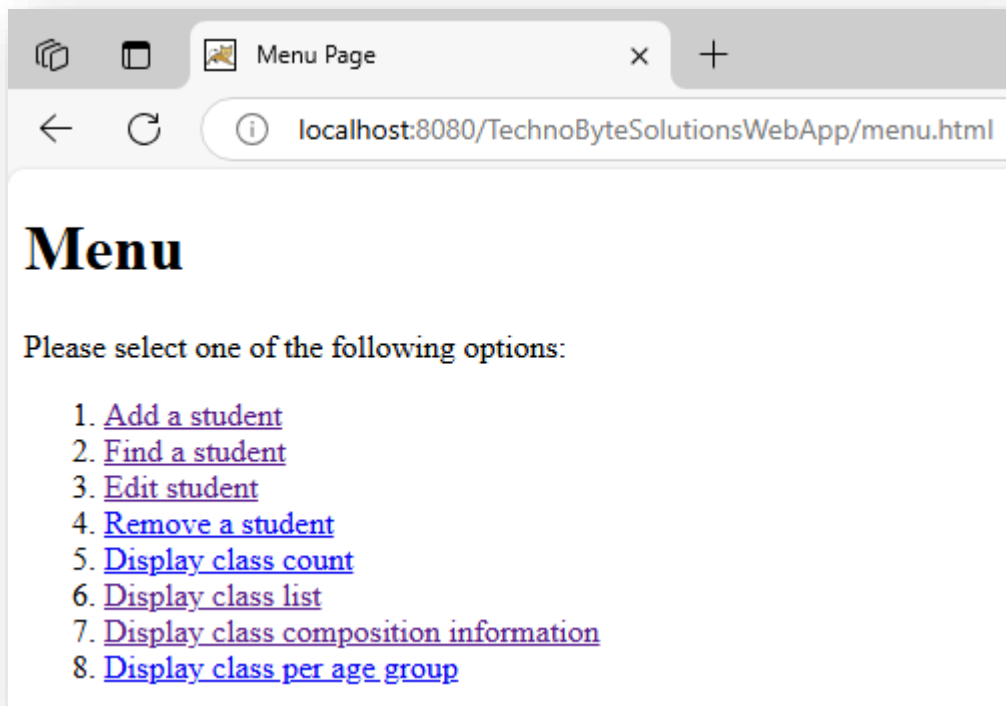
Persist the following records to the database

No.	Student number	Age	Gender	Name	Surname
1	111	20	M	Tata	Mama
2	222	46	F	Khom	Mashao
3	333	30	F	Titi	Mimi
4	444	33	M	Toto	Momo
5	555	33	M	Tutu	Mumu
6	666	30	M	Vuyi	Memani
7	777	25	F	Thato	Ranamane
8	888	22	F	Lira	Nkosi
9	999	40	M	Tebza	Mabale
10	101	50	M	Rendani	Musetha

Deploy and run the web application.



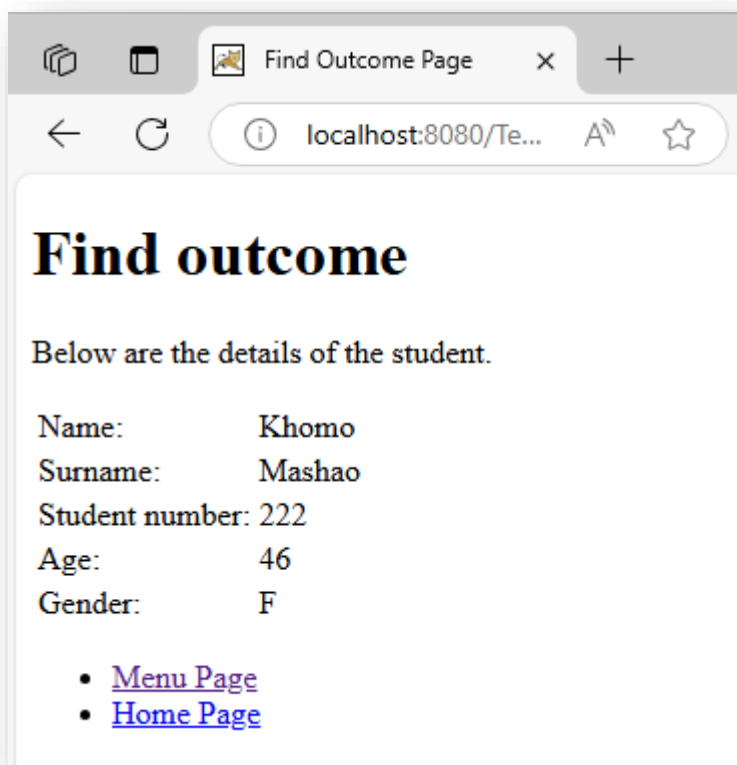
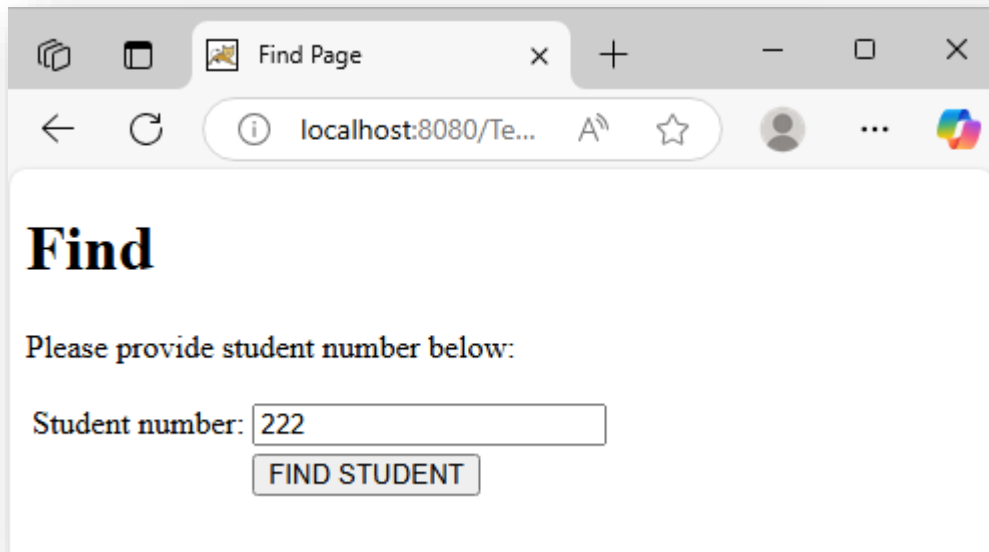
Click on the link



Click the first link to add the records.

#	STUDENT_NUMBER	AGE	CREATIONDATE	GENDER	NAME	SURNAME
1	111	20	2025-05-23 22:33:39.852	M	Tata	Mama
2	222	46	2025-05-23 23:20:16.474	F	Khomo	Mashao
3	333	30	2025-05-23 22:34:23.869	F	Titi	Mimi
4	444	33	2025-05-23 22:34:50.946	M	Toto	Momo
5	555	33	2025-05-23 22:35:12.415	M	Tutu	Mumu
6	666	30	2025-05-24 13:08:18.763	M	Vuyi	JMemani
7	777	25	2025-05-24 13:08:46.059	F	Thato	Ranamane
8	888	22	2025-05-24 13:09:11.448	F	Lira	Nkosi
9	999	40	2025-05-24 13:09:38.192	M	Tebza	Mabale
10	101	50	2025-05-24 13:10:05.059	M	Rendani	Musetha

Click the second link to find a student



Click on the third link to edit a student

The screenshot shows a web browser window with the title 'Get Student to Edit Page'. The address bar displays 'localhost:8080/Te...'. The main content area has a heading 'Get a student to edit' and a prompt 'Please provide student number below:'. Below this is a text input field containing '222' and a button labeled 'GET STUDENT'.

Get a student to edit

Please provide student number below:

Student number:

The screenshot shows a web browser window with the title 'Edit Page'. The address bar displays 'localhost:8080/Te...'. The main content area has a heading 'Edit' and a prompt 'Below are the details of the student.'. Below this is a form with five rows, each with a label, a text input field, and a second empty text input field. The input fields contain the following values: Name: Khomo, Surname: Mashao, Student number: 222, Age: 46, and Gender: F. At the bottom of the form is a button labeled 'EDIT'.

Edit

Below are the details of the student.

Name:	Khomo	
Surname:	Mashao	
Student number:	222	222
Age:	46	
Gender:	F	

Edit

Below are the details of the student.

Name:	Khomo	Khomotjo
Surname:	Mashao	Memani
Student number:	222	222
Age:	46	45
Gender:	F	F

Edit outcome

The student has been successfully edited.

- [Menu Page](#)
- [Home Page](#)

SELECT * FROM APP.STUDENT ... X

Max. rows: 100 | Fetched Rows: 10 | Matching Rows:

#	STUDENT_NUMBER	AGE	CREATIONDATE	GENDER	NAME	SURNAME
1	111	20	2025-05-23 22:33:39.852	M	Tata	Mama
2	222	45	2025-05-24 13:16:36.866	F	Khomotjo	Memani
3	333	30	2025-05-23 22:34:23.869	F	Titi	Mimi
4	444	33	2025-05-23 22:34:50.946	M	Toto	Momo
5	555	33	2025-05-23 22:35:12.415	M	Tutu	Mumu
6	666	30	2025-05-24 13:08:18.763	M	Vuyi	JMemani
7	777	25	2025-05-24 13:08:46.059	F	Thato	Ranamane
8	888	22	2025-05-24 13:09:11.448	F	Lira	Nkosi
9	999	40	2025-05-24 13:09:38.192	M	Tebza	Mabale
10	101	50	2025-05-24 13:10:05.059	M	Rendani	Musetha

Click on the fourth link to remove a student from the classlist

Remove Page

localhost:8080/Te...

Remove

Please provide student number below:

Student number:

Remove Outcome Page

localhost:8080/Te...

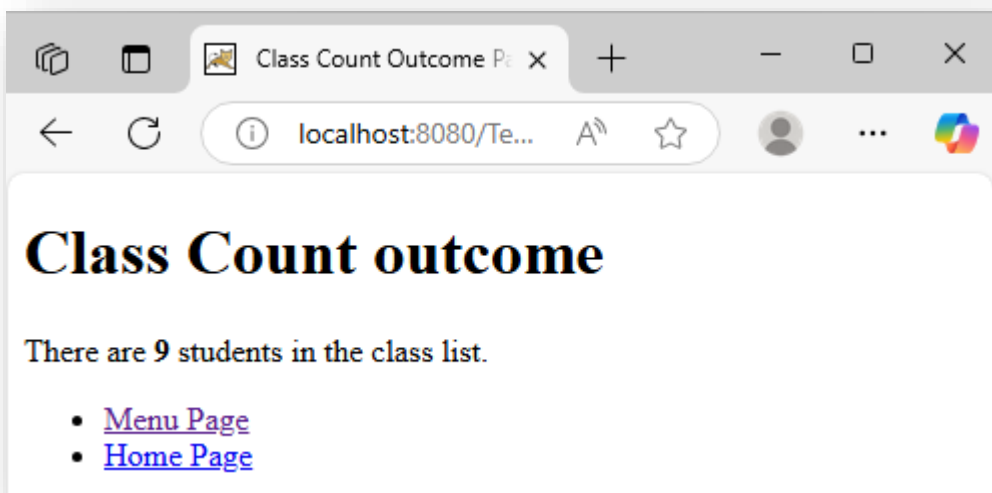
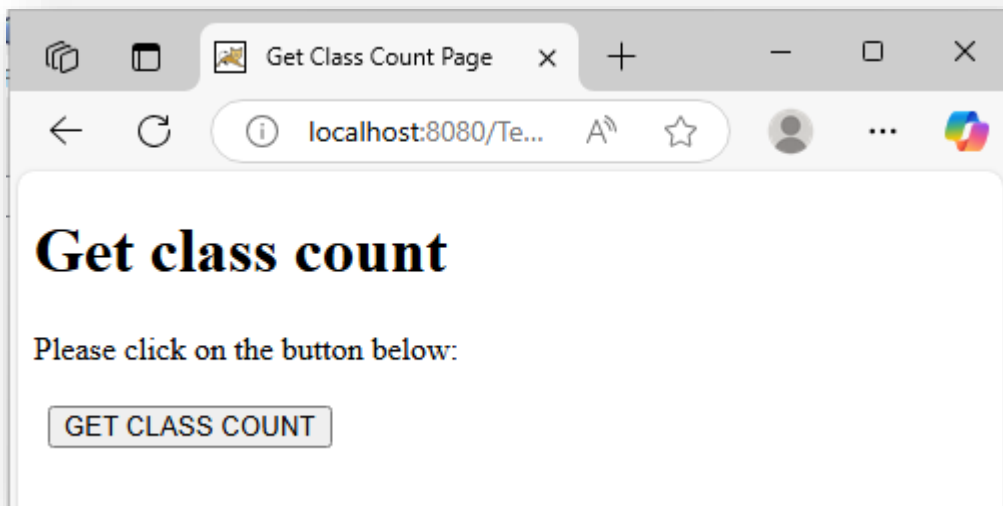
Remove outcome

The student has been successfully removed.

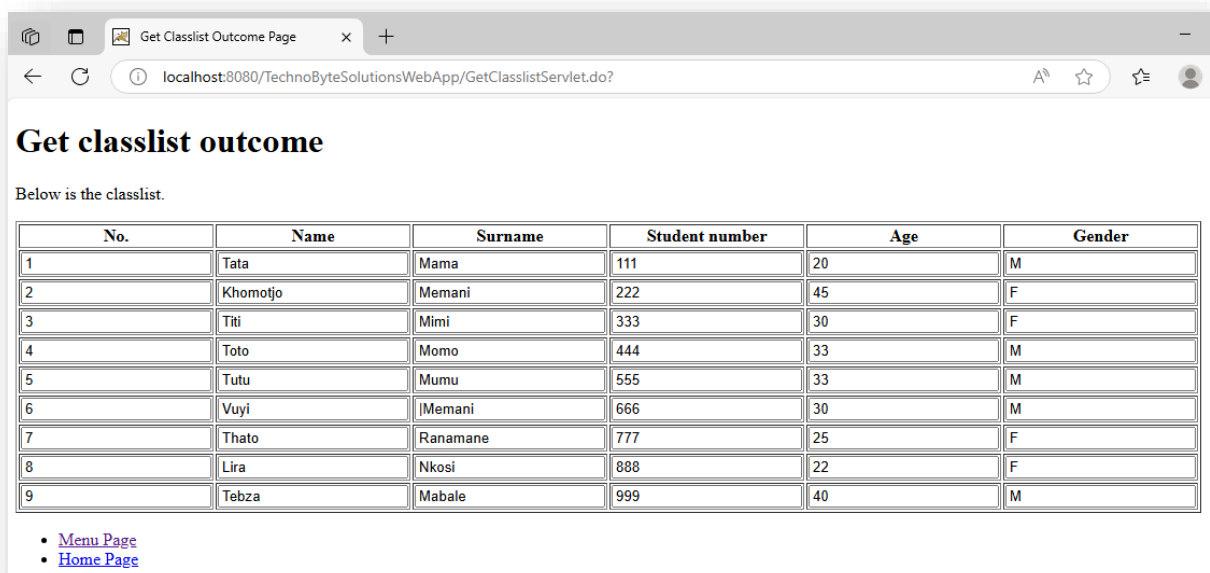
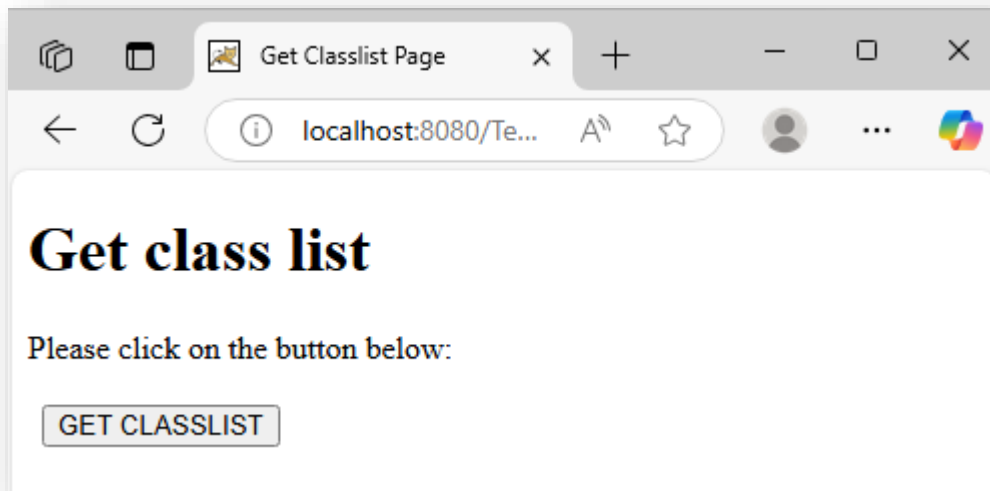
- [Menu Page](#)
- [Home Page](#)

SELECT * FROM APP.STUDENT ... X						
		Max. rows: 100	Fetched Rows: 9		Matching Rows:	
#	STUDENT_NUMBER	AGE	CREATIONDATE	GENDER	NAME	SURNAME
1	111	20	2025-05-23 22:33:39.852	M	Tata	Mama
2	222	45	2025-05-24 13:16:36.866	F	Khomotjo	Memani
3	333	30	2025-05-23 22:34:23.869	F	Titi	Mimi
4	444	33	2025-05-23 22:34:50.946	M	Toto	Momo
5	555	33	2025-05-23 22:35:12.415	M	Tutu	Mumu
6	666	30	2025-05-24 13:08:18.763	M	Vuyi	JMemani
7	777	25	2025-05-24 13:08:46.059	F	Thato	Ranamane
8	888	22	2025-05-24 13:09:11.448	F	Lira	Nkosi
9	999	40	2025-05-24 13:09:38.192	M	Tebza	Mabale

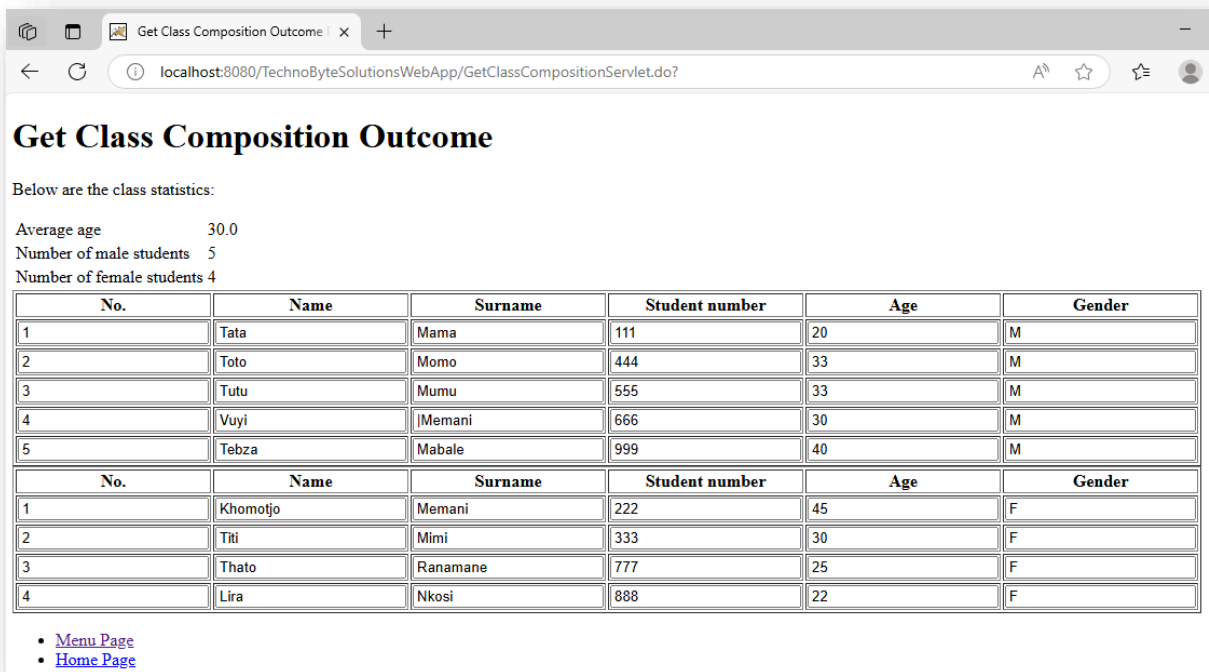
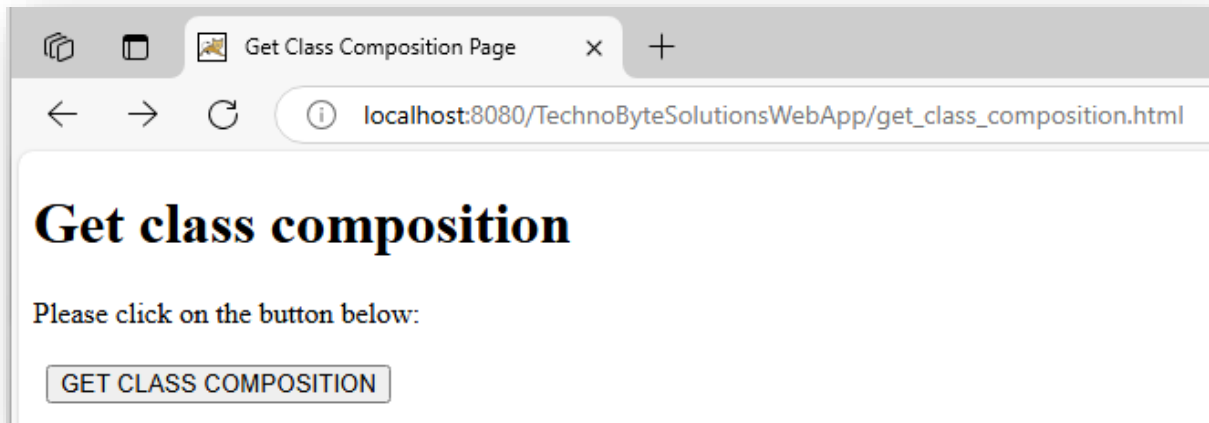
Click the fifth link to display the class count



Click the sixth link to display the classlist



Click the seventh link to display class composition information



Click the eight link to display the classlist according to age group

Get Classlist Per Age Group

Please provide the following information:

Minimum age:

Maximum age:

Get classlist outcome

There are 6 students falling under the specified age group.

No.	Name	Surname	Student number	Age	Gender
1	Khomotjo	Memani	222	45	F
2	Titi	Mimi	333	30	F
3	Toto	Momo	444	33	M
4	Tutu	Mumu	555	33	M
5	Vuyi	Memani	666	30	M
6	Tebza	Mabale	999	40	M

- [Menu Page](#)
- [Home Page](#)

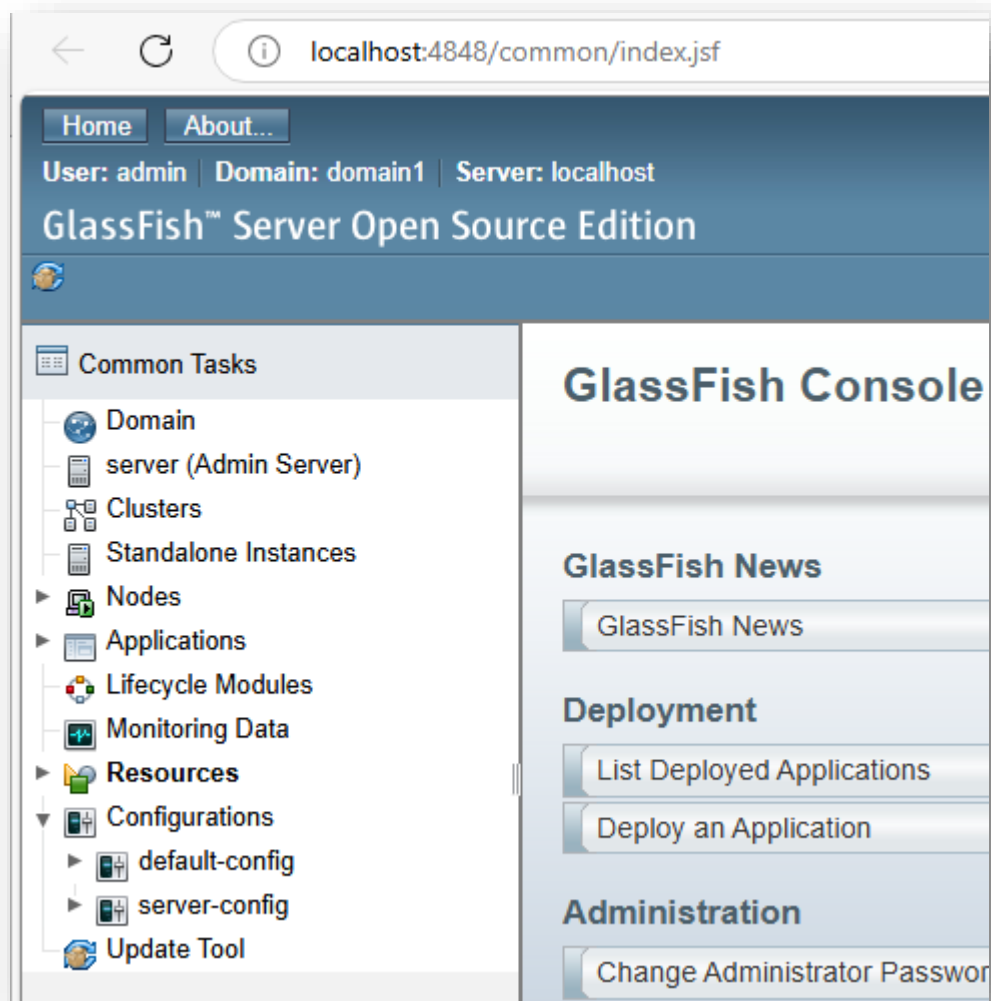
Step 5: Include security

1. Create the roles on the server (Glassfish).

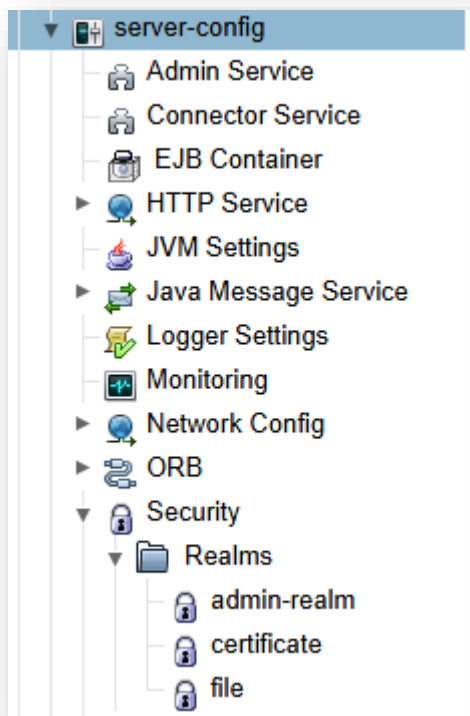
The users to create are as shown in the table below

No.	Username	Password	Role(s)
1	Nyiko	123	hod
2	Vuyi	456	lecturer, assistant
3	Thato	678	assistant

Open the **Admin Console** of the server



Under **Configurations**, click on **server-config | Security | Realms**



Click on **file**

Edit Realm

Edit an existing security (authentication) realm.

[Manage Users](#)

Configuration Name: server-config

Realm Name: file

Class Name: com.sun.enterprise.security.auth.realm.file.FileRealm

Properties specific to this Class

JAAS Context: *
Identifier for the login module to use for this realm

Key File: *
Full path and name of the file where the server will store all user, group, and password information for this realm

Assign Groups:
Comma-separated list of group names

Click on **Manage Users**

File Users

Manage user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

File Users (0)

New...Delete

Select	User ID	Group List:
No items found.		

Click on **New**

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name:

file

User ID: *

Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:

Separate multiple groups with colon

New Password:

Confirm New Password:

Populate the form with the details of the first user (Nyiko, who is an HoD)

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

User ID: *

Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:

Separate multiple groups with colon

New Password:

Confirm New Password:

Click OK

File Users

Manage user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

File Users (1)

[New...](#)[Delete](#)

Select	User ID	Group List:
☐	nyiko	hod

Add the second user. Click on **New**

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

User ID: *

Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:

Separate multiple groups with colon

New Password:

Confirm New Password:

Provide the details of user **vuyi**. This user wears two hats, that of a **lecturer** and **student assistant**.

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

User ID: *

Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:

Separate multiple groups with colon

New Password:

Confirm New Password:

Click OK

File Users

Manage user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

File Users (2)		
New... Delete		
Select	User ID	Group List:
☐	nyiko	hod
☐	vuyi	lecturer assistant

Click **New**

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

User ID: *
Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:
Separate multiple groups with colon

New Password:

Confirm New Password:

Provide details of the third user (student assistant).

New File Realm User

Create new user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

User ID: *

thato

Name can be up to 255 characters, must contain only letters, digits, underscore, dash, or dot characters

Group List:

assistant

Separate multiple groups with colon

New Password:

...

Confirm New Password:

...

Click OK

File Users

Manage user accounts for the currently selected security realm.

Configuration Name: server-config

Realm Name: file

File Users (3)

New...

Delete

Select	User ID	Group List:
☐	nyiko	hod
☐	vuyi	lecturer assistant
☐	thato	assistant

2. Perform security measures on the session beans.

Open the session bean

```
1 package za.ac.tut.model.bl;
2
3 import java.util.List;
4 import javax.ejb.Stateless;
5 import javax.persistence.EntityManager;
6 import javax.persistence.PersistenceContext;
7 import javax.persistence.Query;
8 import za.ac.tut.model.entity.Student;
9
10 /**
11  *
12  * @author MemaniV
13  */
14 @Stateless
15 public class StudentFacade extends AbstractFacade<Student> implements StudentFacadeLocal {
16     @PersistenceContext(unitName = "TechnoByteSolutionsEJBModulePU")
17     private EntityManager em;
18 }
```

Apply security Annotations to each method

```
30 @RolesAllowed("lecturer")
31 @Override
32 public Long getNumberOfStudentsPerGender(Character gender) {
33     String queryStr = "SELECT count(s) FROM Student s WHERE s.gender = ?1";
34     Query query = em.createQuery(queryStr);
35     query.setParameter(1, gender);
36     Long cnt = (Long)query.getSingleResult();
37     return cnt;
38 }
39
40 @RolesAllowed("lecturer")
41 @Override
42 public List<Student> getStudentsPerGender(Character gender) {
43     String queryStr = "SELECT s FROM Student s WHERE s.gender = ?1";
44     Query query = em.createQuery(queryStr);
45     query.setParameter(1, gender);
46     List<Student> students = (List<Student>)query.getResultList();
47     return students;
48 }
```



```

49  @RolesAllowed("lecturer")
50  @Override
51  public Double getAverageAge() {
52      String queryStr = "SELECT avg(s.age) FROM Student s";
53      Query query = em.createQuery(queryStr);
54      Double cnt = (Double)query.getSingleResult();
55      return cnt;
56  }
57
58  @RolesAllowed("lecturer")
59  @Override
60  public Long getNumberOfStudentsPerAgeRange(Integer minAge, Integer maxAge) {
61      String queryStr = "SELECT count(s) FROM Student s WHERE s.age >= ?1 AND s.age <= ?2";
62      Query query = em.createQuery(queryStr);
63      query.setParameter(1, minAge);
64      query.setParameter(2, maxAge);
65      Long cnt = (Long)query.getSingleResult();
66      return cnt;
67  }

```

```

70  @RolesAllowed("lecturer")
71  @Override
72  public List<Student> getStudentsPerAgeRange(Integer minAge, Integer maxAge) {
73      String queryStr = "SELECT s FROM Student s WHERE s.age >= ?1 AND s.age <= ?2";
74      Query query = em.createQuery(queryStr);
75      query.setParameter(1, minAge);
76      query.setParameter(2, maxAge);
77      List<Student> students = (List<Student>)query.getResultList();
78      return students;
79  }
80  }
81

```

Open the abstract facade class

```
1 package za.ac.tut.model.bl;
2
3 import java.util.List;
4 import javax.persistence.EntityManager;
5
6 /**
7  *
8  * @author MemaniV
9  */
10 public abstract class AbstractFacade<T> {
11
12     private Class<T> entityClass;
13
14     public AbstractFacade(Class<T> entityClass) {
15         this.entityClass = entityClass;
16     }
17 }
```

Apply security annotations to the respective methods

```
22 @RolesAllowed({"lecturer", "assistant"})
23 public void create(T entity) {
24     getEntityManager().persist(entity);
25 }
26
27 @RolesAllowed("lecturer")
28 public void edit(T entity) {
29     getEntityManager().merge(entity);
30 }
31 }
```

```

32     @RolesAllowed("hod")
33     public void remove(T entity) {
34         getEntityManager().remove(getEntityManager().merge(entity));
35     }
36
37     @RolesAllowed({"lecturer", "assistant"})
38     public T find(Object id) {
39         return getEntityManager().find(entityClass, id);
40     }

```

```

42     @PermitAll
43     public List<T> findAll() {
44         javax.persistence.criteria.CriteriaQuery cq =
45             getEntityManager().getCriteriaBuilder().createQuery();
46         cq.select(cq.from(entityClass));
47         return getEntityManager().createQuery(cq).getResultList();
48     }
49
50     @PermitAll
51     public List<T> findRange(int[] range) {
52         javax.persistence.criteria.CriteriaQuery cq =
53             getEntityManager().getCriteriaBuilder().createQuery();
54         cq.select(cq.from(entityClass));
55         javax.persistence.Query q = getEntityManager().createQuery(cq);
56         q.setMaxResults(range[1] - range[0] + 1);
57         q.setFirstResult(range[0]);
58         return q.getResultList();
59     }

```

```

61     @RolesAllowed("lecturer")
62     public int count() {
63         javax.persistence.criteria.CriteriaQuery cq =
64             getEntityManager().getCriteriaBuilder().createQuery();
65         javax.persistence.criteria.Root<T> rt = cq.from(entityClass);
66         cq.select(getEntityManager().getCriteriaBuilder().count(rt));
67         javax.persistence.Query q = getEntityManager().createQuery(cq);
68         return ((Long) q.getSingleResult()).intValue();
69     }
70
71 }
72

```

Clean and Build the EJB module

```

Output x
Java DB Database Process x SQL 1 execution x GlassFish Server 4.1.1 x SQL 2 execution x SQL 3 execution x SQL 4 execution x TechnoByteSolutionsEJBModule (clean,dist) x
INFO: Generating META-INF/eclipse-link-orm.xml
WARNING: META-INF/eclipse-link-orm.xml file was not found: META-INF/eclipse-link-orm.xml continuing with generation.
WARNING: META-INF/eclipse-link-orm.xml file was not found: META-INF/orm.xml continuing with generation.
WARNING: META-INF/eclipse-link-orm.xml file was not found: META-INF/eclipse-link-orm.xml continuing with generation.
WARNING: input files use unchecked or unsafe operations.
WARNING: compile with -Xlint:unchecked for details.

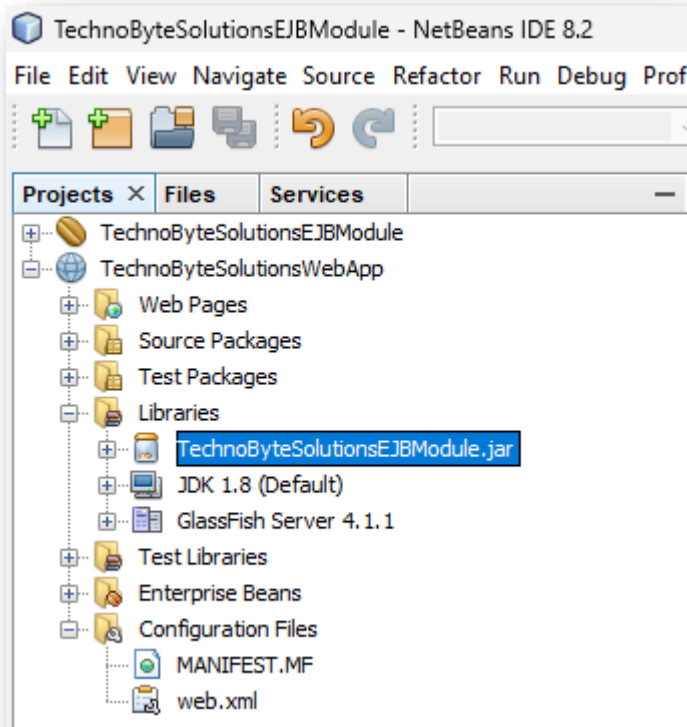
Inclusion-in-archive:
jar: C:\Users\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsEJBModule\dist\TechnoByteSolutionsEJBModule.jar
jar: C:\Users\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsEJBModule\dist\TechnoByteSolutionsEJBModule.jar

SUCCESSFUL (total time: 1 second)

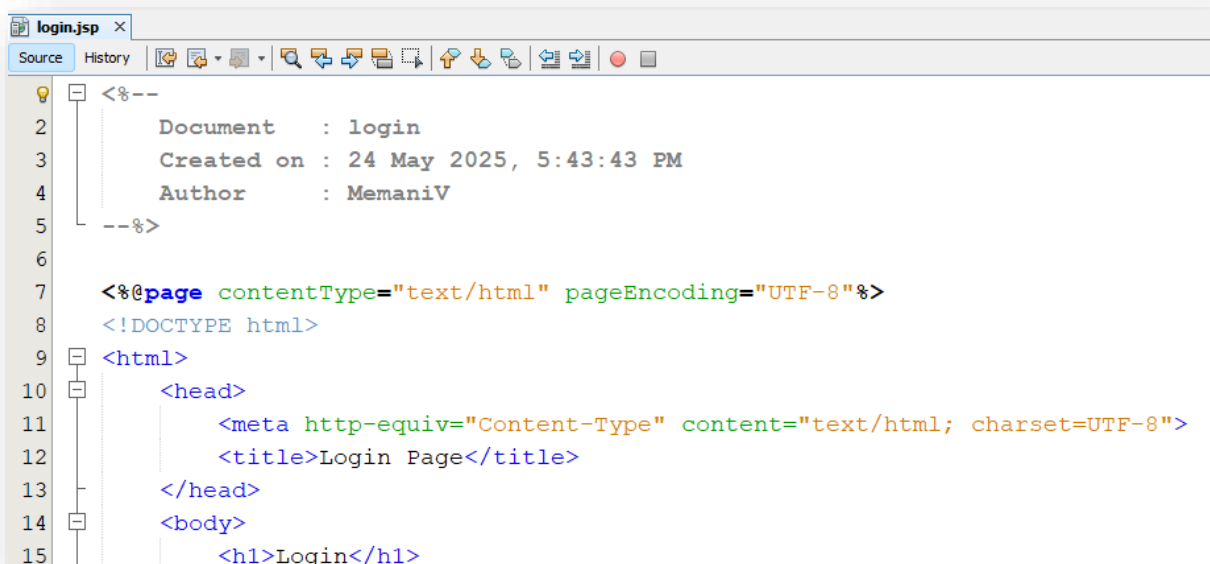
```

3. Perform security measures on the web module.

Include the latest EJB jar file in the web module



Create a jsp file called login



```

19      <form action="j_security_check" method="POST">
20      <table>
21      <tr>
22      <td>Username:</td>
23      <td><input type="text" name="j_username" required=""></td>
24      </tr>
25      <tr>
26      <td>Password:</td>
27      <td><input type="text" name="j_password" required=""></td>
28      </tr>
29      <tr>
30      <td></td>
31      <td><input type="submit" value="LOGIN"></td>
32      </tr>
33      </table>
34      </form>
35      </body>
36      </html>
37

```

Create a login error page

```

login_error.jsp x
Source History
--%>
2      Document    : login_error
3      Created on  : 24 May 2025, 5:53:21 PM
4      Author     : MemaniV
5      --%>
6
7      <%@page contentType="text/html" pageEncoding="UTF-8"%>
8      <!DOCTYPE html>
9      <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Login Error Page</title>
13     </head>
14     <body>
15         <h1>Login error</h1>

```

```

19      <ul>
20          <li><a href="menu.html">Menu Page</a></li>
21          <li><a href="index.html">Home Page</a></li>
22      </ul>          </form>
23  </body>
24 </html>
25

```

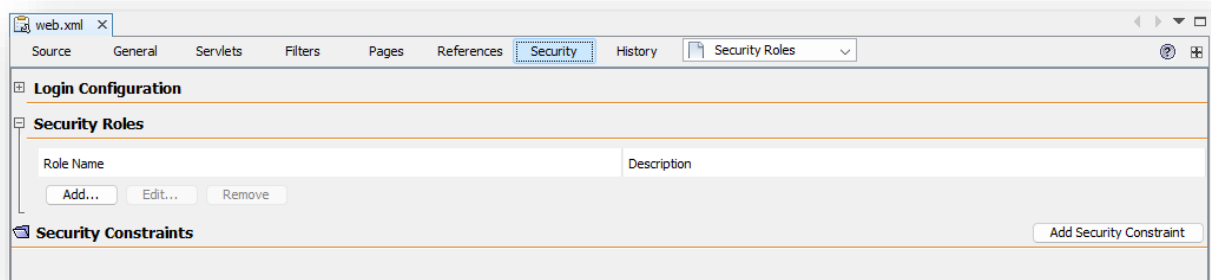
Open the **web.xml** page

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app version="3.1" xmlns="http://xmlns.jcp.org/xml/ns/javaee" xmlns:xsi="http://xmlns.jcp.org/xml/ns/xsi"
3     <servlet>
4         <servlet-name>AddStudentServlet</servlet-name>
5         <servlet-class>za.ac.tut.web.AddStudentServlet</servlet-class>
6     </servlet>
7     <servlet>
8         <servlet-name>FindStudentServlet</servlet-name>
9         <servlet-class>za.ac.tut.web.FindStudentServlet</servlet-class>
10    </servlet>
11    <servlet>
12        <servlet-name>GetStudentServlet</servlet-name>
13        <servlet-class>za.ac.tut.web.GetStudentServlet</servlet-class>
14    </servlet>

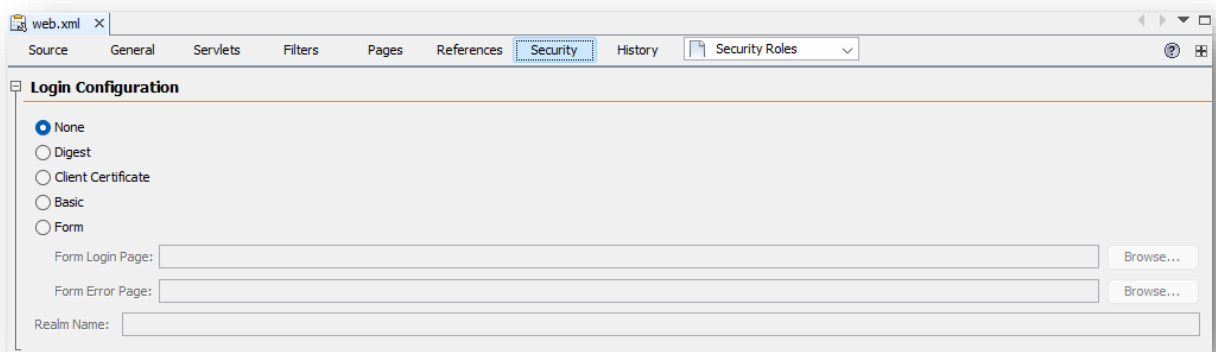
```

Click on **Security**



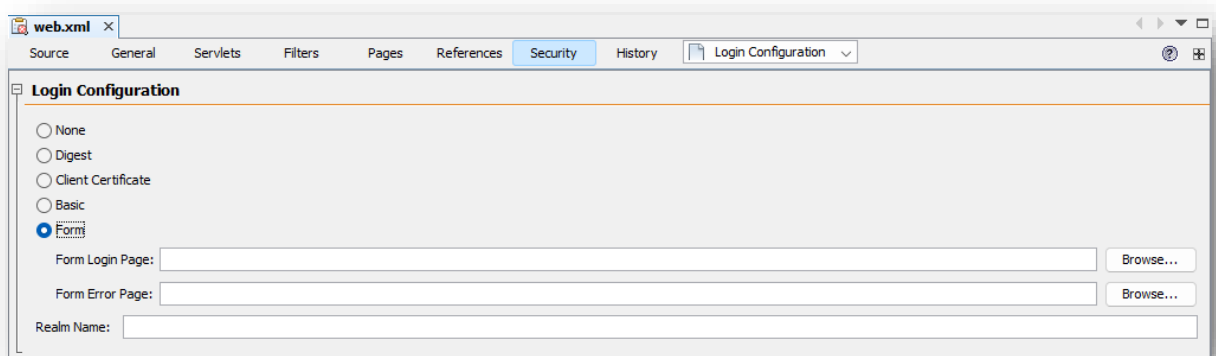
Part A: Login Configuration

Expand Login Configuration



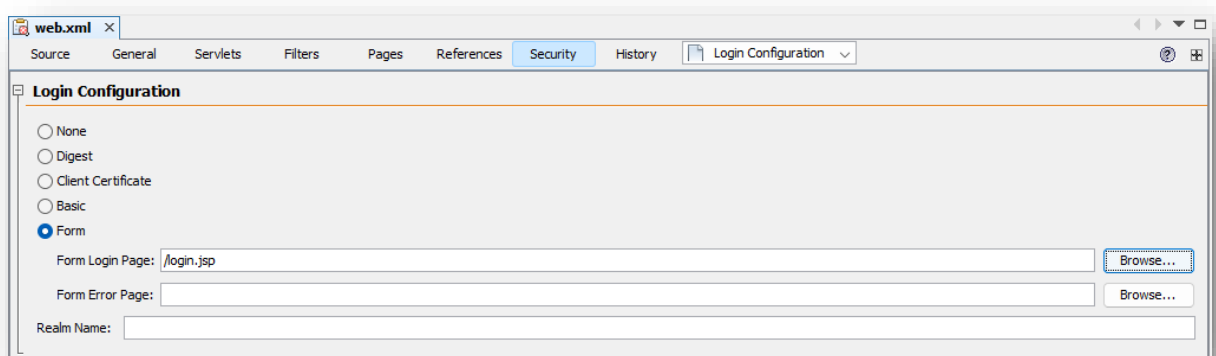
The screenshot shows the 'Login Configuration' dialog box with the 'Security' tab selected. The 'None' radio button is selected. The 'Form Login Page' and 'Form Error Page' fields are empty, each with a 'Browse...' button to its right. The 'Realm Name' field is also empty.

Select Form



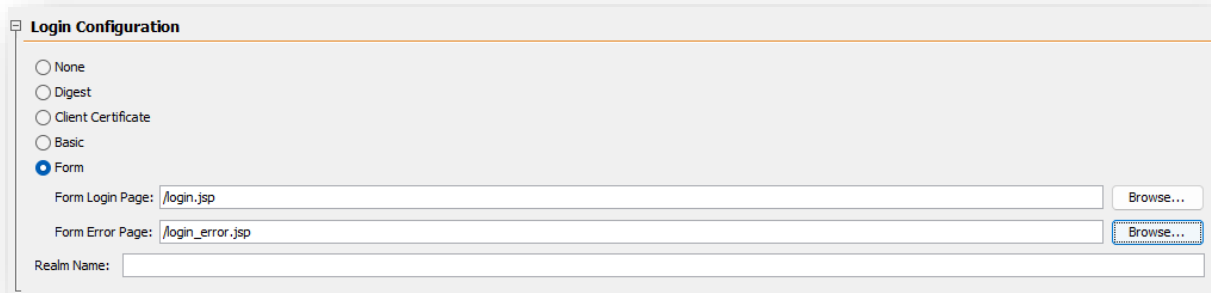
The screenshot shows the 'Login Configuration' dialog box with the 'Form' radio button selected. The 'Form Login Page' and 'Form Error Page' fields are empty, each with a 'Browse...' button to its right. The 'Realm Name' field is also empty.

Click on the Browse button to load the login page



The screenshot shows the 'Login Configuration' dialog box with the 'Form' radio button selected. The 'Form Login Page' field now contains the path '/login.jsp', and the 'Browse...' button next to it is highlighted with a blue border. The 'Form Error Page' field is still empty, and the 'Realm Name' field is also empty.

Do the same for the login error page



Login Configuration

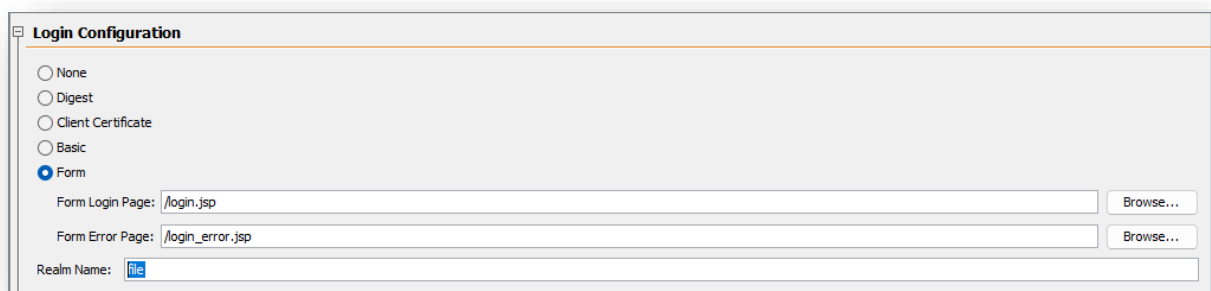
☐ None
☐ Digest
☐ Client Certificate
☐ Basic
☒ Form

Form Login Page:

Form Error Page:

Realm Name:

Enter **file** under **Realm name**



Login Configuration

☐ None
☐ Digest
☐ Client Certificate
☐ Basic
☒ Form

Form Login Page:

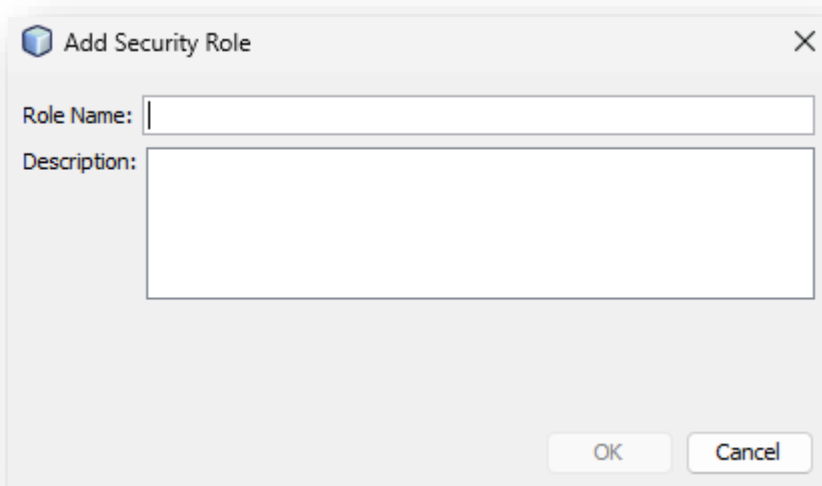
Form Error Page:

Realm Name:

Part B: Security Roles

In this section we define the roles.

Click **Add** under Security Roles

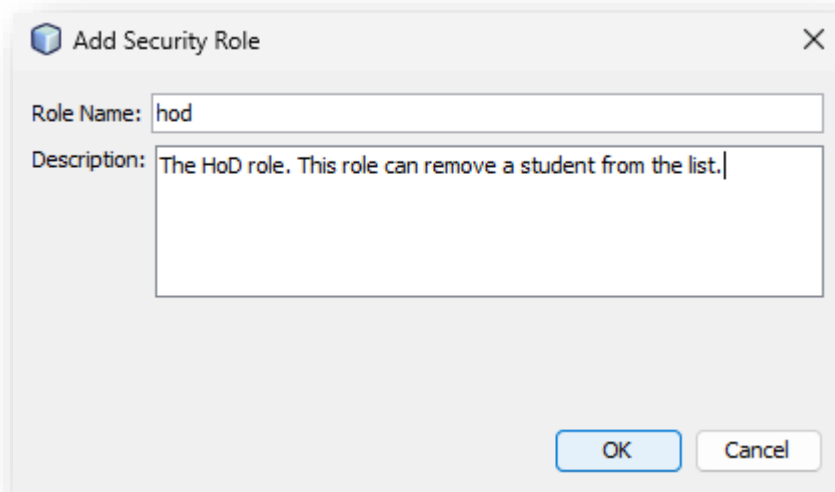


Add Security Role

Role Name:

Description:

Provide role information



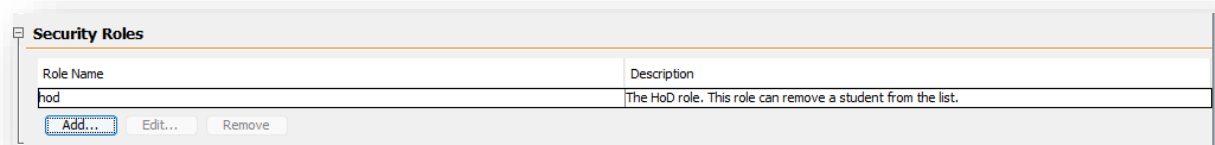
The 'Add Security Role' dialog box is shown. It has a title bar with a cube icon and a close button. The 'Role Name' field contains the text 'hod'. The 'Description' field contains the text 'The HoD role. This role can remove a student from the list.' There are 'OK' and 'Cancel' buttons at the bottom right.

Role Name:

Description:

OK Cancel

Click OK

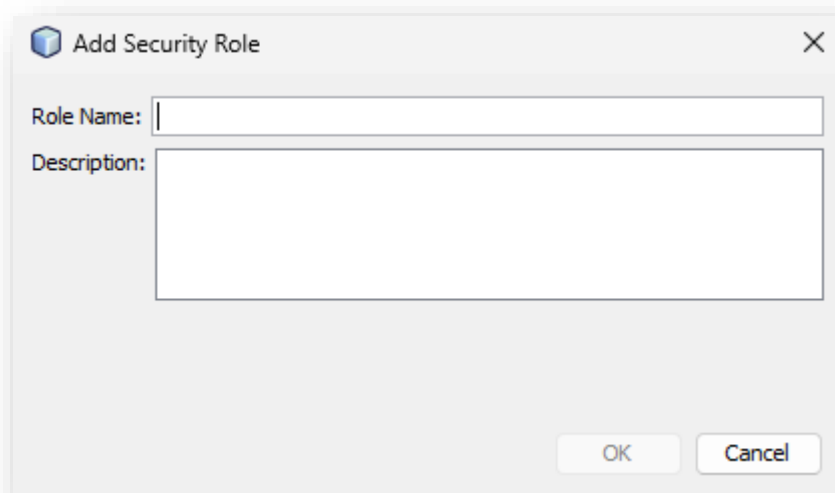


The 'Security Roles' table is shown. It has two columns: 'Role Name' and 'Description'. The first row contains the role 'hod' and its description 'The HoD role. This role can remove a student from the list.' Below the table are buttons for 'Add...', 'Edit...', and 'Remove'.

Role Name	Description
hod	The HoD role. This role can remove a student from the list.

Add... Edit... Remove

Click on **Add**.



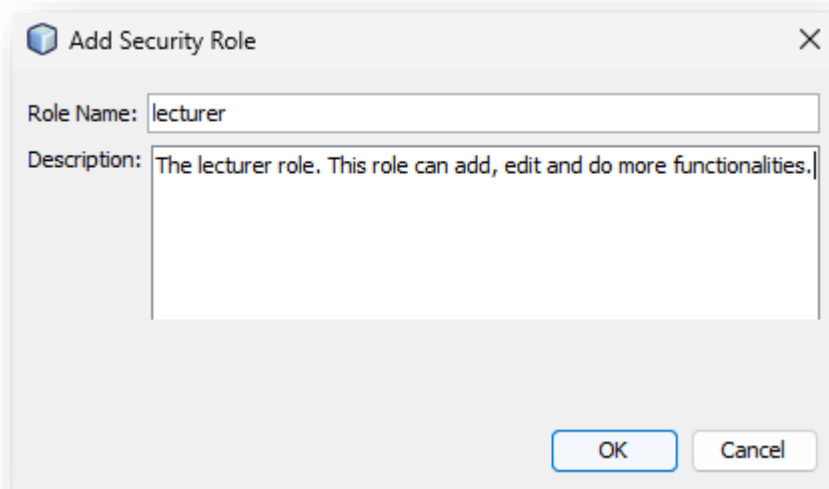
The 'Add Security Role' dialog box is shown again. The 'Role Name' field is empty. The 'Description' field is empty. There are 'OK' and 'Cancel' buttons at the bottom right.

Role Name:

Description:

OK Cancel

Provide role information



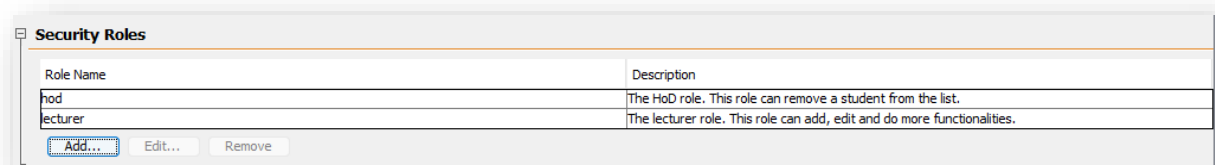
The 'Add Security Role' dialog box is shown. It has a title bar with a blue cube icon and a close button. The 'Role Name' field contains the text 'lecturer'. The 'Description' field contains the text 'The lecturer role. This role can add, edit and do more functionalities.'. At the bottom right are 'OK' and 'Cancel' buttons.

Role Name: lecturer

Description: The lecturer role. This role can add, edit and do more functionalities.

OK Cancel

Click OK

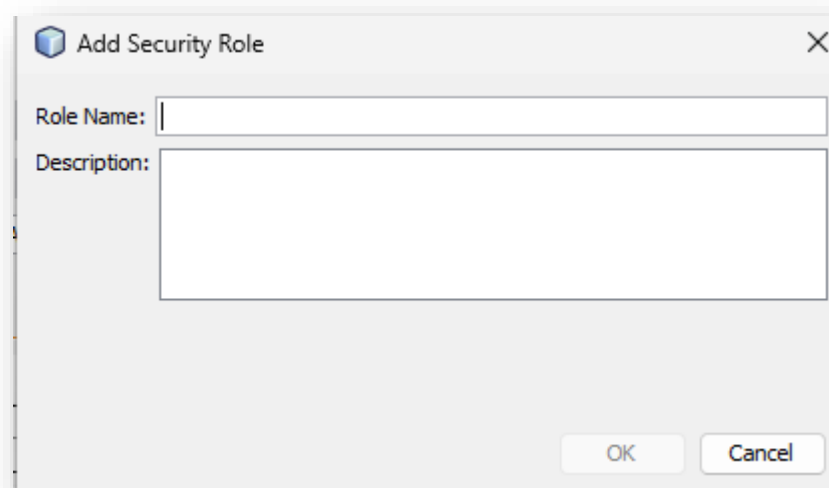


The 'Security Roles' table is shown. It has two columns: 'Role Name' and 'Description'. The first row has 'hod' in the 'Role Name' column and 'The HoD role. This role can remove a student from the list.' in the 'Description' column. The second row has 'lecturer' in the 'Role Name' column and 'The lecturer role. This role can add, edit and do more functionalities.' in the 'Description' column. Below the table are 'Add...', 'Edit...', and 'Remove' buttons.

Role Name	Description
hod	The HoD role. This role can remove a student from the list.
lecturer	The lecturer role. This role can add, edit and do more functionalities.

Add... Edit... Remove

Click on Add



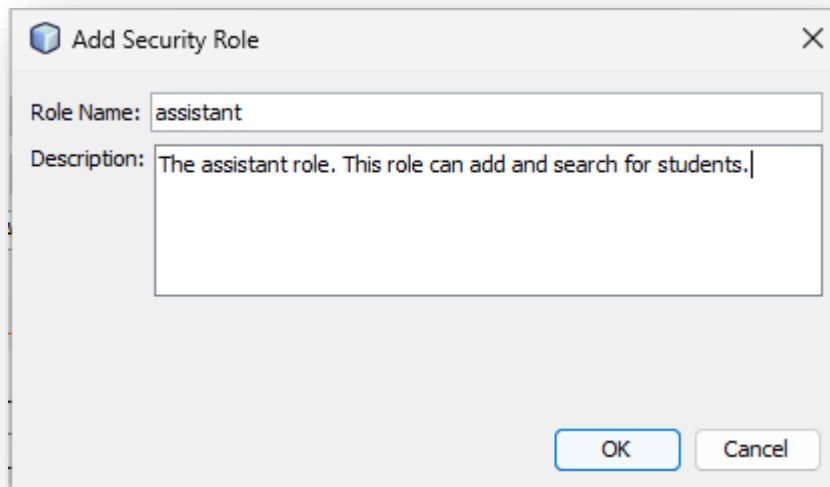
The 'Add Security Role' dialog box is shown. It has a title bar with a blue cube icon and a close button. The 'Role Name' field is empty. The 'Description' field is empty. At the bottom right are 'OK' and 'Cancel' buttons.

Role Name:

Description:

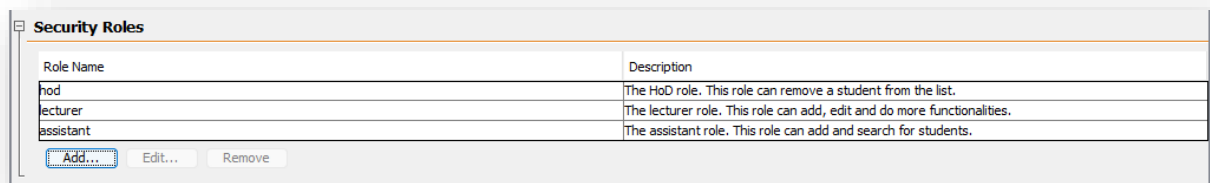
OK Cancel

Provide role information



The 'Add Security Role' dialog box is shown. It has a title bar with a blue icon and the text 'Add Security Role'. Inside, there are two text input fields. The first is labeled 'Role Name:' and contains the text 'assistant'. The second is labeled 'Description:' and contains the text 'The assistant role. This role can add and search for students.' At the bottom right, there are two buttons: 'OK' and 'Cancel'.

Click OK



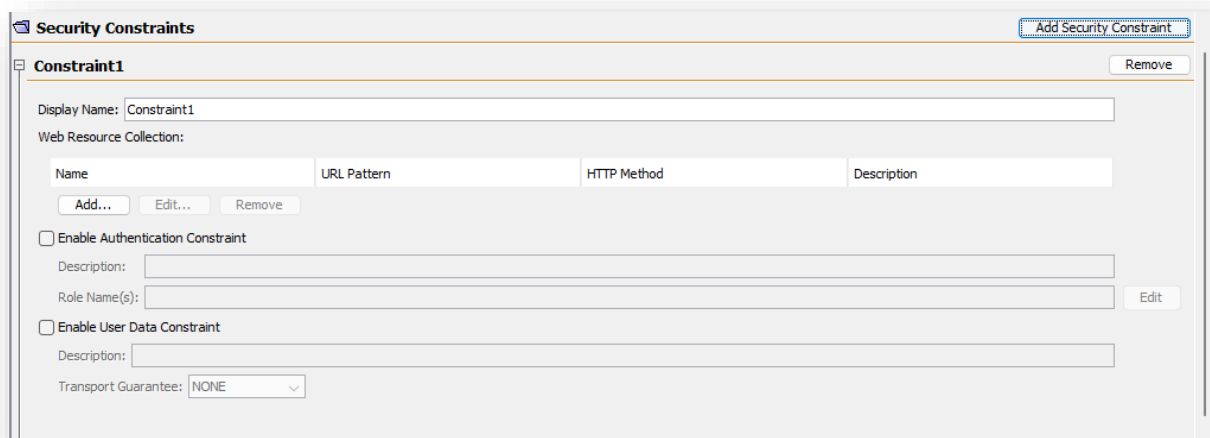
The 'Security Roles' table is shown. It has a title bar with a blue icon and the text 'Security Roles'. Below the title bar is a table with two columns: 'Role Name' and 'Description'. The table contains three rows: 'hod', 'lecturer', and 'assistant'. Below the table are three buttons: 'Add...', 'Edit...', and 'Remove'.

Role Name	Description
hod	The HoD role. This role can remove a student from the list.
lecturer	The lecturer role. This role can add, edit and do more functionalities.
assistant	The assistant role. This role can add and search for students.

Part C: Security Constraints

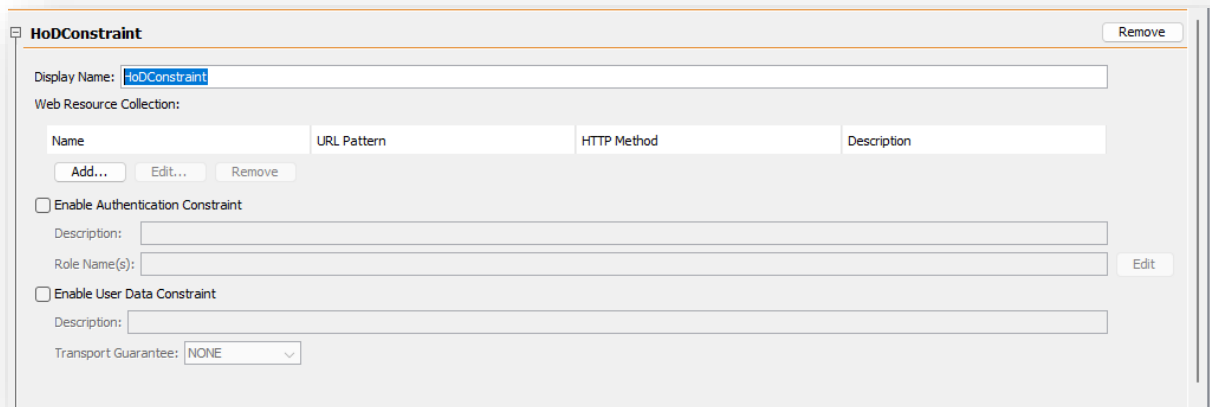
In this section we state what each role can do.

Under **Security Constraints**, click on **Add Security Constraint**



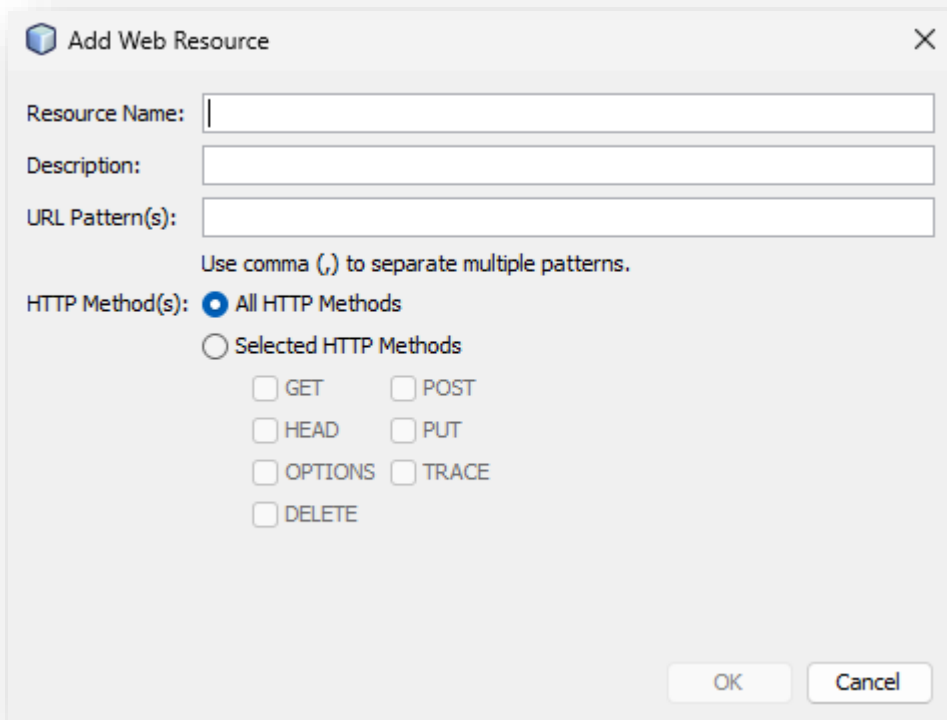
The 'Security Constraints' dialog box is shown. It has a title bar with a blue icon and the text 'Security Constraints'. Below the title bar is a button 'Add Security Constraint'. Below that is a section titled 'Constraint1' with a 'Remove' button. Inside this section, there is a 'Display Name:' field with the value 'Constraint1'. Below that is a 'Web Resource Collection:' section with a table. The table has four columns: 'Name', 'URL Pattern', 'HTTP Method', and 'Description'. Below the table are three buttons: 'Add...', 'Edit...', and 'Remove'. Below the table are two checkboxes: 'Enable Authentication Constraint' and 'Enable User Data Constraint'. Below 'Enable Authentication Constraint' are two text input fields: 'Description:' and 'Role Name(s):'. Below 'Enable User Data Constraint' are two text input fields: 'Description:' and 'Transport Guarantee:'. The 'Transport Guarantee:' field has a dropdown menu with the value 'NONE'.

Name the constraint as **HoDConstraint**



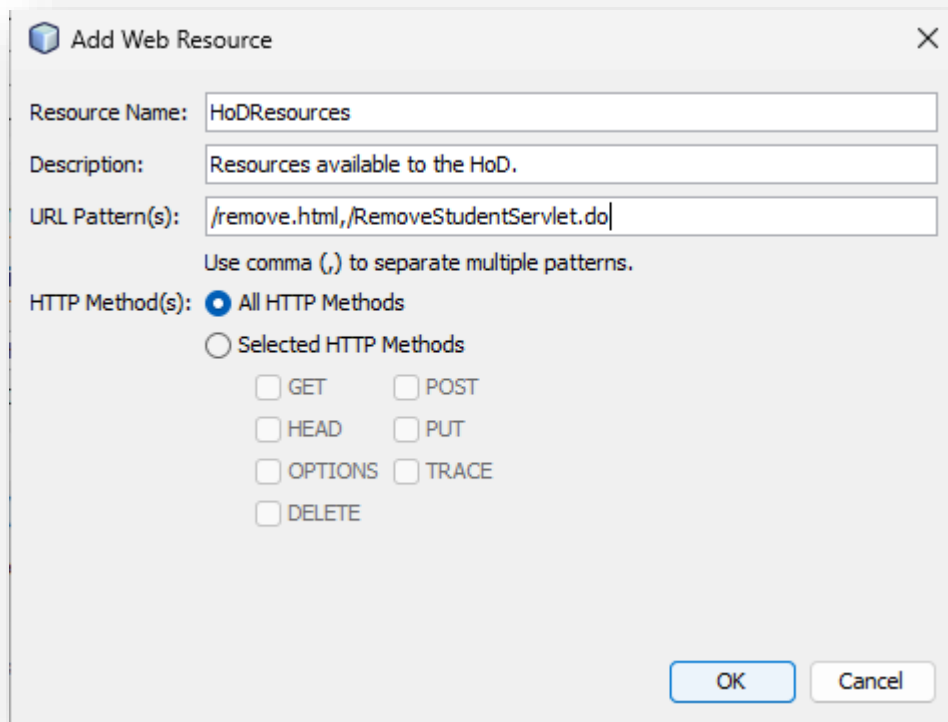
The screenshot shows a configuration window titled "HoDConstraint" with a "Remove" button in the top right corner. The "Display Name" field is set to "HoDConstraint". Below this is a "Web Resource Collection" section containing a table with columns: Name, URL Pattern, HTTP Method, and Description. Under the table are buttons for "Add...", "Edit...", and "Remove". Below the table are two sections: "Enable Authentication Constraint" and "Enable User Data Constraint". Each section has a "Description" field and a "Role Name(s)" field. The "Enable Authentication Constraint" section also has an "Edit" button. At the bottom, there is a "Transport Guarantee" dropdown menu set to "NONE".

Click on Add



The screenshot shows a dialog box titled "Add Web Resource" with a close button (X) in the top right corner. It contains the following fields and options: "Resource Name:" (text field), "Description:" (text field), "URL Pattern(s):" (text field) with a note "Use comma (,) to separate multiple patterns.", "HTTP Method(s):" with two radio buttons: "All HTTP Methods" (selected) and "Selected HTTP Methods". Under "Selected HTTP Methods" are checkboxes for GET, POST, HEAD, PUT, OPTIONS, TRACE, and DELETE. At the bottom are "OK" and "Cancel" buttons.

Complete the form

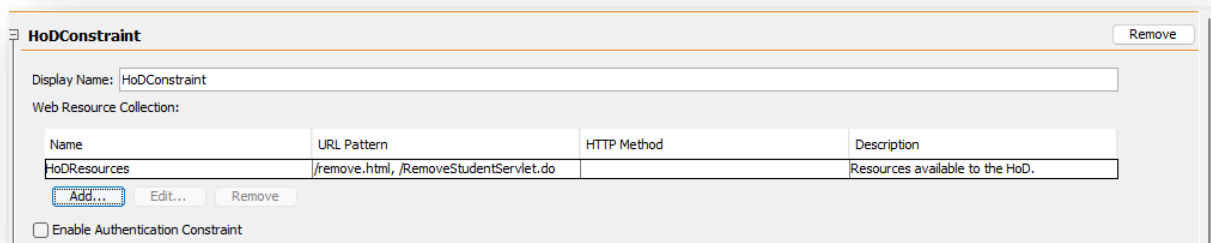


The 'Add Web Resource' dialog box contains the following fields and options:

- Resource Name:** HoDResources
- Description:** Resources available to the HoD.
- URL Pattern(s):** /remove.html,/RemoveStudentServlet.do
- Use comma (,) to separate multiple patterns.**
- HTTP Method(s):**
 - ☒ All HTTP Methods
 - ☐ Selected HTTP Methods
 - ☐ GET ☐ POST
 - ☐ HEAD ☐ PUT
 - ☐ OPTIONS ☐ TRACE
 - ☐ DELETE

Buttons: OK, Cancel

Click OK



The 'HoDConstraint' configuration window shows the following details:

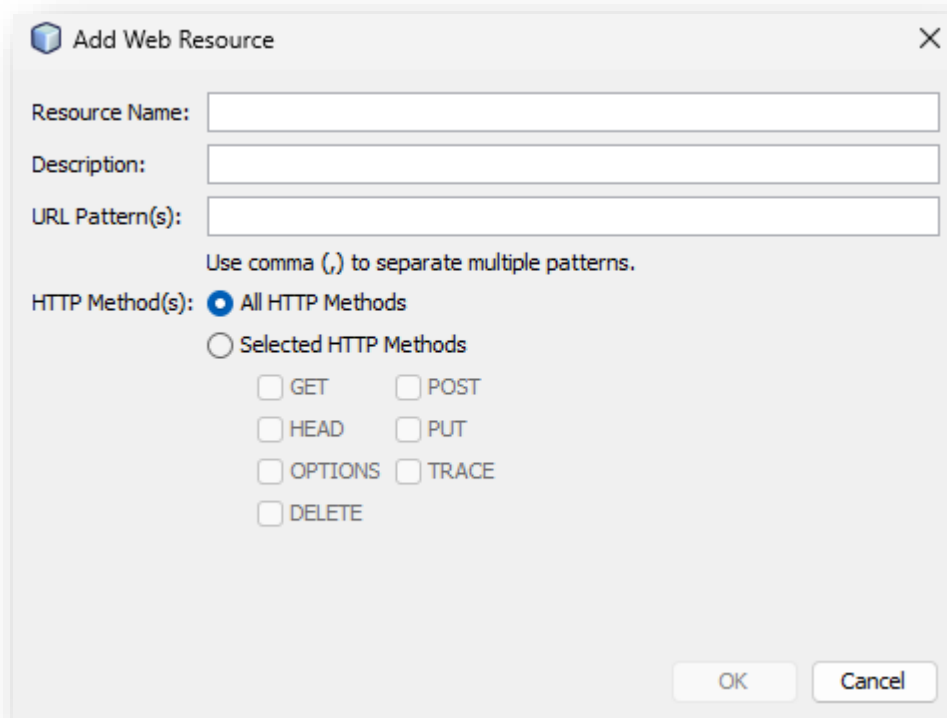
- Display Name:** HoDConstraint
- Web Resource Collection:**

Name	URL Pattern	HTTP Method	Description
HoDResources	/remove.html, /RemoveStudentServlet.do		Resources available to the HoD.

Buttons: Add..., Edit..., Remove

☐ Enable Authentication Constraint

Click Add

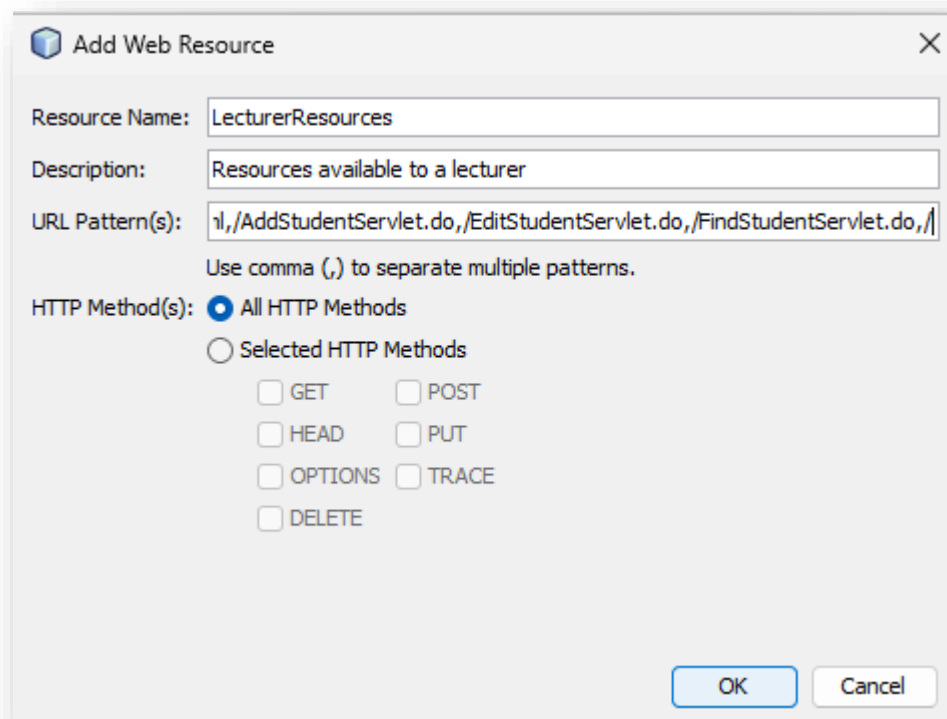


The dialog box is titled "Add Web Resource" and contains the following fields and options:

- Resource Name:** An empty text input field.
- Description:** An empty text input field.
- URL Pattern(s):** An empty text input field.
- HTTP Method(s):** A group of radio buttons and checkboxes.
 - ☒ All HTTP Methods
 - ☐ Selected HTTP Methods
 - ☐ GET
 - ☐ POST
 - ☐ HEAD
 - ☐ PUT
 - ☐ OPTIONS
 - ☐ TRACE
 - ☐ DELETE

Buttons at the bottom right: OK, Cancel.

Complete the form

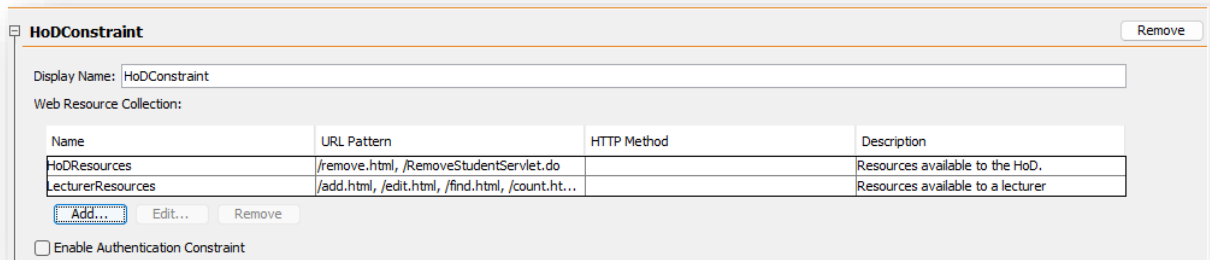


The dialog box is titled "Add Web Resource" and contains the following fields and options:

- Resource Name:** LecturerResources
- Description:** Resources available to a lecturer
- URL Pattern(s):** /, /AddStudentServlet.do, /EditStudentServlet.do, /FindStudentServlet.do, /
- HTTP Method(s):** A group of radio buttons and checkboxes.
 - ☒ All HTTP Methods
 - ☐ Selected HTTP Methods
 - ☐ GET
 - ☐ POST
 - ☐ HEAD
 - ☐ PUT
 - ☐ OPTIONS
 - ☐ TRACE
 - ☐ DELETE

Buttons at the bottom right: OK, Cancel.

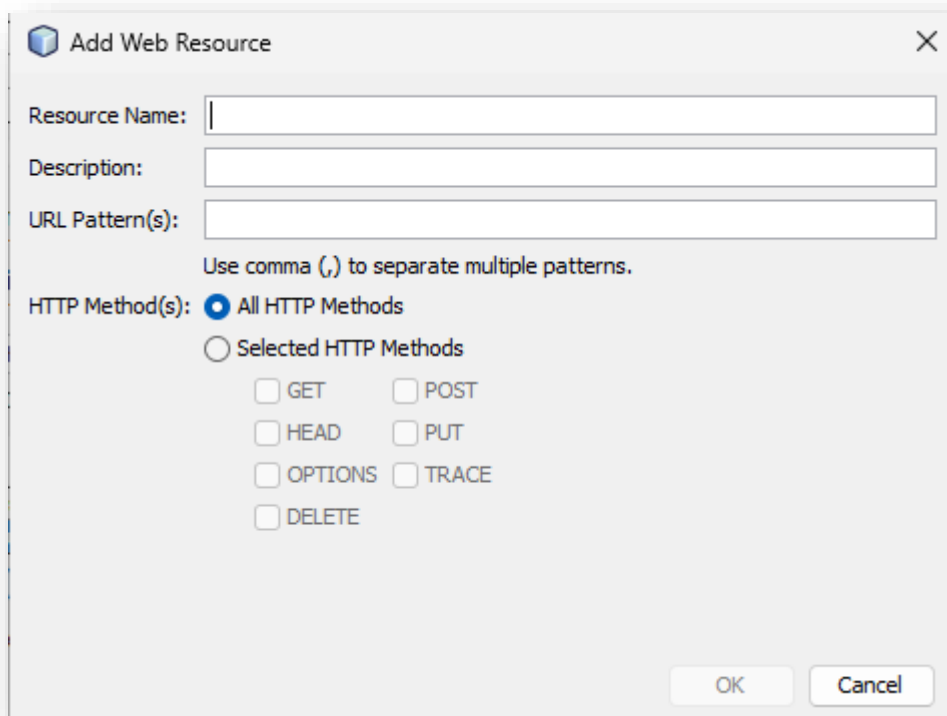
Click OK



The **HoDConstraint** dialog box is shown. It has a title bar with a minimize button, a maximize button, and a close button. The main area contains a **Display Name** field with the text "HoDConstraint". Below it is a **Web Resource Collection** section with a table. The table has four columns: **Name**, **URL Pattern**, **HTTP Method**, and **Description**. There are two rows of data. Below the table are three buttons: **Add...**, **Edit...**, and **Remove**. At the bottom, there is a checkbox labeled **Enable Authentication Constraint**.

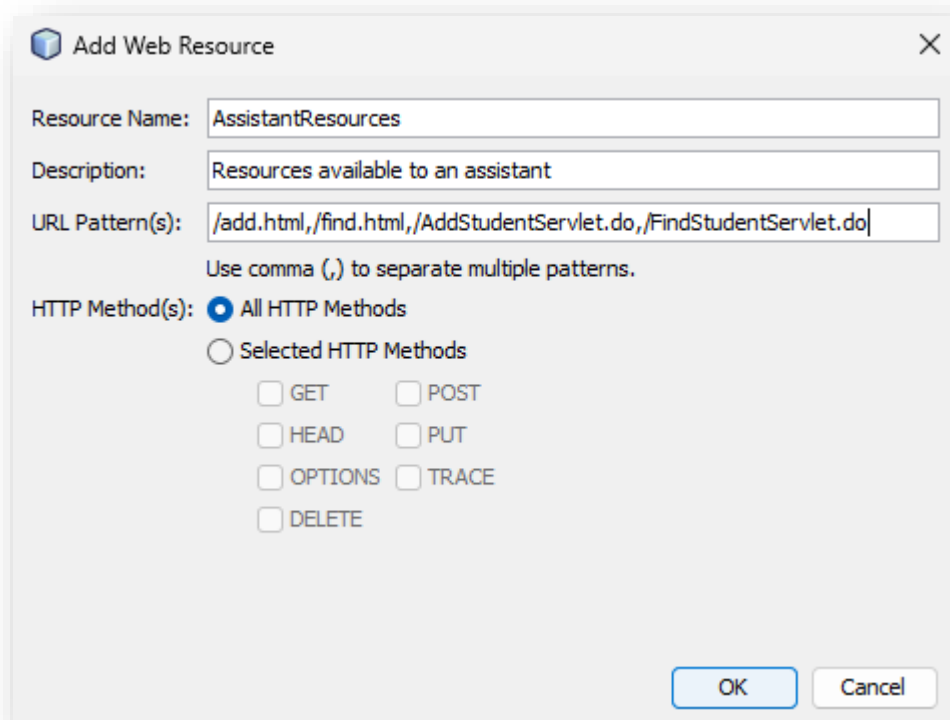
Name	URL Pattern	HTTP Method	Description
HoDResources	/remove.html, /RemoveStudentServlet.do		Resources available to the HoD.
LecturerResources	/add.html, /edit.html, /find.html, /count.ht...		Resources available to a lecturer

Click Add



The **Add Web Resource** dialog box is shown. It has a title bar with a minimize button, a maximize button, and a close button. The main area contains four text input fields: **Resource Name**, **Description**, **URL Pattern(s)**, and **HTTP Method(s)**. Below the **URL Pattern(s)** field is a note: "Use comma (,) to separate multiple patterns." Below the **HTTP Method(s)** field are two radio buttons: **All HTTP Methods** (selected) and **Selected HTTP Methods**. Below the **Selected HTTP Methods** radio button are six checkboxes: **GET**, **POST**, **HEAD**, **PUT**, **OPTIONS**, **TRACE**, and **DELETE**. At the bottom right are two buttons: **OK** and **Cancel**.

Complete the form



Add Web Resource

Resource Name: AssistantResources

Description: Resources available to an assistant

URL Pattern(s): /add.html,/find.html,/AddStudentServlet.do,/FindStudentServlet.do

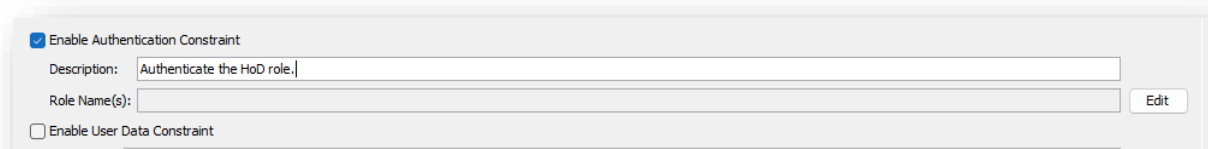
Use comma (,) to separate multiple patterns.

HTTP Method(s): ☒ All HTTP Methods
☐ Selected HTTP Methods

☐ GET ☐ POST
☐ HEAD ☐ PUT
☐ OPTIONS ☐ TRACE
☐ DELETE

OK Cancel

Select **Enable Authentication Constraint**



☒ Enable Authentication Constraint

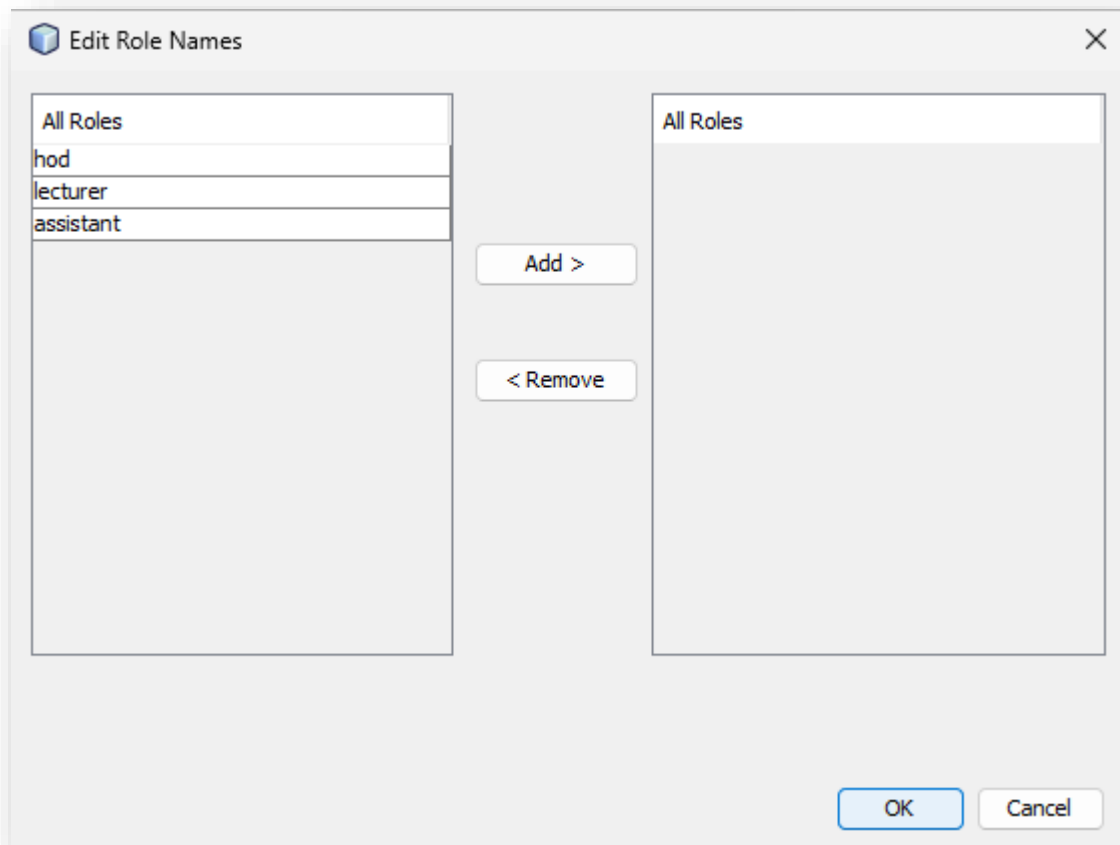
Description: Authenticate the HoD role.

Role Name(s):

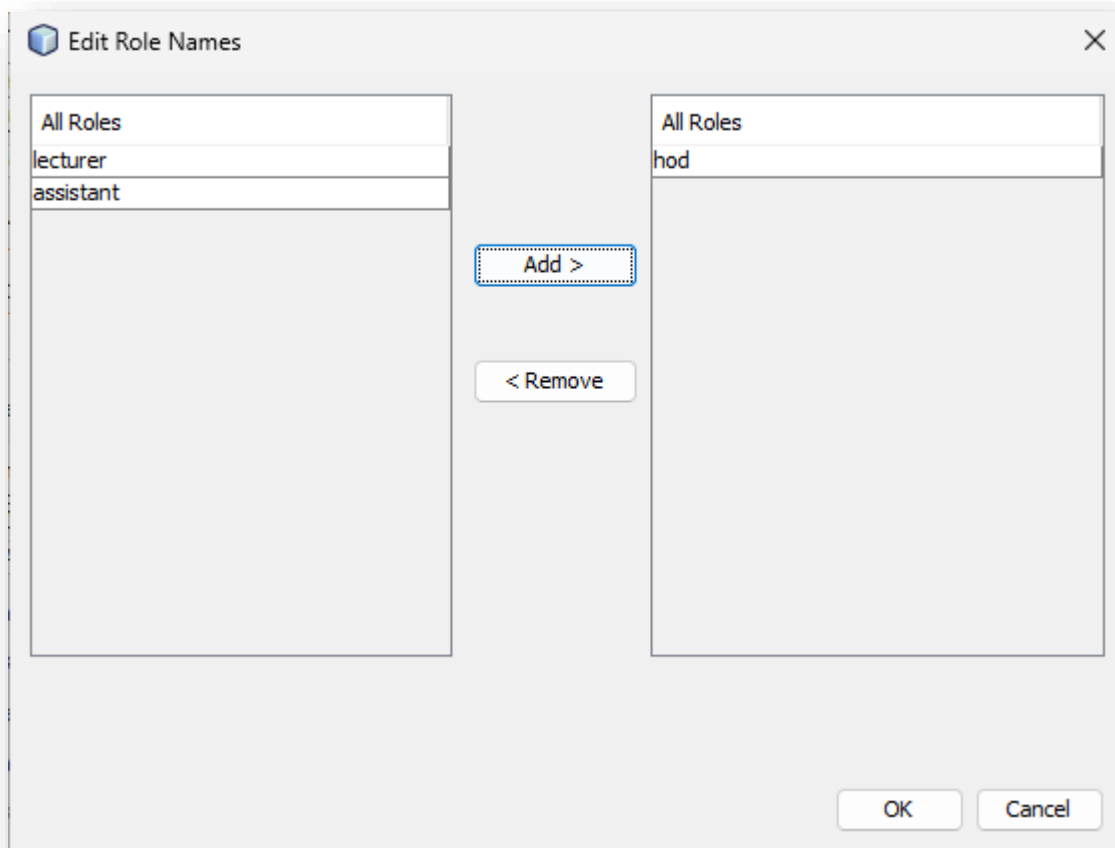
Edit

☐ Enable User Data Constraint

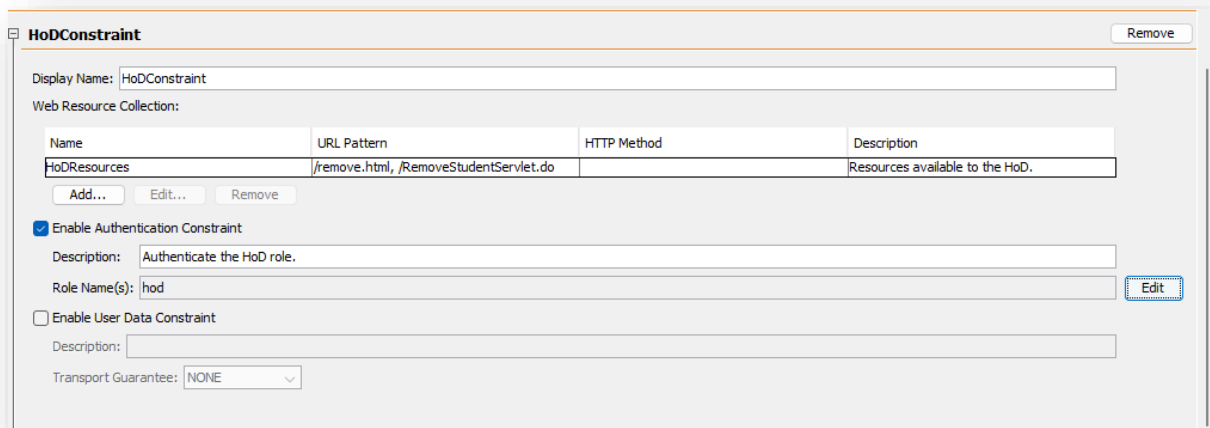
Click on Edit (opposite Role Name(s))



Select all the hod role and click on Add



Click OK



Follow the same procedure to add two more constraints, LecturerConstraint)

Add Web Resource

Resource Name:

Description:

URL Pattern(s):

Use comma (,) to separate multiple patterns.

HTTP Method(s):

☒ All HTTP Methods
 ☐ Selected HTTP Methods

☐ GET
 ☐ POST
 ☐ HEAD
 ☐ PUT
 ☐ OPTIONS
 ☐ TRACE
 ☐ DELETE

OK

Cancel

LecturerConstraint

Remove

Display Name:

Web Resource Collection:

Name	URL Pattern	HTTP Method	Description
LecturerConstraint	/add.html, /edit.html, /find.html, /get_class...		Resources available to the lecturer

Add...

Edit...

Remove

☐ Enable Authentication Constraint

Description:

Role Name(s):

Edit

☐ Enable User Data Constraint

Description:

Transport Guarantee:

NONE

☒ Enable Authentication Constraint

Description:

Role Name(s):

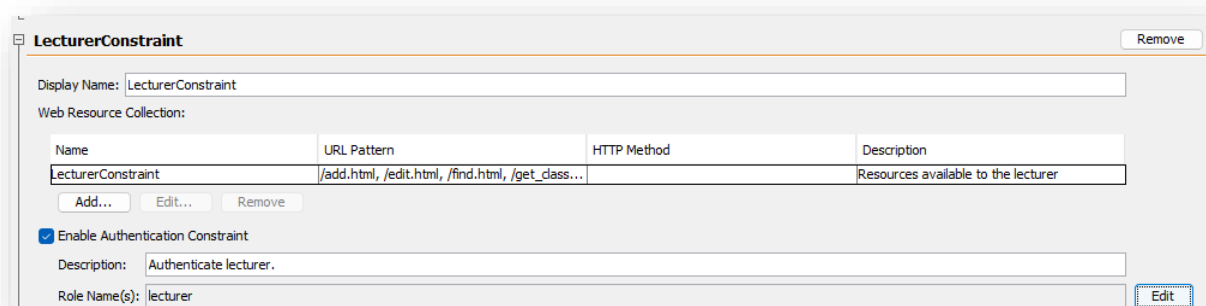
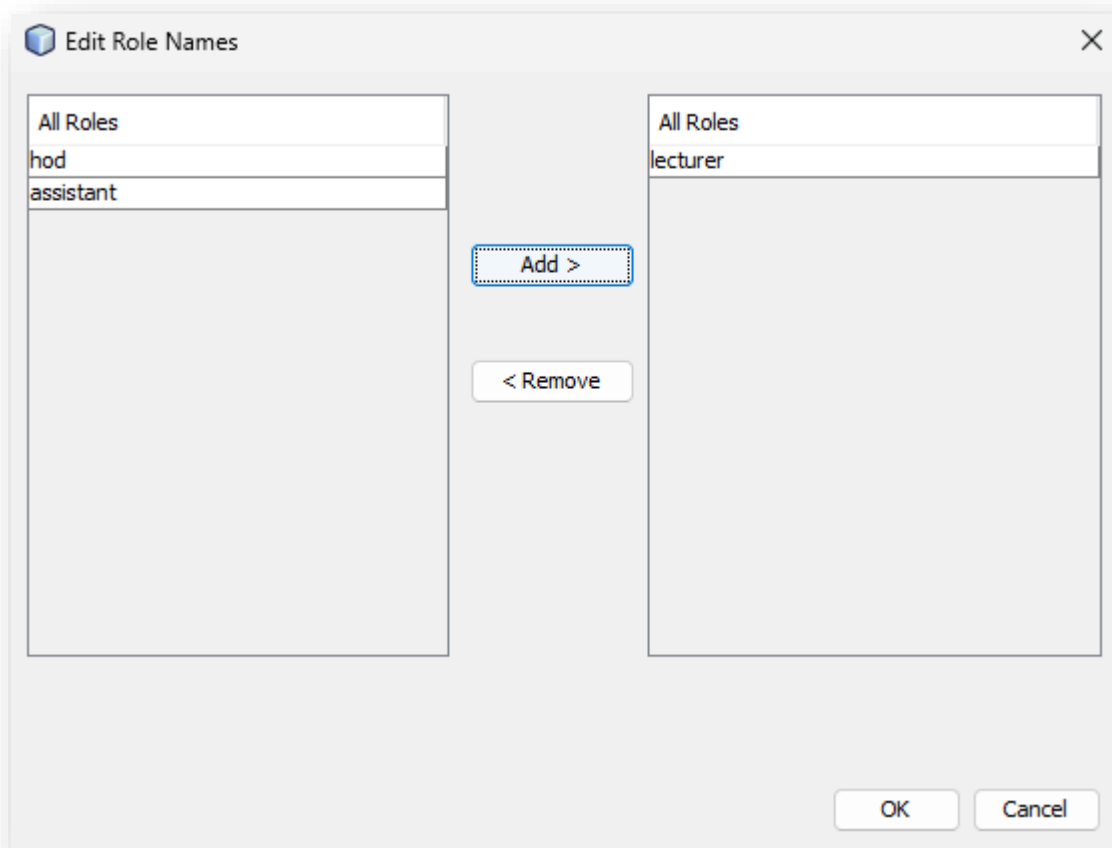
Edit

☐ Enable User Data Constraint

Description:

Transport Guarantee:

NONE



Assistant

Add Web Resource

Resource Name:

Description:

URL Pattern(s):

Use comma (,) to separate multiple patterns.

HTTP Method(s):

☒ All HTTP Methods
 ☐ Selected HTTP Methods

☐ GET
 ☐ POST
 ☐ HEAD
 ☐ PUT
 ☐ OPTIONS
 ☐ TRACE
 ☐ DELETE

OK

Cancel

AssistantConstraint

Remove

Display Name:

Web Resource Collection:

Name	URL Pattern	HTTP Method	Description
AssistantResources	/add.htm, /find.html, /AddStudentServlet.d...		Resources available to an assistant

Add...

Edit...

Remove

☐ Enable Authentication Constraint

Description:

Role Name(s):

Edit

☐ Enable User Data Constraint

Description:

Transport Guarantee:

NONE

▼

Edit Role Names

All Roles

hod

lecturer

Add >

< Remove

All Roles

assistant

OK

Cancel

AssistantConstraint

Remove

Display Name: AssistantConstraint

Web Resource Collection:

Name	URL Pattern	HTTP Method	Description
AssistantResources	/add.htm, /find.html, /AddStudentServlet.d...		Resources available to an assistant

Add...

Edit...

Remove

☒ Enable Authentication Constraint

Description: Authenticate an assistant

Role Name(s): assistant

Edit

☐ Enable User Data Constraint

Description:

Transport Guarantee: NONE

View the security information on the web.xml. Click on **Source**.

```
72 <security-constraint>
73     <display-name>HoDConstraint</display-name>
74     <web-resource-collection>
75         <web-resource-name>HoDResources</web-resource-name>
76         <description>Resources available to the HoD.</description>
77         <url-pattern>/remove.html</url-pattern>
78         <url-pattern>/RemoveStudentServlet.do</url-pattern>
79     </web-resource-collection>
80     <auth-constraint>
81         <description>Authenticate the HoD role.</description>
82         <role-name>hod</role-name>
83     </auth-constraint>
84 </security-constraint>
85 <security-constraint>
86     <display-name>LecturerConstraint</display-name>
87     <web-resource-collection>
88         <web-resource-name>LecturerConstraint</web-resource-name>
89         <description>Resources available to the lecturer</description>
90         <url-pattern>/add.html</url-pattern>
```

```

91      <url-pattern>/edit.html</url-pattern>
92      <url-pattern>/find.html</url-pattern>
93      <url-pattern>/get_class_per_age_group.html</url-pattern>
94      <url-pattern>/get_class_composition.html</url-pattern>
95      <url-pattern>/GetStudentServlet.do</url-pattern>
96      <url-pattern>/GetClassCountServlet.do</url-pattern>
97      <url-pattern>/GetClassCompositionServlet.do</url-pattern>
98      <url-pattern>/GetClasslistPerAgeGroupServlet.do</url-pattern>
99      <url-pattern>/AddStudentServlet.do</url-pattern>
100     <url-pattern>/FindStudentServlet.do</url-pattern>
101     <url-pattern>/EditStudentServlet.do</url-pattern>
102   </web-resource-collection>
103   <auth-constraint>
104     <description>Authenticate lecturer.</description>
105     <role-name>lecturer</role-name>
106   </auth-constraint>
107 </security-constraint>
108 <security-constraint>
109   <display-name>AssistantConstraint</display-name>

```

```

110   <web-resource-collection>
111     <web-resource-name>AssistantResources</web-resource-name>
112     <description>Resources available to an assistant</description>
113     <url-pattern>/add.htm</url-pattern>
114     <url-pattern>/find.html</url-pattern>
115     <url-pattern>/AddStudentServlet.do</url-pattern>
116     <url-pattern>/FindStudentServlet.do</url-pattern>
117   </web-resource-collection>
118   <auth-constraint>
119     <description>Authenticate an assistant</description>
120     <role-name>assistant</role-name>
121   </auth-constraint>
122 </security-constraint>
123 <login-config>
124   <auth-method>FORM</auth-method>
125   <realm-name>Click Add</realm-name>
126   <form-login-config>
127     <form-login-page>/login.jsp</form-login-page>
128     <form-error-page>/login_error.jsp</form-error-page>

```



```

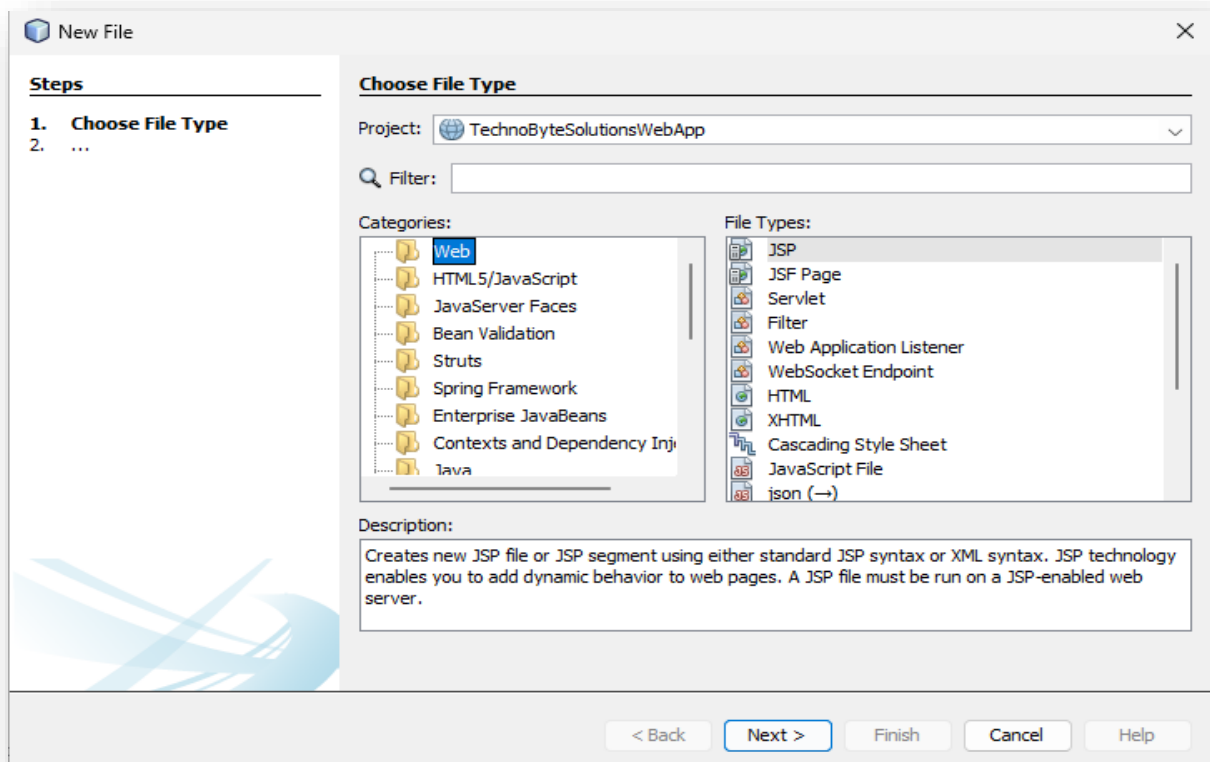
129     </form-login-config>
130 </login-config>
131 <security-role>
132     <description>The HoD role. This role can remove a student from the list.</description>
133     <role-name>hod</role-name>
134 </security-role>
135 <security-role>
136     <description>The lecturer role. This role can add, edit and do more functions.</description>
137     <role-name>lecturer</role-name>
138 </security-role>
139 <security-role>
140     <description>The assistant role. This role can add and search for students.</description>
141     <role-name>assistant</role-name>
142 </security-role>
143 </web-app>
144

```

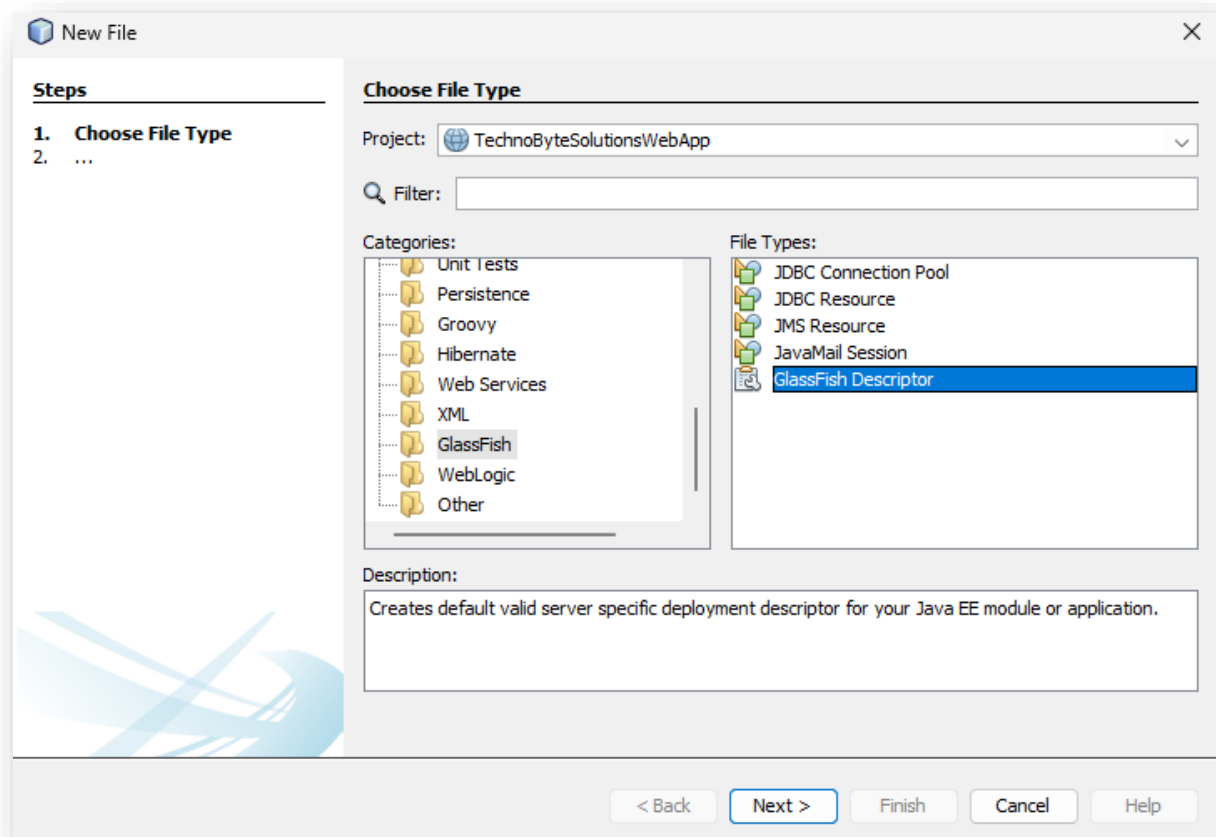
Part D: Role mapping

In this section we map the roles we have defined on web.xml to those stated under the file security realm in the server.

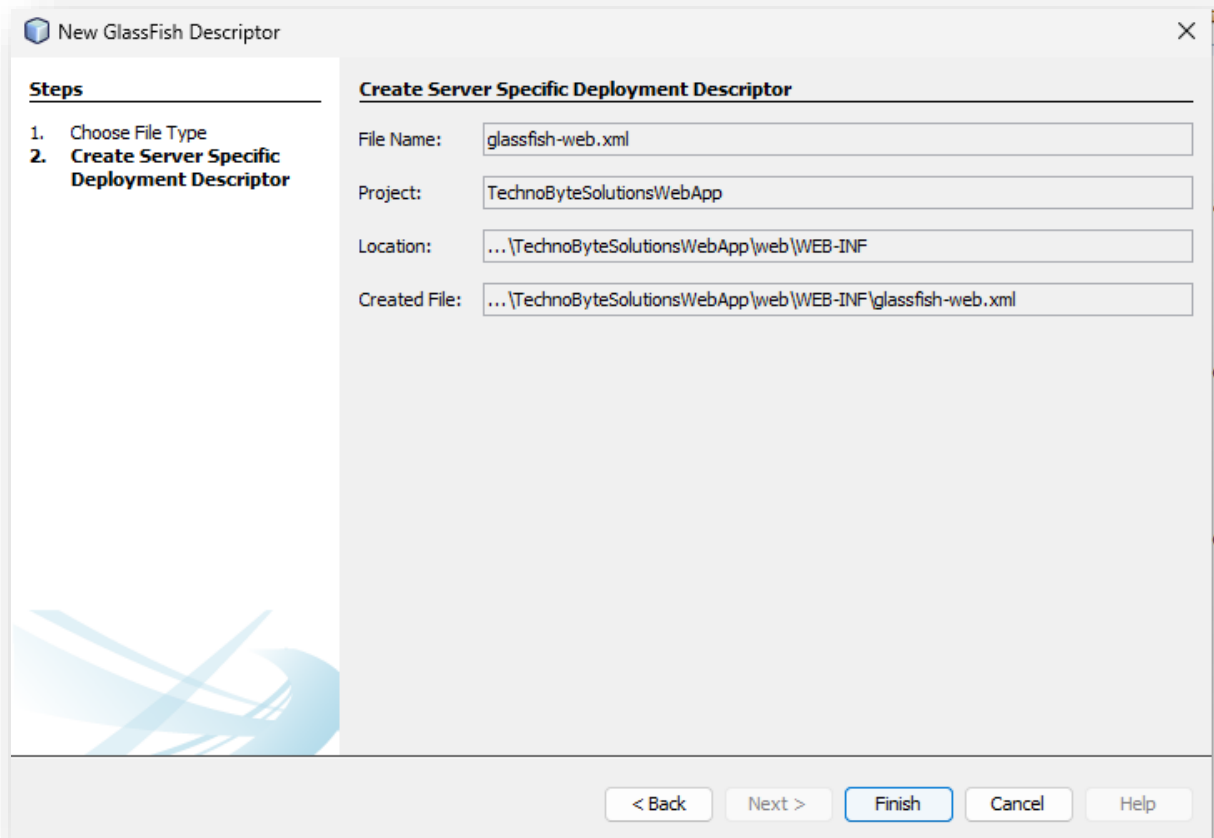
On the web project, right-click and select New | Other



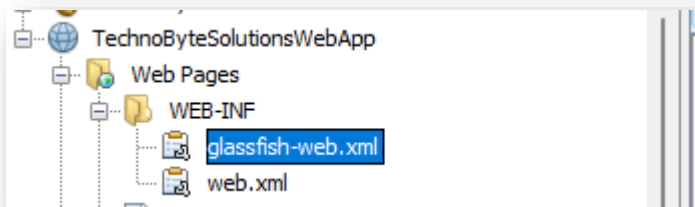
Under Categories, see



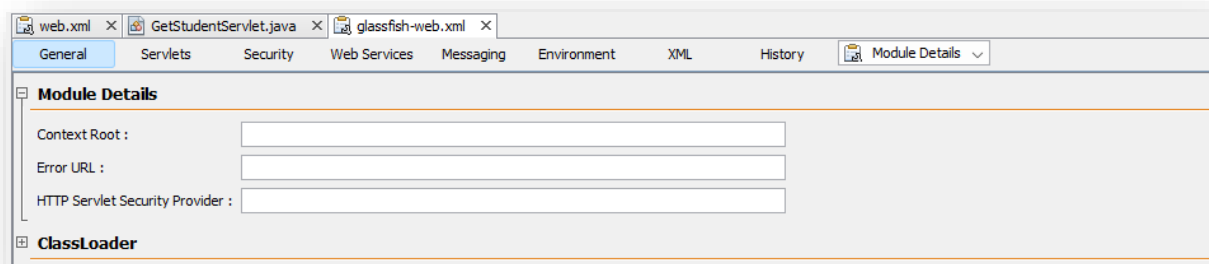
Click Next



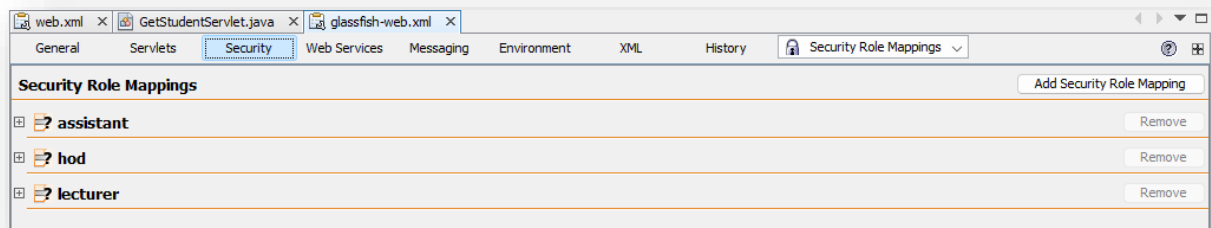
Click Finish



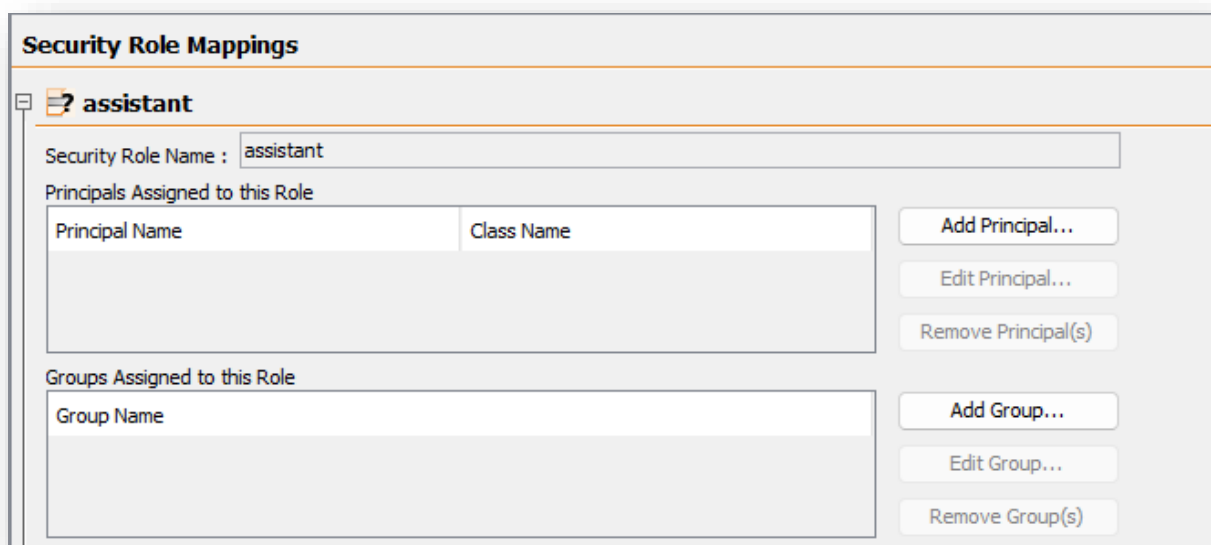
Open the glassfish xml file



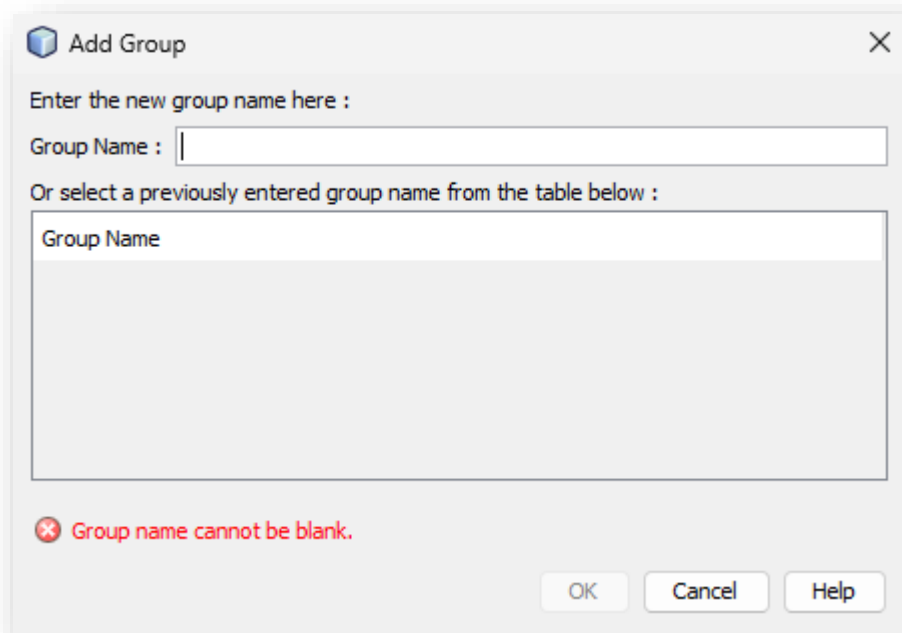
Click on Security



Expand **assistant**



Click on **Add Group**



The image shows a Windows-style dialog box titled "Add Group". It has a close button (X) in the top right corner. The main content area contains the text "Enter the new group name here :" followed by a text input field labeled "Group Name :". Below this is the text "Or select a previously entered group name from the table below :" followed by a table with a single header "Group Name" and an empty body. At the bottom left, there is a red error icon and the text "Group name cannot be blank.". At the bottom right, there are three buttons: "OK", "Cancel", and "Help".

Enter the new group name here :

Group Name :

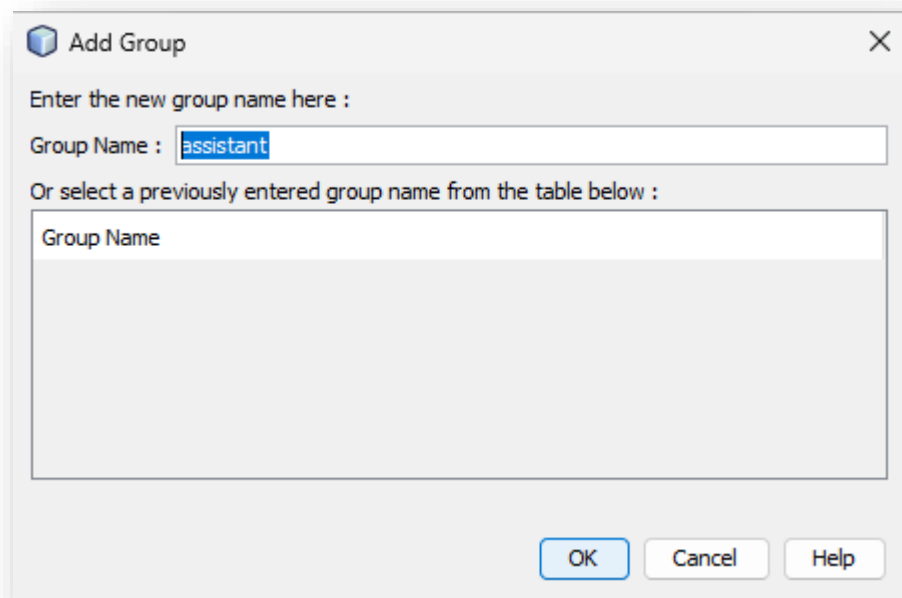
Or select a previously entered group name from the table below :

Group Name

❌ Group name cannot be blank.

OK Cancel Help

Enter **“assistant”**



The image shows the same "Add Group" dialog box as before, but now the text "assistant" is entered in the "Group Name" input field and is highlighted with a blue selection box. The error message and other elements remain the same.

Enter the new group name here :

Group Name :

Or select a previously entered group name from the table below :

Group Name

OK Cancel Help

Click OK

Security Role Mappings

 **assistant**

Security Role Name :

Principals Assigned to this Role

Principal Name	Class Name

Groups Assigned to this Role

Group Name	
assistant	

Expand **hod** and add **hod** to the group

 **Add Group** ✕

Enter the new group name here :

Group Name :

Or select a previously entered group name from the table below :

Group Name	
assistant	

hod

Security Role Name :

Principals Assigned to this Role

Principal Name	Class Name

Add Principal...
Edit Principal...
Remove Principal(s)

Groups Assigned to this Role

Group Name	
hod	

Add Group...
Edit Group...
Remove Group(s)

Expand lecturer and add the value to the group

Add Group

Enter the new group name here :

Group Name :

Or select a previously entered group name from the table below :

Group Name	
assistant	
hod	

OK Cancel Help

Add Group [X]

Enter the new group name here :

Group Name :

Or select a previously entered group name from the table below :

Group Name
assistant
hod

[OK] [Cancel] [Help]

Click on XML to view the output

web.xml x GetStudentServlet.java x glassfish-web.xml x

General Servlets Security Web Services Messaging Environment **XML** History

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE glassfish-web-app PUBLIC "-//GlassFish.org/
3 <glassfish-web-app error-url="">
4   <security-role-mapping>
5     <role-name>assistant</role-name>
6     <group-name>assistant</group-name>
7   </security-role-mapping>
8   <security-role-mapping>
9     <role-name>hod</role-name>
10    <group-name>hod</group-name>
11  </security-role-mapping>
12  <security-role-mapping>
13    <role-name>lecturer</role-name>
14    <group-name>lecturer</group-name>
15  </security-role-mapping>
16  <class-loader delegate="true"/>

```

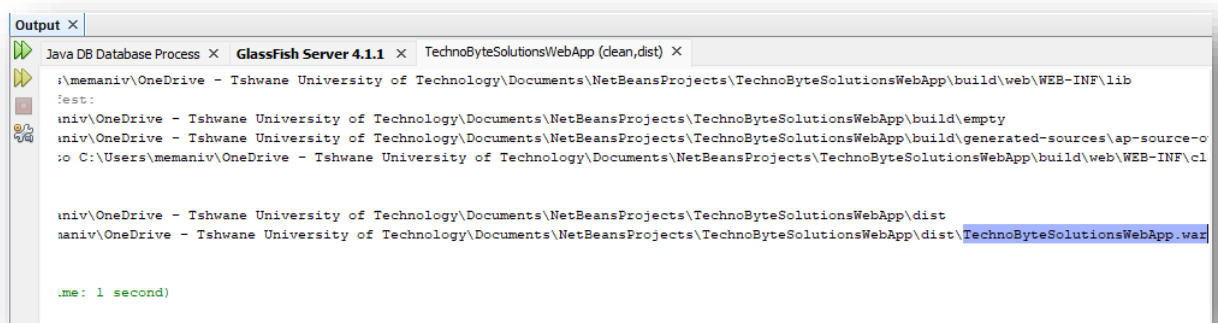


```

17 <jsp-config>
18   <property name="keepgenerated" value="true">
19     <description>Keep a copy of the generated servlet
20   </property>
21 </jsp-config>
22 </glassfish-web-app>
23

```

Clean and build the project



```

Output x
Java DB Database Process x GlassFish Server 4.1.1 x TechnoByteSolutionsWebApp (clean,dist) x
\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\web\WEB-INF\lib
Test:
univ\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\empty
univ\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\generated-sources\ap-source-o
:o C:\Users\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\web\WEB-INF\cl

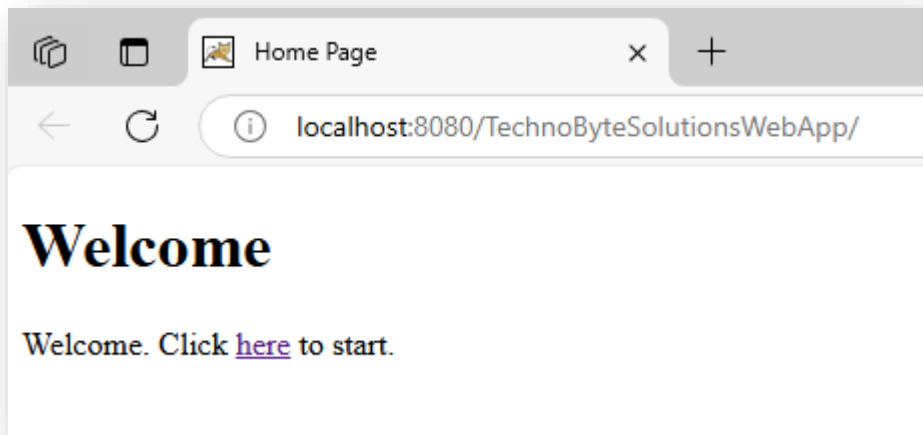
univ\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist
aniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist\TechnoByteSolutionsWebApp.war

.me: 1 second)

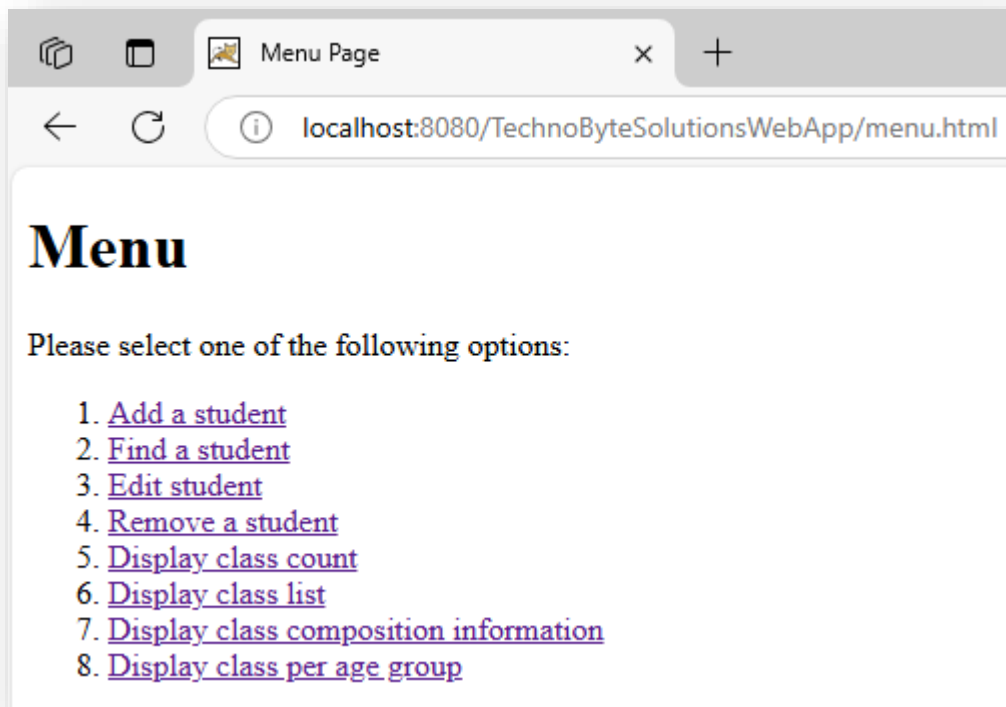
```

Step 6: Test the web application

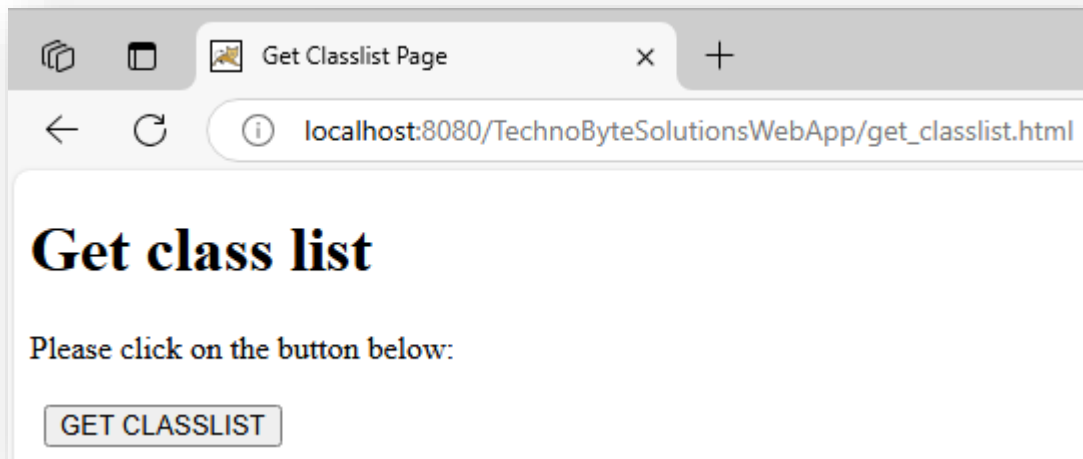
Deploy and run the web project



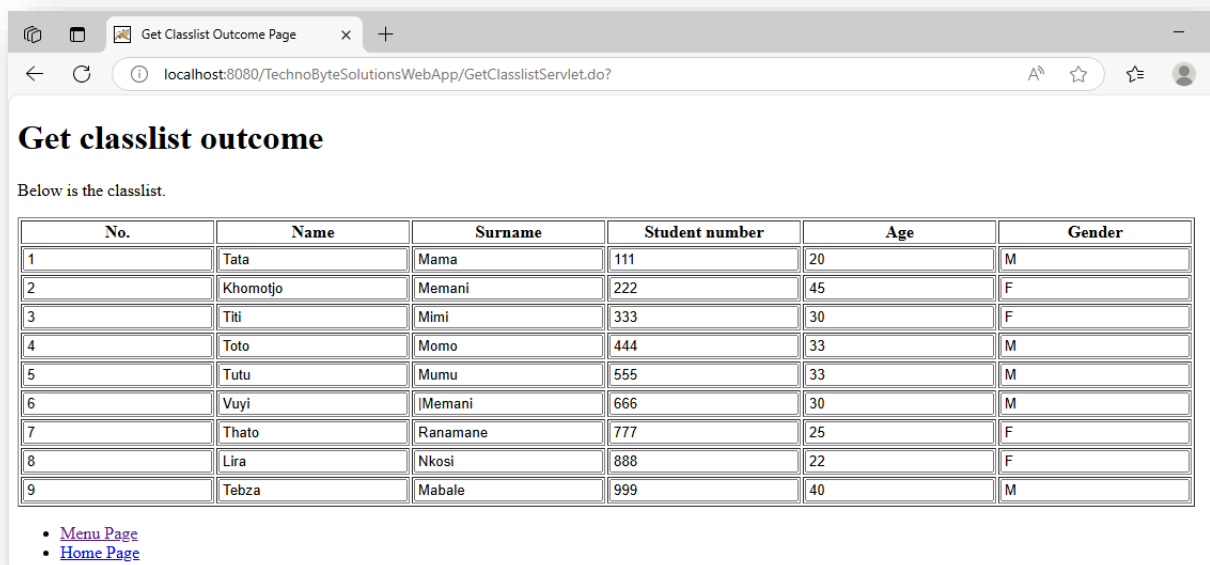
Click on the link



Click on the 6th link.



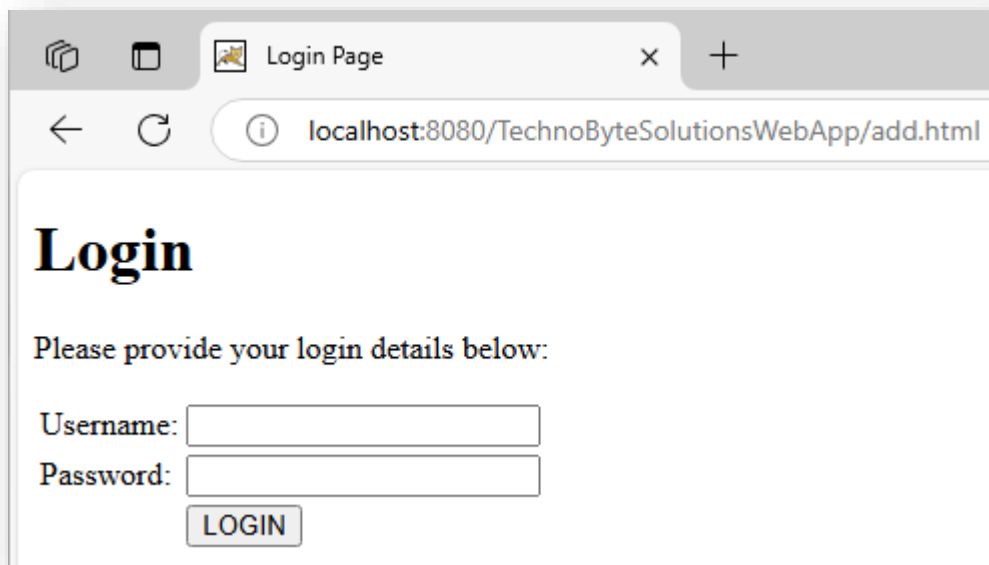
Click on the button



Pause and learn

The above functionality is possible because anyone can access the functionality. The functionality doesn't require users to be authenticated.

Click on the first link



Login Page

localhost:8080/TechnoByteSolutionsWebApp/add.html

Login

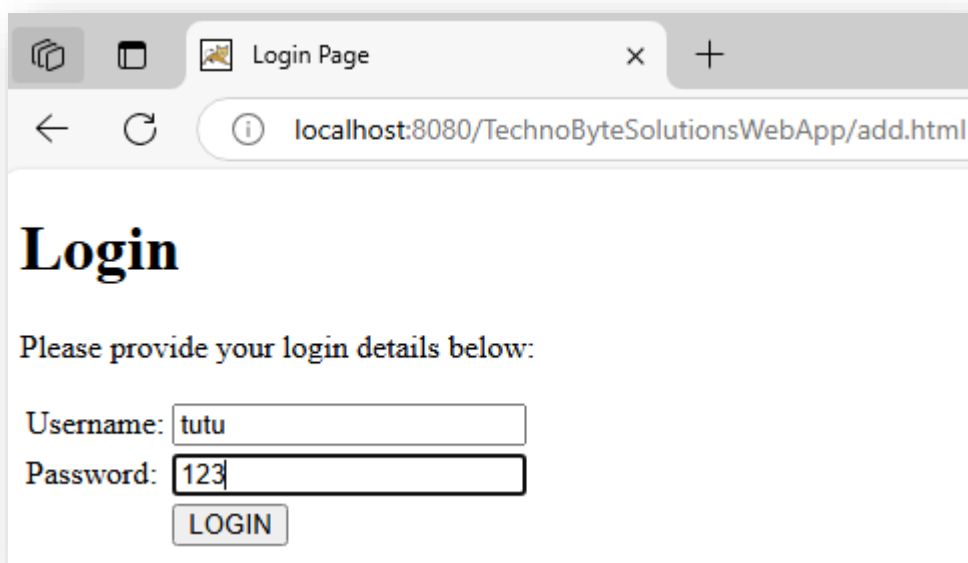
Please provide your login details below:

Username:

Password:

LOGIN

Oops!!! This is a protected functionality. It requires a user to be authenticated. Enter invalid details (tutu, 123)



Login Page

localhost:8080/TechnoByteSolutionsWebApp/add.html

Login

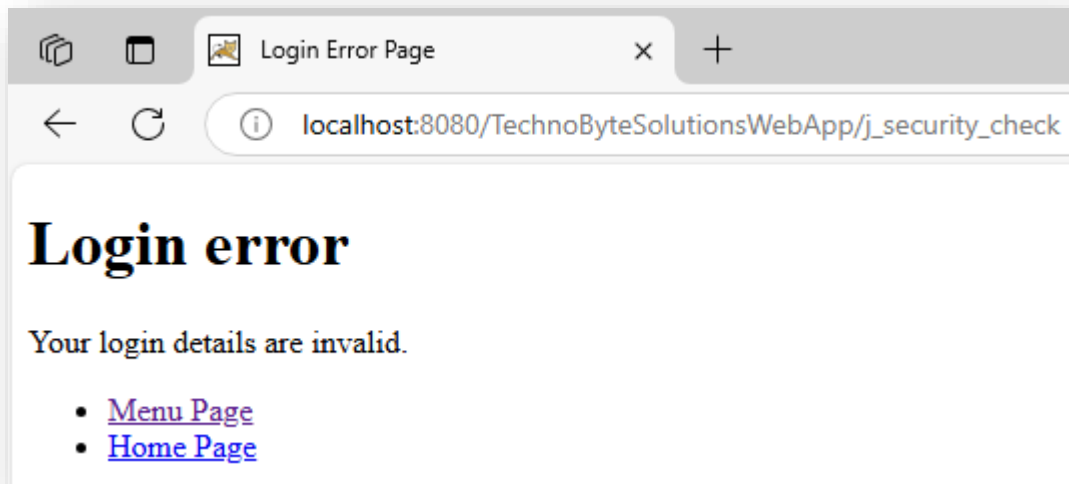
Please provide your login details below:

Username:

Password:

LOGIN

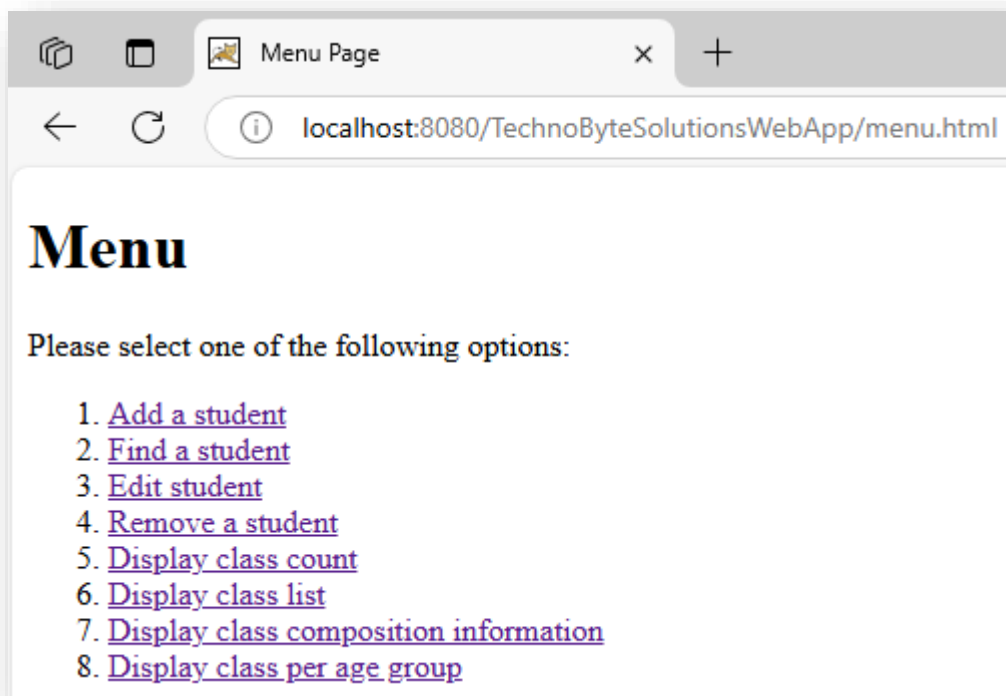
Login



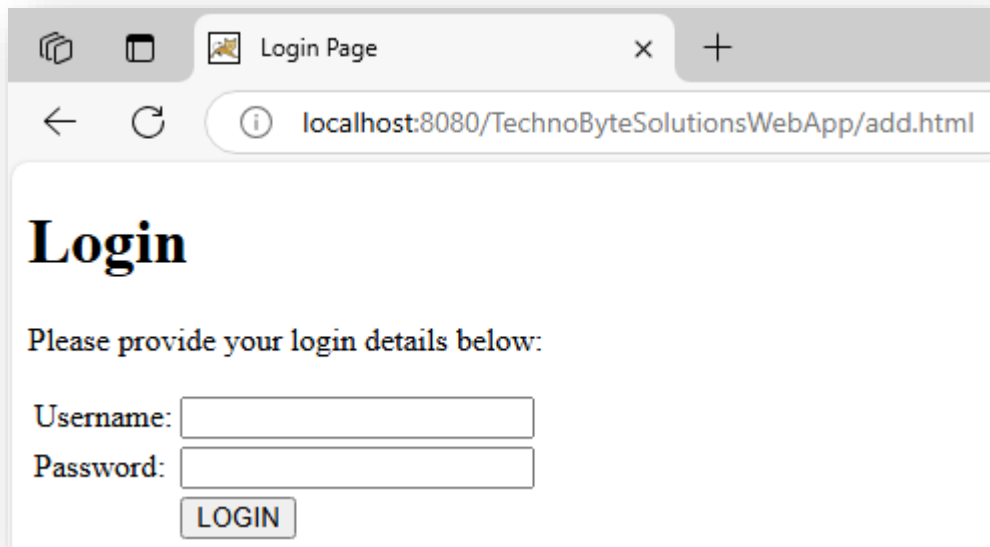
Pause and learn

The error is thrown because tutu is not a registered user on the system.

Go back to the menu page



Click again on the first link



Browser tab: Login Page

Address bar: localhost:8080/TechnoByteSolutionsWebApp/add.html

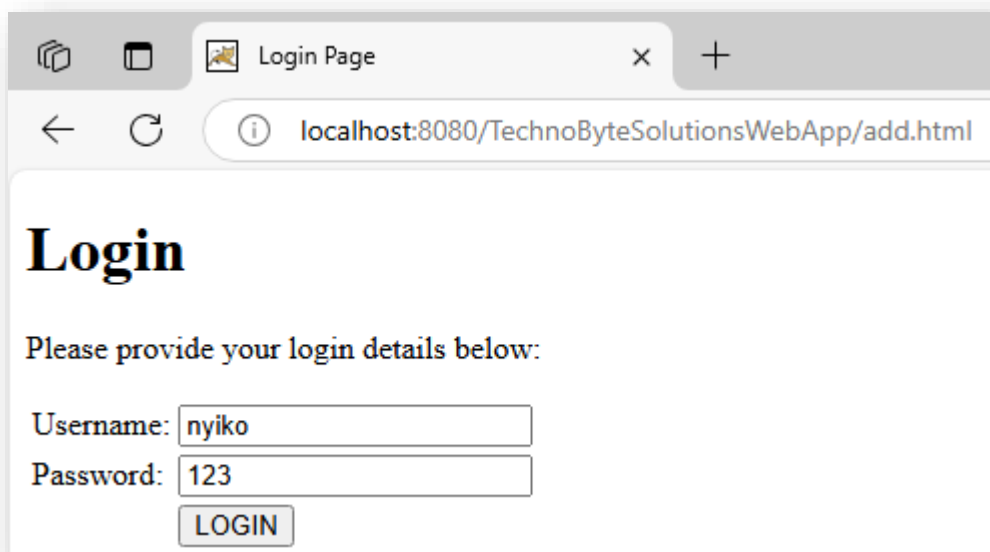
Login

Please provide your login details below:

Username:

Password:

Enter the details of the HoD (though we know that an HoD can only remove a student from the classlist)



Browser tab: Login Page

Address bar: localhost:8080/TechnoByteSolutionsWebApp/add.html

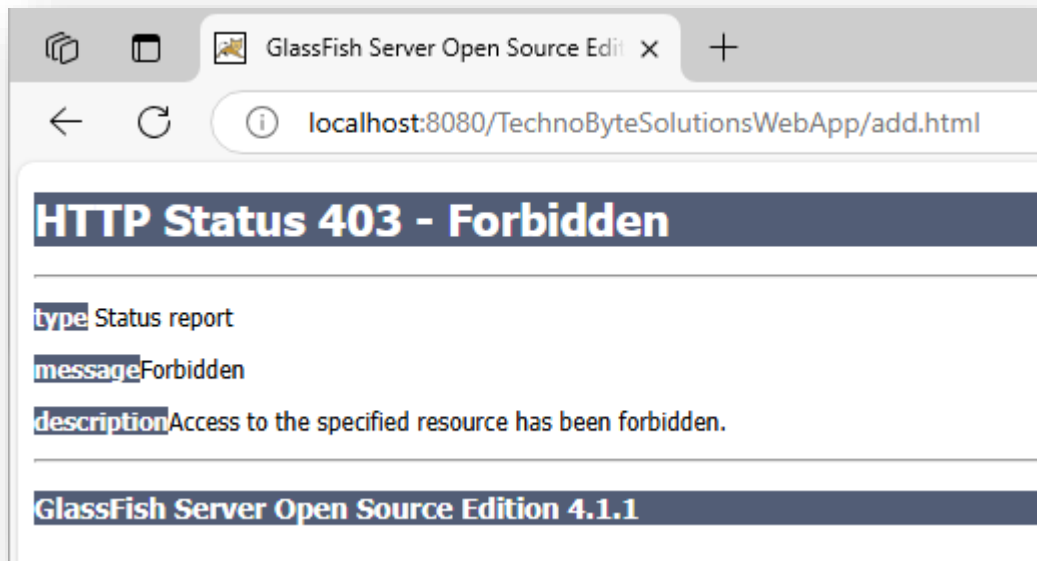
Login

Please provide your login details below:

Username:

Password:

Login

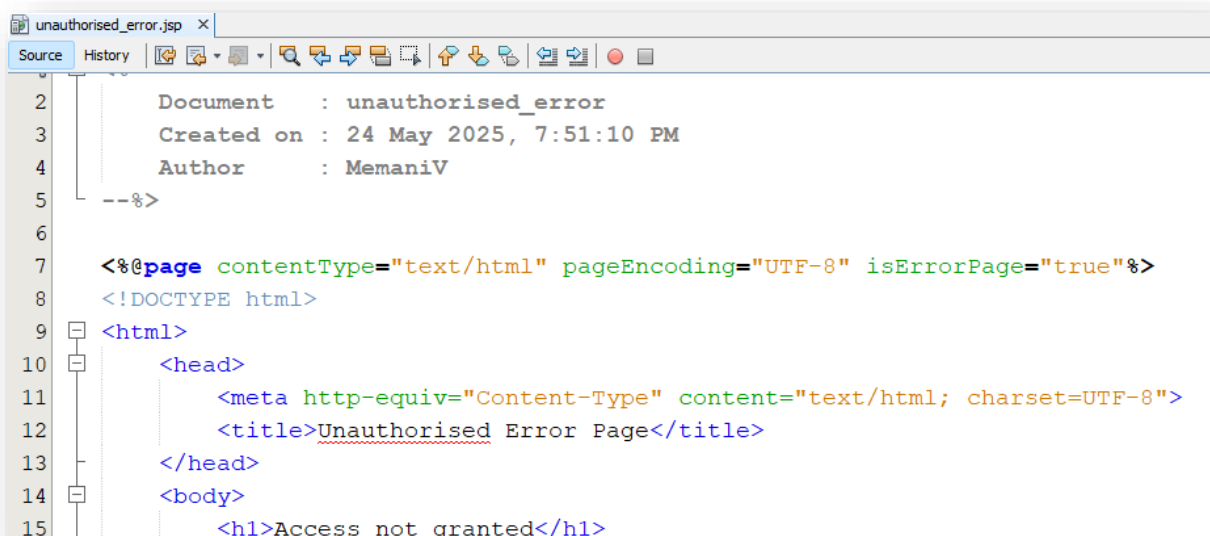


Pause and learn

A 403 error was thrown. This error says to things:

1. The user is authenticated (value).
2. But the user is not authorised to access the functionality. Of which is true in this case, an HoD can only perform a remove functionality.

Let's quickly create an error page to handle the 403 error. Create an error page called **unauthorised_error**



```

16      <p>
17          You are not allowed to access the requested functionality.
18      </p>
19      <ul>
20          <li><a href="menu.html">Menu Page</a></li>
21          <li><a href="index.html">Home Page</a></li>
22          <li><a href="LogoutServlet.do">Logout</a></li>
23      </ul>
24  </body>
25 </html>
26

```

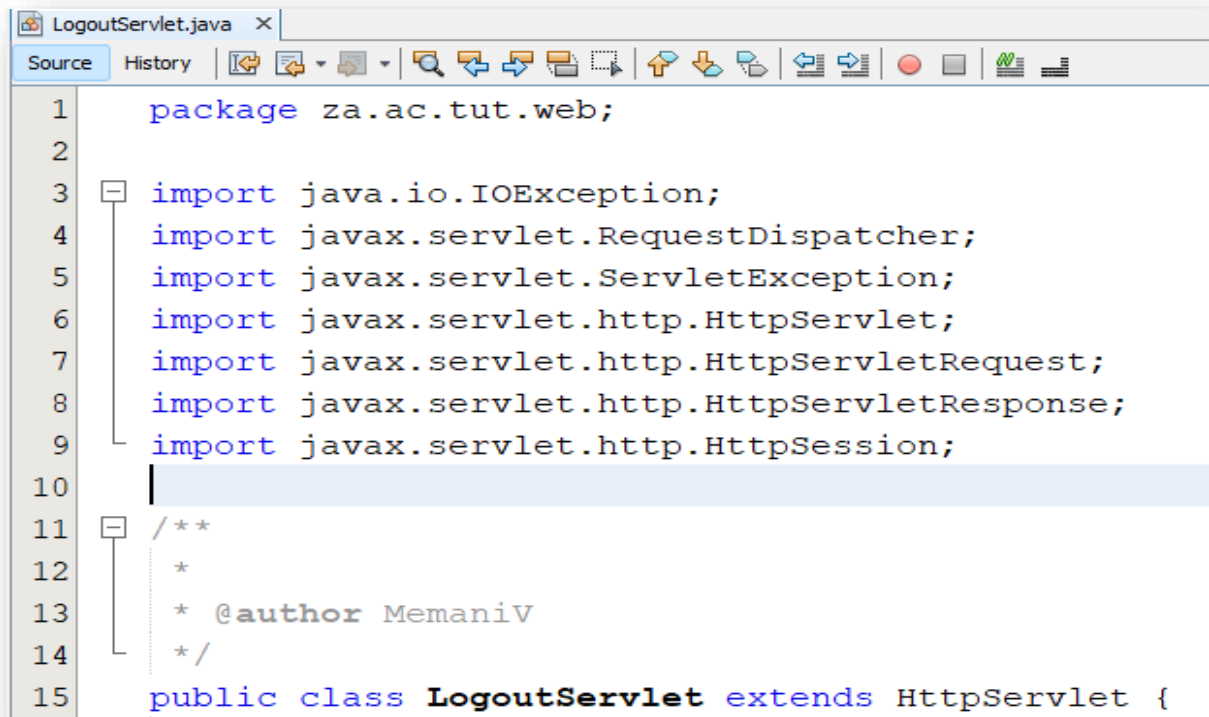
Define the error in the **web.xml** file

```

143      <error-page>
144          <error-code>403</error-code>
145          <location>/unauthorised_error.jsp</location>
146      </error-page>
147  </web-app>
148

```

Create the LogoutServlet class

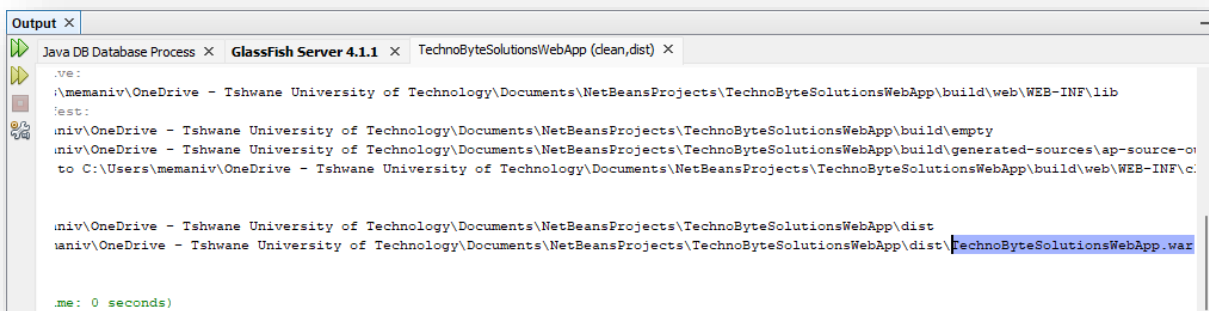


```
1 package za.ac.tut.web;
2
3 import java.io.IOException;
4 import javax.servlet.RequestDispatcher;
5 import javax.servlet.ServletException;
6 import javax.servlet.http.HttpServlet;
7 import javax.servlet.http.HttpServletRequest;
8 import javax.servlet.http.HttpServletResponse;
9 import javax.servlet.http.HttpSession;
10
11 /**
12  *
13  * @author MemaniV
14  */
15 public class LogoutServlet extends HttpServlet {
```



```
16 @Override
17 protected void doGet(HttpServletRequest request, HttpServletResponse response)
18     throws ServletException, IOException {
19     HttpSession session = request.getSession();
20     session.invalidate();
21
22     RequestDispatcher disp = request.getRequestDispatcher("index.html");
23     disp.forward(request, response);
24 }
25
26 }
```

Clean and Build

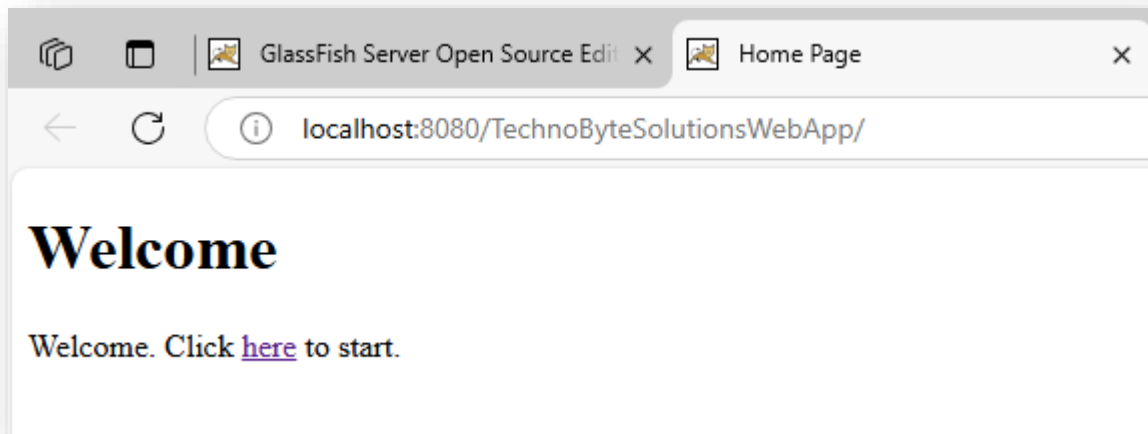


```
Output X
Java DB Database Process X GlassFish Server 4.1.1 X TechnoByteSolutionsWebApp (clean, dist) X
-ve:
i:\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\web\WEB-INF\lib
Test:
niv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\empty
niv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\generated-sources\ap-source-o
to C:\Users\memaniv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\build\web\WEB-INF\c

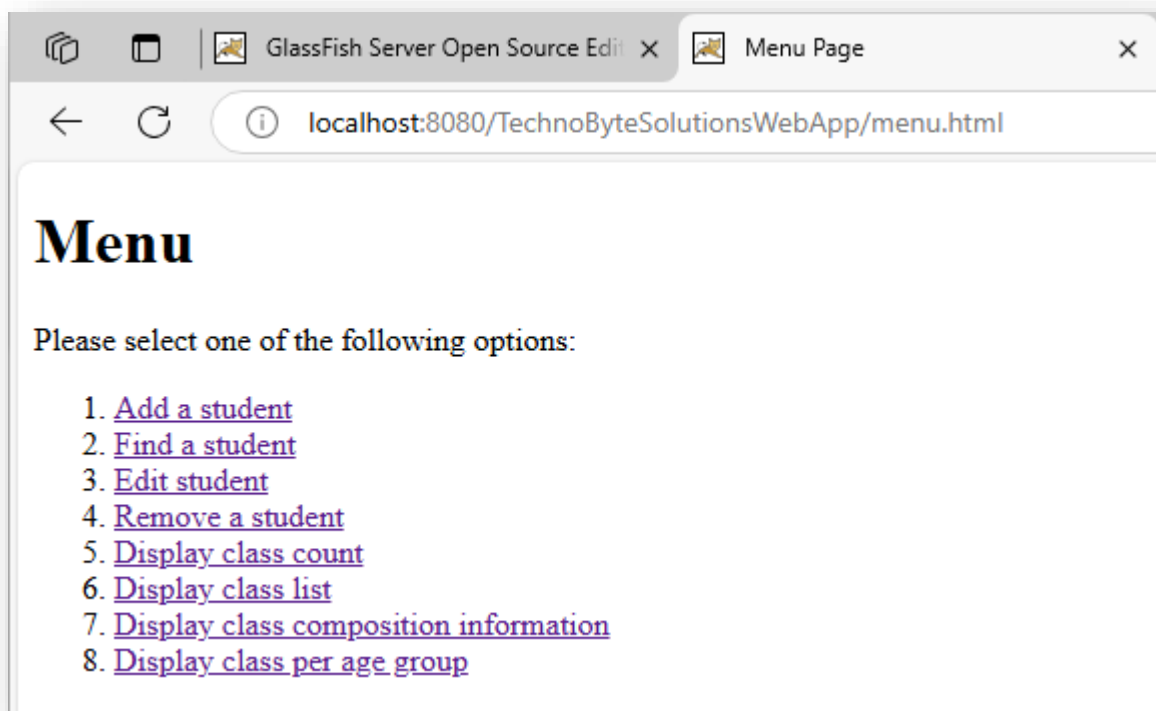
niv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist
niv\OneDrive - Tshwane University of Technology\Documents\NetBeansProjects\TechnoByteSolutionsWebApp\dist\TechnoByteSolutionsWebApp.war

.me: 0 seconds)
```

Deploy and run the project



Click on the link



Click on the first link

Login

Please provide your login details below:

Username:

Password:

Enter the details of the HoD

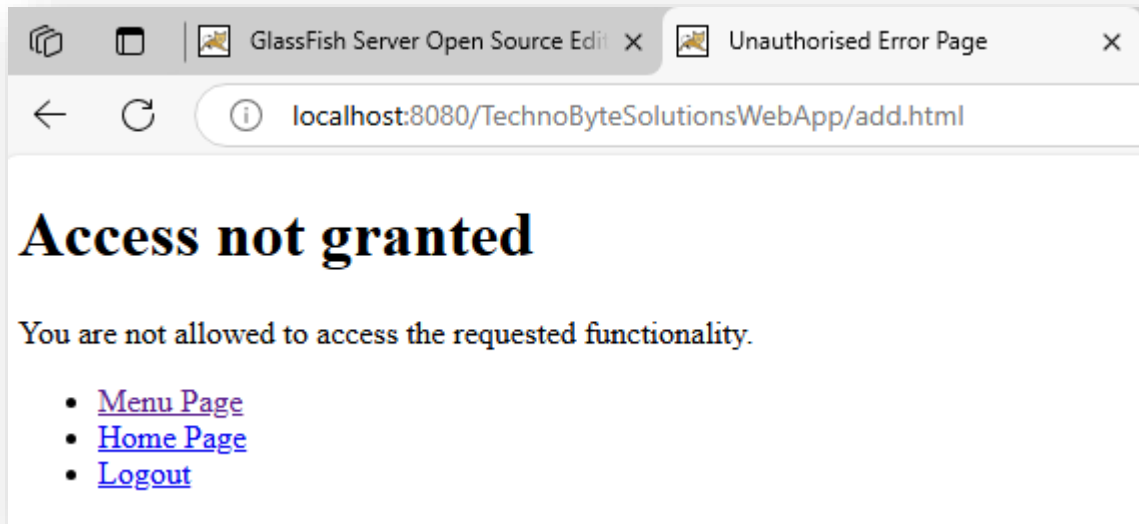
Login

Please provide your login details below:

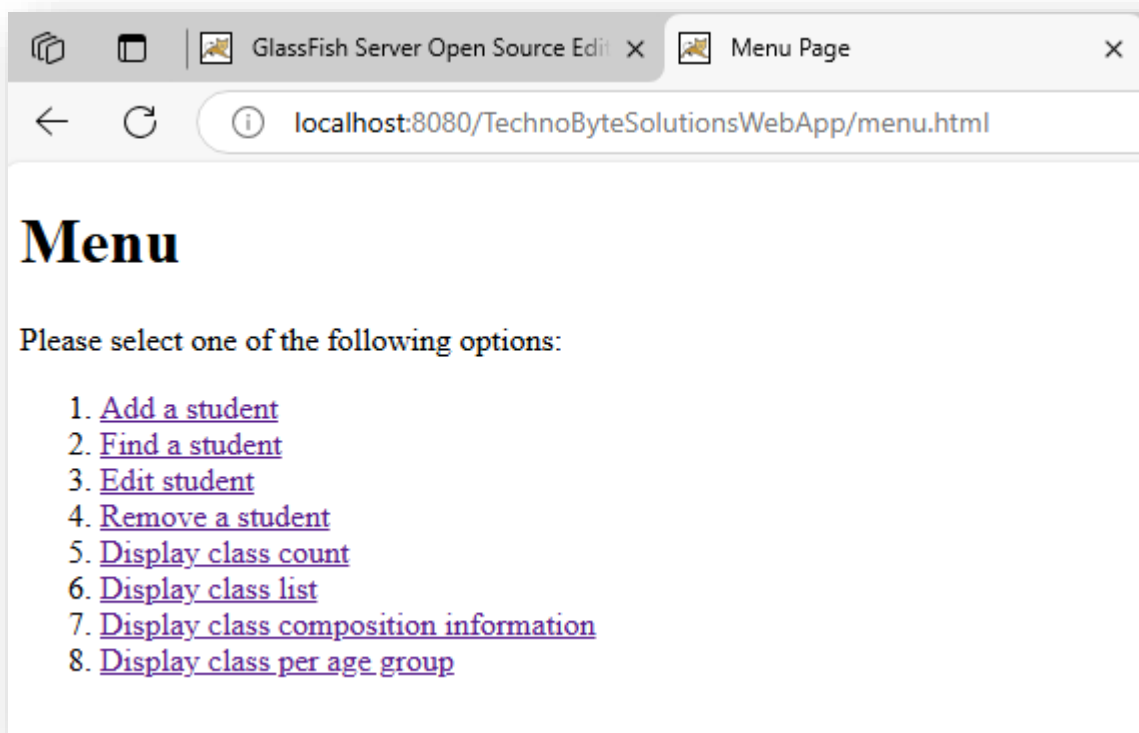
Username:

Password:

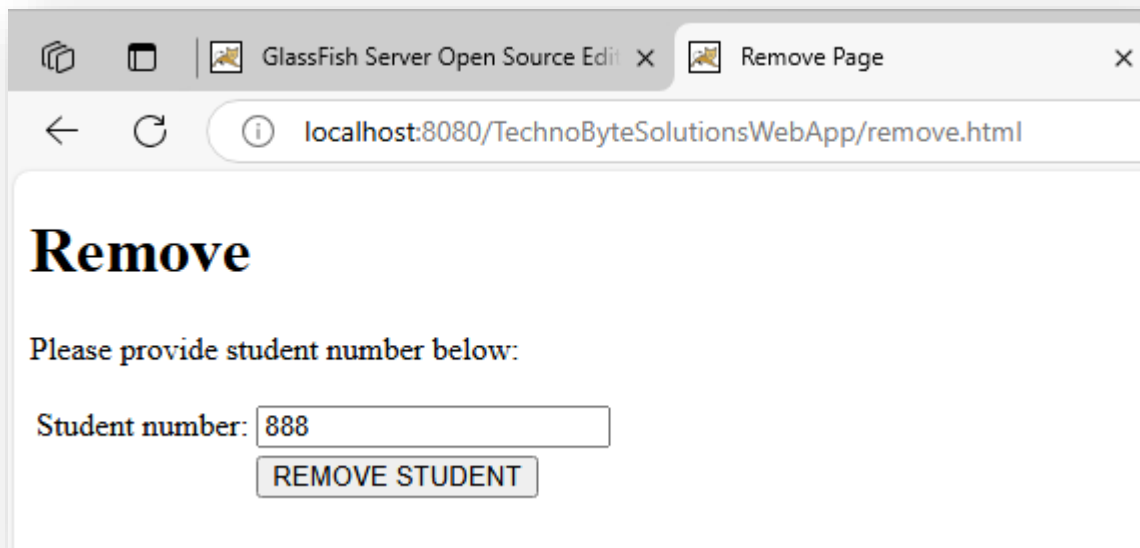
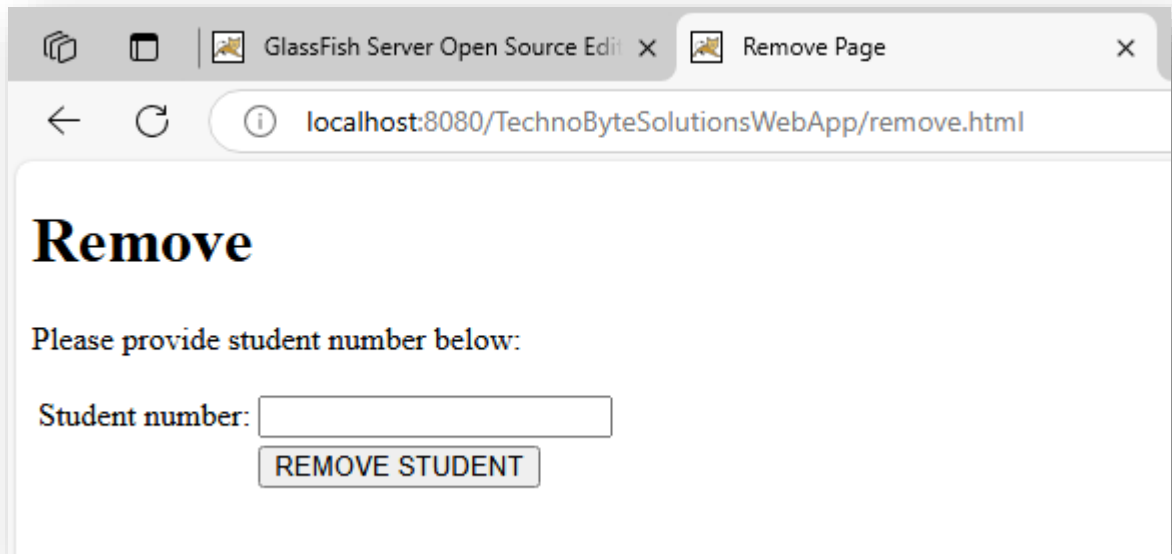
Login



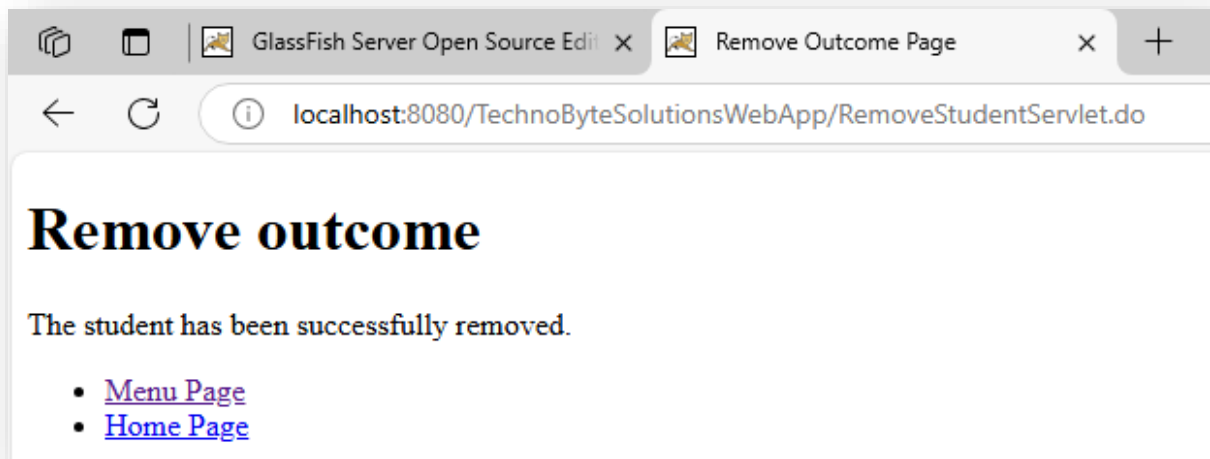
Go to the menu page



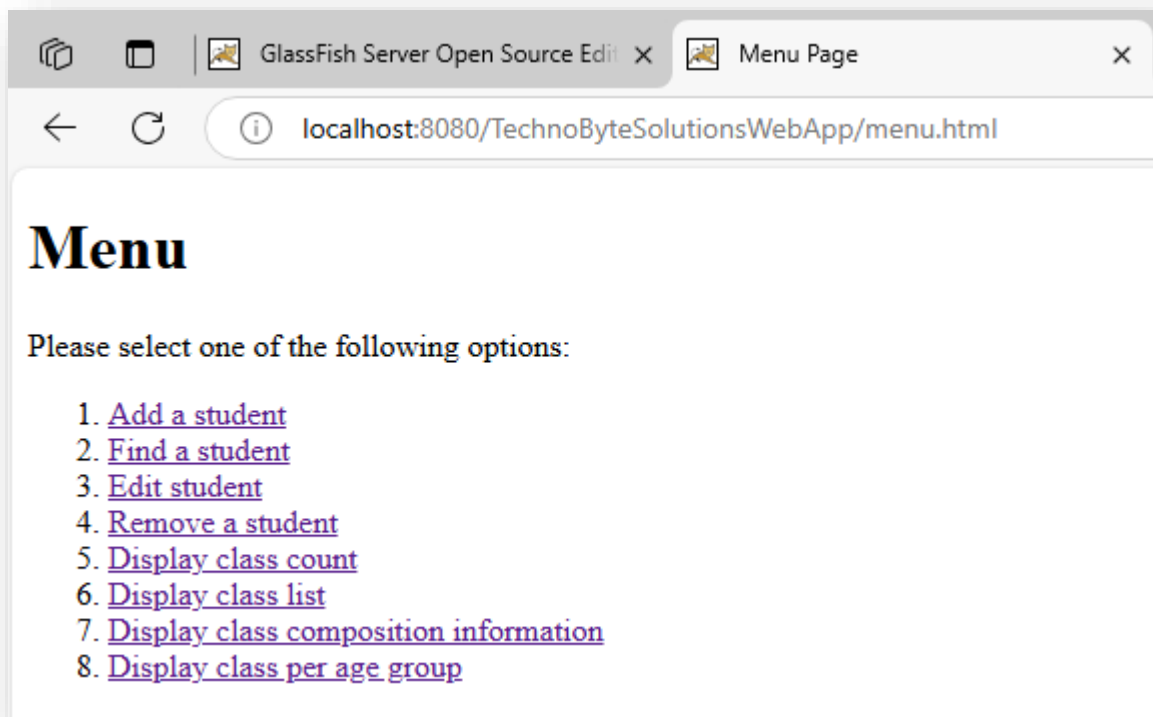
Click on the 4th link



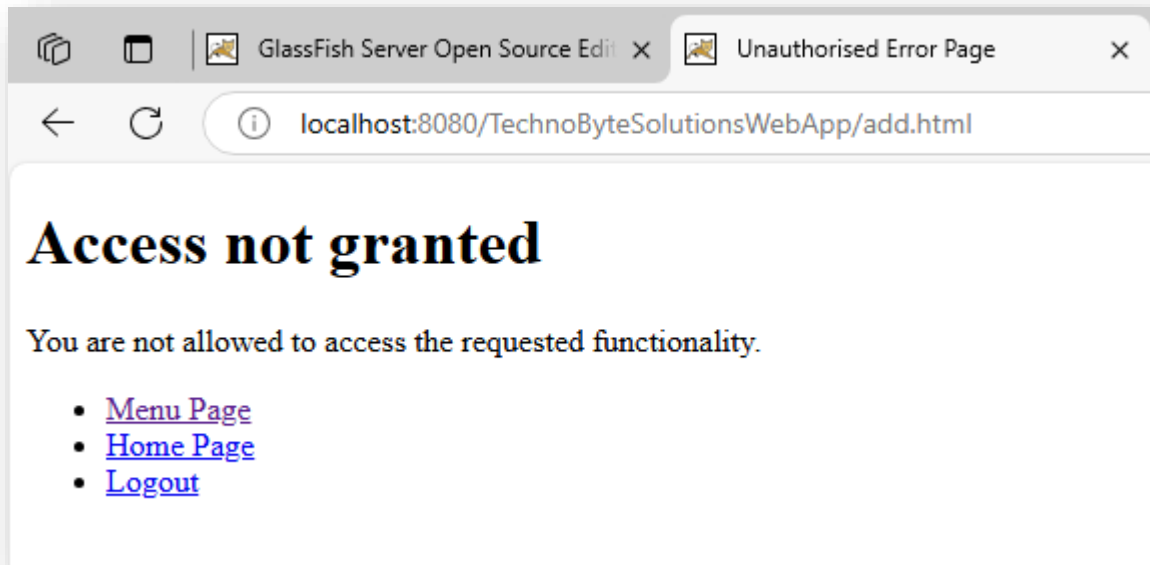
Remove the student



Click on the menu



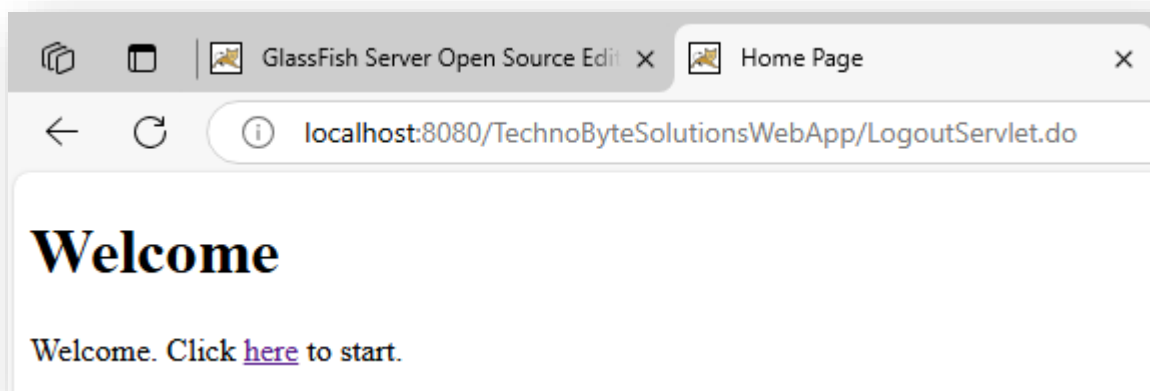
Click on the first link



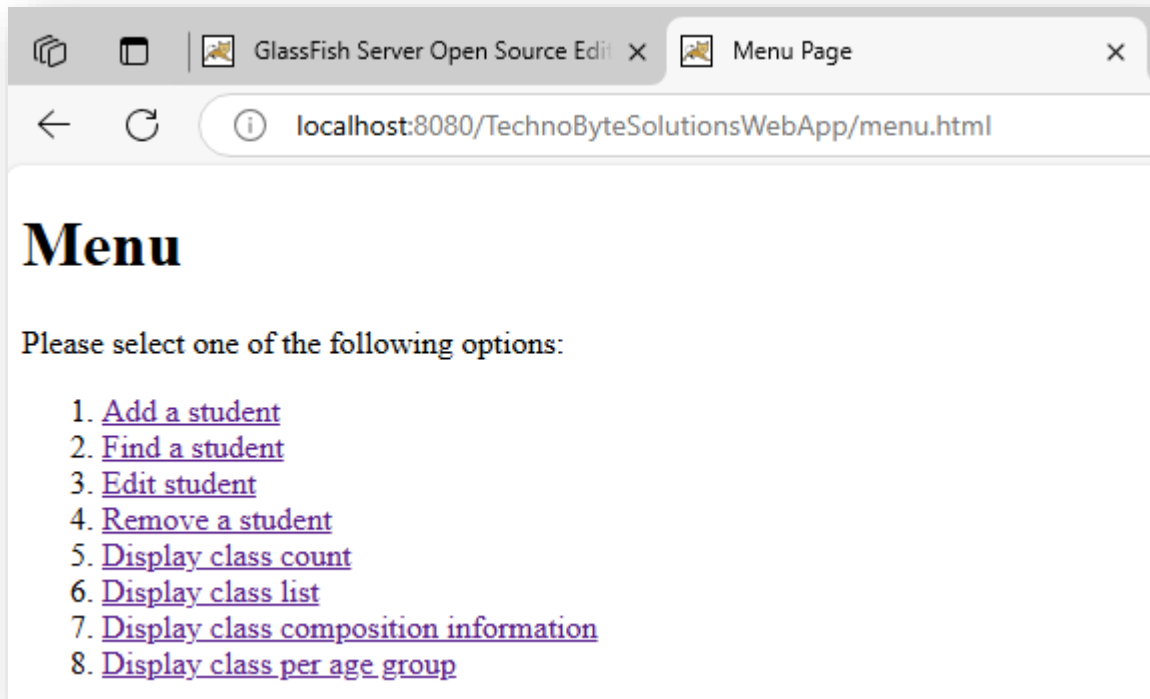
Pause and learn

The system recognizes that the HoD is still logged on. As a result, the HoD will only access what has been assigned to them, nothing more.

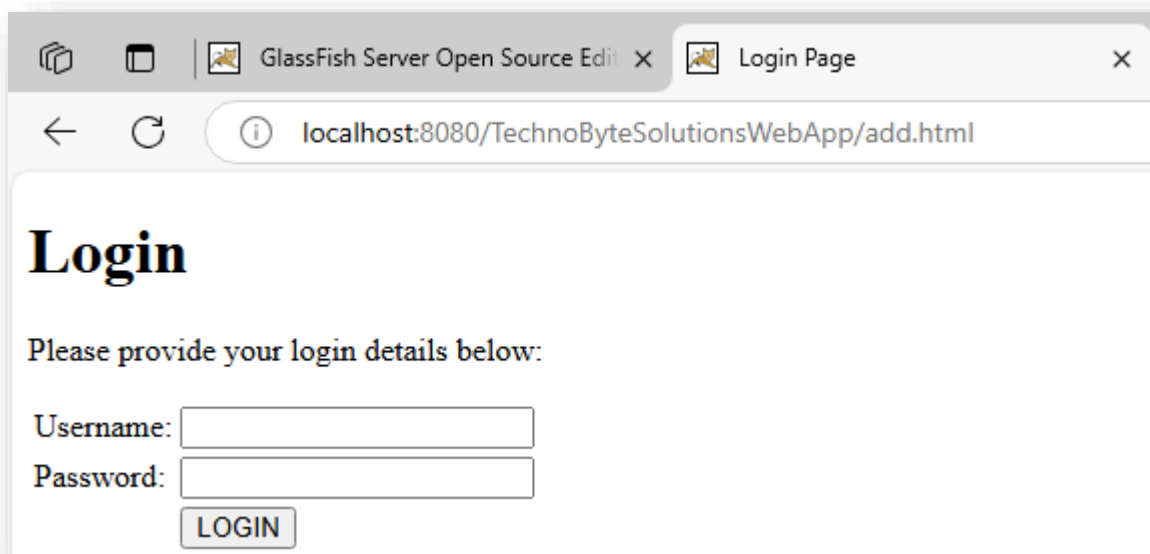
Logout as an HoD



Click on the link



Click on the first link



Enter the details of an assistant

Username: vuyi

Password: 456

LOGIN

Login

Enter the details

Name: Onkgopotse

Surname: Tiro

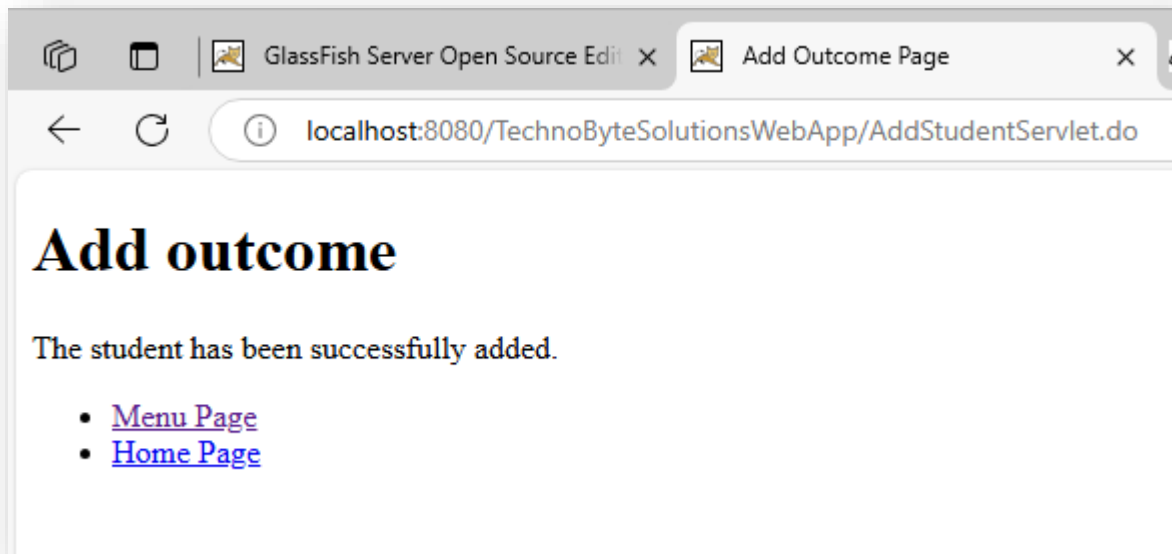
Student number: 567

Age: 50

Gender: Male

ADD STUDENT

Add the student



SELECT * FROM APP.STUDENT... X

Max. rows: 100 | Fetched Rows: 9 | Matching Rows:

#	STUDENT_NUMBER	AGE	CREATIONDATE	GENDER	NAME	SURNAME
1	111	20	2025-05-23 22:33:39.852	M	Tata	Mama
2	222	45	2025-05-24 13:16:36.866	F	Khomotjo	Memani
3	333	30	2025-05-23 22:34:23.869	F	Titi	Mimi
4	444	33	2025-05-23 22:34:50.946	M	Toto	Momo
5	555	33	2025-05-23 22:35:12.415	M	Tutu	Mumu
6	666	30	2025-05-24 13:08:18.763	M	Vuyi	IMemani
7	777	25	2025-05-24 13:08:46.059	F	Thato	Ranamane
8	999	40	2025-05-24 13:09:38.192	M	Tebza	Mabale
9	567	50	2025-05-24 20:25:17.23	M	Onkgopotse	Tiro

Conclusion

Thank you for your time. See you later. God bless you.

